

FACTORIZATION AND VANISHING OF SCHUR POLYNOMIALS TWISTED BY ROOTS OF UNITY AND RECIPROCAL PAIRS

CARLES MARÍN

ABSTRACT. Let $\mu_t = \{1, \zeta, \dots, \zeta^{t-1}\}$ be the full set of t -th roots of unity and let (z, z^{-1}) be a free reciprocal pair. We evaluate $s_\lambda(\mu_t, z, z^{-1})$ for every $t \geq 2$ and every partition λ with at most $t + 2$ parts, and no hypothesis on its shape: it is a signed product of exactly three factors over a fixed denominator, or zero, and all three arguments are read off the residue profile and the t -quotient of λ . Equivalently, and with no exponential in sight, it is a ratio of $\mathfrak{sl}_2$ characters: a triple product over the square of one. What the identity exposes is a structure, and the structure is the part we expect to outlast it: everything this evaluation can see of λ is a residue profile, an interval triple and a sign, and we isolate that datum as an *evaluation invariant*. The value factors through it; it is complete, two partitions with any sizes that share the invariant sharing the value; and, read as a multiset together with the sign, it is also minimal, two partitions sharing a nonzero value only if they share it. So an arbitrarily large partition is compressed to three integers and a sign, exactly. The proof is a Laplace expansion along the t frozen rows of the bialternant together with one cancellation lemma in the symmetric group, and it delivers the sign, which is the sorting sign already present in Littlewood's evaluation of $s_\lambda(\mu_t)$ and which we also give in closed form in the notation of Ayer and Kumari. Three consequences are recorded. First, a vanishing criterion: the character vanishes exactly when some residue class modulo t is empty, or when two distinguished classes are concentric as intervals — and the second case occurs only for t even, so for odd t an empty class is the only way to vanish. Second, an extension of a recent independence criterion of Ayer and Kumari: their theorem determines when a character is independent of the variables it is twisted by, and we show that for two-row shapes on the reciprocal locus the criterion acquires exactly one extra family, which we classify by its core and quotient. Their theorem is stated for free variables and a reciprocal pair is not free, so the locus lies outside its hypotheses rather than against them, and what we determine is what the criterion becomes there. Third, an enumerative reading: the identity is a product formula for a root-of-unity weighted count of semistandard tableaux in which a parameter is still free, and at $t = 2$ this is a (-1) -enumeration of plane partitions in a box refined by that parameter; a sign-reversing proof of it has to carry that parameter, and we supply the half of the cancellation that crosses the splitting of the alphabet. Finally we show the factorization is isolated: it fails under each of the four deformations of the alphabet — more reciprocal pairs, more orbit variables, the orbit replaced by a coset, the reciprocal pair replaced by a free pair — and a single reading of the proof accounts for all four failures at once.

What survives the first of those deformations is the *zero locus*, and the last section is about it. Write $\Psi_r = s_\lambda(1, -1, z_1, z_1^{-1}, \dots, z_r, z_r^{-1})$. Two conditions make Ψ_r vanish: the beta set of λ having constant parity, and λ being self-complementary of odd width. *That* implication we prove for every r and every λ , and we claim nothing for it: it is a short corollary of the complementation identity over an index family Ayer and Behrend already single out, and at the same alphabet *without* the letter -1 their theorems make those very shapes factorize instead — the determinant of the alphabet is what decides. The new content is the *converse*, that nothing else vanishes, and we prove it for every r and every λ by an extremal argument in the degree filtration whose only external input is a rigidity theorem for products of Schur polynomials. Read on the group rather than on the alphabet,

Date: August 17, 2026.

2020 Mathematics Subject Classification. Primary 05E05; Secondary 05A15, 05E10, 20G05.

Key words and phrases. Schur polynomials, roots of unity, reciprocal variables, factorization, cores and quotients, signed enumeration.

that criterion classifies exactly the irreducible polynomial $GL(N)$ -modules whose restriction to $O(N)$ is stable under twisting by the determinant. The same argument settles a second line of the (t, r) plane and settles it outright, with no external input at all: for every *odd* t and every r , the character vanishes if and only if a residue class is absent from the beta set, so the concentric branch is a phenomenon of even t alone. That leaves $t \geq 4$ even and $r \geq 2$ as the one region where the zero locus is open, and there we record the necessary condition the argument does give. One pair also turns out not to be typical: there Ψ_1 is a genuine character up to sign, so a single order-two specialization decides whether it vanishes, and from two pairs on that stops being true.

1. Introduction

A Schur polynomial is a character. Written $s_\lambda(x_1, \dots, x_n)$, it is the trace of a matrix with eigenvalues $x_1, \dots, x_n$ acting on the irreducible representation of GL_n labelled by λ , and reading it that way turns every choice of the variables into a question about a group element rather than about a polynomial. Two such questions are as old as the subject. At the identity, all $x_i = 1$, the trace is a dimension, and it grows with λ . At an element of finite order it stops growing: if the variables run over a full orbit of t -th roots of unity the value is 0 or ± 1 , and which of the three it is depends on a divisibility property of λ and not on its size. Something that could have been arbitrarily large has been compressed to almost nothing, and the reason is arithmetic. Everything below is about how far that compression reaches: how much of an alphabet one may add before the collapse stops, and what exactly survives of a partition when it does not.

Two structural reasons for such a collapse are known, and each has produced its own literature. The first is that the alphabet is an *orbit*: a full set of t -th roots of unity is the eigenvalue list of an element of order t , and orbits are what the operators of that line are built to see. The second is that the alphabet is *closed under* $x \mapsto x^{-1}$: such a list is the spectrum of an orthogonal matrix, so the GL character is being restricted to a smaller group, and it is the restriction that factorizes. The two reasons are independent of one another, and so are the two literatures; the paragraphs that follow trace each, and §7 shows that neither survives being pushed into the other's territory. The alphabet of this paper is the smallest one that carries one reason of each kind.

Figure 1 is the paper read as a single chain: each answer is what raises the next question, and each pearl carries the section that settles it. Figure 2 draws the alphabet itself, which is the one object everything below is about.

A partition can be arbitrarily large, and an alphabet of $N = t + 2$ letters can only see so much of it. This paper settles, for one alphabet, exactly how much. The full set of t -th roots of unity together with one free reciprocal pair sees a multiset of three integers and a sign, and nothing else whatever: two partitions of any sizes that agree on that datum take the same value there. The closed form of Theorem 3.1 is how we prove it, and the compression is the part we expect to outlast the formula.

The evaluation of Schur polynomials at roots of unity is classical and remarkably clean. Littlewood and Richardson [LR34] showed that s_λ specializes to 0 or ± 1 when the variables are taken to be all the t -th roots of unity, the sign being a sorting sign and the vanishing being governed by the t -core of λ ; and in the same work they generalized this to an alphabet carrying a free variable in every letter. The latter result was rediscovered by Prasad [Pra16], given a new proof by Karmakar [Kar22], extended to all classical groups by Ayyer and

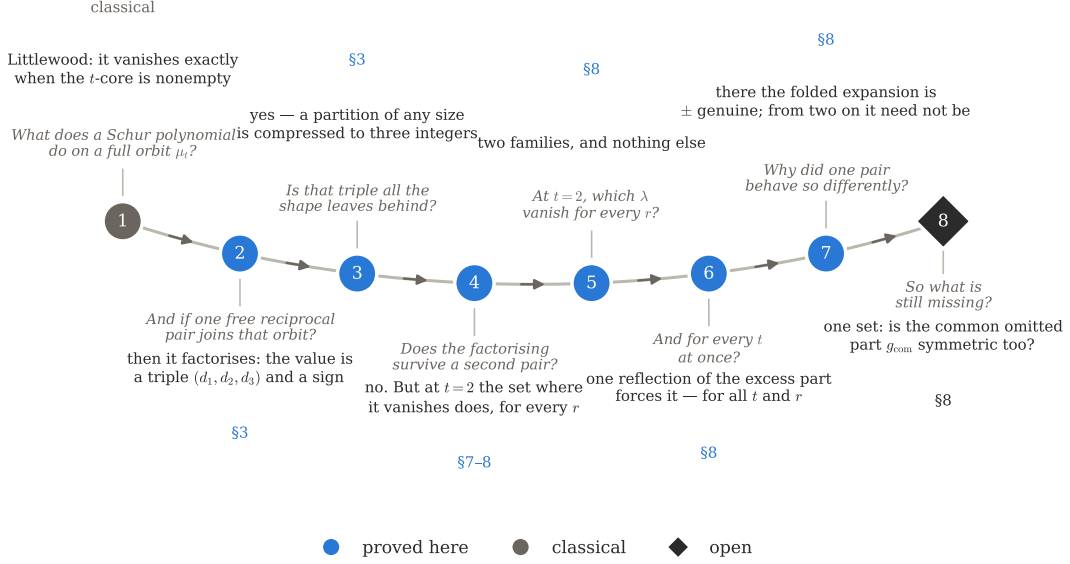


Figure 1. The paper as one thread. Each question is answered by the pearl below it, and each answer is what makes the next question the natural one to ask: the orbit invites the free pair, the factorisation invites the triple, the triple invites the second pair, and the second pair turns a formula into a locus. Colour marks what is classical, what is proved here, and what is left open, and the tag under each answer is the section that settles it. Drawn by `anc/fig-intro.py`.

Kumari [AK22] — who also identified the earlier, almost unnoticed work of Lecouvey [Lec09] — and carried to the universal characters by Albion [Alb23]. Kumari [Kum24] extended the family further, Karmakar [Kar24] treated elements of order two, and Ayyer and Kumari [AK25] added several more specializations, and Albion [Alb25] related the factorizations to z -asymmetric partitions and plethysm. Sathish Kumar [SK23] carried the factorization to flagged skew shapes, where it produces a family of Demazure characters; the flags there are required to be multiples of t , which is the same demand for root-of-unity structure in another guise. Read together, that line does more than accumulate cases. The statement migrates from a group to an alphabet: once the characters are universal, evaluating one is a question about which letters are written down, and the classical types stop being separate theorems — [AK25, Theorem 5.3] is one statement for s , sp and o at once. That migration is what makes the question of this paper askable, because the alphabet below belongs to no group in the list; and the tools we borrow in §8 are the ones that migration produced. We refer to Albion [Alb23] for an account of the whole line.

Read for its *value*, Littlewood’s theorem is a factorization. Read for its *zeros* it is something else — a classification of the λ that die at an element of order t — and that second reading has a descendant of its own, which is worth naming here because §8 turns out to be its cousin. Prasad [Pra16, Theorem 2] asks the vanishing question not at a point but along a whole family: on the non-identity component of $GL_m^n \rtimes \mathbb{Z}/n$ that paper determines when the character is identically zero *as a function of the free parameter*, and the answer is again read off the residue classes of $\lambda + \delta$ — each class must be represented exactly m times. At

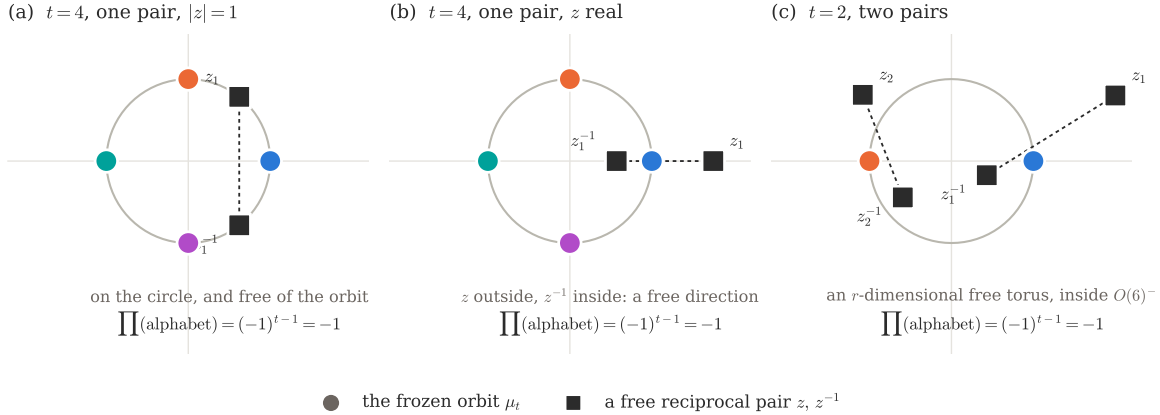


Figure 2. The alphabet, in the complex plane. The orbit μ_t is frozen — a regular t -gon on the unit circle, coloured by residue — and each reciprocal pair is one free complex direction, drawn joined by a dashed segment. (a) the pair may sit on the circle; it is a free variable and not part of the orbit, and for all but finitely many z it misses μ_t , which is why an operator built for ζ -orbits has nothing to say about it. (b) off the circle, z outside and z^{-1} inside: a free direction. (c) r pairs give an r -dimensional free torus. In every case the alphabet is closed under inversion, so it is the eigenvalue list of an element of $O(N, \mathbb{C})$ — panel (b) shows why the group has to be the complex one — and the product of its letters is $\zeta^{t(t-1)/2} = (-1)^{t-1}$: for even t that matrix lies in the non-identity component. Computed by `anc/fig_intro.py`.

$m = 1$ that is Littlewood’s condition. So one 1934 theorem has two descendants, and they part company over what is allowed to stay free: there, complete root-of-unity orbits; here, as we shall see, a single frozen orbit together with free *reciprocal* directions. Neither statement contains the other.

A second line of factorization results concerns alphabets closed under $x \mapsto x^{-1}$. Okada [Oka98] and Ciucu–Krattenthaler [CK09] treated rectangular shapes; Behrend, Fischer and Konvalinka, and Ayer, Behrend and Fischer, treated double staircases, for which we follow the attributions given in [AB19]; and Ayer–Behrend [AB19] treated self-dual shapes, with bijective proofs of the latter given by Ayer and Fischer [AF20]. There the payoff is enumerative: the factorizations produce product formulas for symmetry classes of plane partitions and for rhombus tilings. The self-complementary shapes of [AB19] reappear in §8 from the other side: adding the letter -1 turns the determinant of the alphabet from $+1$ to -1 , and the shapes that factorize there are exactly the shapes that vanish here.

The two lines have never shared an alphabet, and the obstruction is structural in both directions. The engine of the first is an operator — the Verschiebung φ_t , in Albion’s formulation [Alb23], acting on universal characters by $\varphi_t h_r = h_{r/t}$ if $t \mid r$ and 0 otherwise — which requires the alphabet to be a union of ζ -orbits; every extension in that line therefore adds *more* root-of-unity structure, never less. The engine of the second requires the alphabet to be entirely reciprocal, or the shape to be self-complementary. A free reciprocal pair is not a ζ -orbit, and a root-of-unity orbit is not a union of free pairs.

The word *smallest* below can be made precise, and inversion-closure is what makes it so. An inversion-closed extension of μ_t can adjoin only reciprocal pairs $\{x, x^{-1}\}$ and the fixed points of inversion, which are 1 and -1 ; and 1 lies in μ_t for every t , while -1 lies in μ_t

exactly when t is even. The only extension smaller than a pair is therefore $\mu_t \cup \{-1\}$ with t odd, and it is frozen: over $t = 3, 5, 7$ and $|\lambda| \leq 14$, all 1092 shapes, $s_\lambda(\mu_t, -1)$ takes only the values 0 and ± 1 , with nothing left to depend on. A free reciprocal pair is the smallest extension that leaves anything to evaluate.

In this paper we take the smallest alphabet containing one letter of each kind,

$$\Phi_t(\lambda; z) := s_\lambda(\underbrace{1, \zeta, \dots, \zeta^{t-1}}_{\mu_t}, z, z^{-1}), \quad \zeta = e^{2\pi i/t}, \quad (1)$$

a Schur polynomial in $N = t + 2$ variables.

Described that way (1) is a compromise between two lines, which is how we found it. It is better described as one object, and that description is what returns in every section below: (1) is a *torus element of $O(N)$ with a single free direction*. It is closed under inversion, so it already lives where the second line lives; its determinant is $(-1)^{t+1}$, and that determinant is what separates factorizing from vanishing in §8; the ζ -orbit is what places it in the non-identity component when t is even, where a character is a twining character, which is Remark 7.2; and the one free pair is the one variable that survives the signed enumeration of §6. The three results of this paper are three readings of that sentence.

Our main result, Theorem 3.1, evaluates it for every t and every $\lambda \in \mathcal{P}_N$ — every partition with at most N parts, with no restriction whatever on its shape or its size. Write $\beta = \beta(\lambda, N)$ for the shifted partition and let $n_i(\lambda)$ be the number of parts of β congruent to i modulo t , as in [AK22]. Since there are t residue classes and $N = t + 2$ parts, $\sum_i (n_i - 1) = 2$; so either some class is empty, or, every class being occupied, the excess above one each is exactly two. Only three profiles occur: some n_i vanishes, or two classes have size two, or one class has size three. In the first case $\Phi_t = 0$; in the other two, if $A = \{a_1 > a_2\}$ and $B = \{b_1 > b_2\}$ denote the two distinguished classes and

$$d_1 = a_1 - a_2, \quad d_2 = b_1 - b_2, \quad \tilde{d}_3 = a_1 + a_2 - b_1 - b_2, \quad d_3 = |\tilde{d}_3|,$$

then $\Phi_t = 0$ if the *oriented coupling* $\tilde{d}_3$ vanishes, and otherwise, with $z = e^\theta$,

$$\Phi_t(\lambda; z) = \varepsilon_\lambda \frac{\sinh(d_1\theta/2) \sinh(d_2\theta/2) \sinh(d_3\theta/2)}{\sinh^2(t\theta/2) \sinh \theta}$$

with $\varepsilon_\lambda = \pm 1$ given explicitly. *Three factors, never more, and no hypothesis on the shape of λ .* With the reciprocal pair removed the statement degenerates to Littlewood's, so the content is what that pair adds: exactly one coupling term, namely d_3 . Read with $\mathfrak{sl}_2$ characters the same statement carries no exponential at all: it is the triple product $\chi_{d_1/2-1}\chi_{d_2/2-1}\chi_{d_3/2-1}$ over $\chi_{t/2-1}^2$, uniformly in t (Corollary 3.3), with half-integer indices exactly when t is odd.

The alphabet is one object, and so is the answer. The residue profile, the triple (d_1, d_2, d_3) and the sign are the whole of what the theorem needs, and in §3 we collect them into a single *evaluation invariant* $I_t(\lambda)$, (7), through which the value factors. It is complete but not minimal: (5) is symmetric in the three integers, so what is seen is the multiset and the sign, which is the compression announced above. What is *not* seen is then a well-posed question. Section 3.2 answers it for the three integers — the partitions sharing them form a lattice, and we write its generating function down — and what the sign adds to that is Problem 10.7.

The three integers have a reading that makes the formula transparent, and we state it here because it is how we think about the result. Regard A and B as intervals $[a_2, a_1]$ and

$[b_2, b_1]$ on the beta line. Then

$$d_1 = |A|, \quad d_2 = |B|, \quad d_3 = 2 |\text{centre}(A) - \text{centre}(B)| : \quad (2)$$

the two lengths, and twice the distance between the centres (Figure 5). The vanishing criterion becomes geometric — a residue class is empty, or the two intervals are concentric — and it explains at once why the size-three profile, where the intervals share an endpoint, never vanishes. Concentric intervals turn out to require t even (Proposition 3.5), so for odd t an empty residue class is the only way to vanish at all — and that stays true for every number of reciprocal pairs, which is Corollary 8.22. Equivalently, in the language of the first line, d_1 and d_2 are t times the $\mathfrak{sl}_2$ contents of the two distinguished quotient components, and the oriented coupling $\tilde{d}_3$ is an *affine* functional on the vector of quotient *sizes*, the two residues entering as a shift, with $d_3 = |\tilde{d}_3|$ (Proposition 3.4). It is not a functional on the residue vector, and hence not one on the root lattice of type A_{t-1} under the Garvan–Kim–Stanton correspondence [GKS90]: already at $t = 2$ the single two-class profile carries ten different values of d_3 over $|\lambda| \leq 16$. Which half of the core–quotient decomposition each clause of the vanishing criterion lives in is settled in Proposition 3.5.

Two consequences follow. [AK25, Theorem 5.3] determines when a universal character is independent of the variables by which it is twisted, uniformly in the three classical types, and read on (1) it would say that $s_\lambda(z, z^{-1}, \mu_t) = s_\lambda(z, z^{-1})$ exactly on the t -cores, for $\ell(\lambda) \leq 2$. That criterion is stated for free variables, and a reciprocal pair is not free; on that locus one further family appears, and Theorem 5.2 classifies it by its core and quotient. In §6 we read the identity as an enumeration: a product formula for a root-of-unity weighted count of semistandard tableaux in which the parameter z is still free, and at $t = 2$ a (-1) -enumeration of plane partitions in a box refined by z .

The last two sections ask a different question, not about what the identity gives but about how far it reaches. Section 7 tests four natural deformations of the alphabet, each of which destroys the identity, and one reading of the proof accounts for all four. Then §8 isolates what does survive. Adding further reciprocal pairs costs the product formula of Theorem 3.1, but not the *zero locus*: for every number r of pairs, the character $s_\lambda(1, -1, z_1, \bar{z}_1, \dots, z_r, \bar{z}_r)$ vanishes exactly when the beta set has constant parity or λ is self-complementary of odd width (Theorem 8.6). Read on the group rather than on the alphabet the same statement is a classification: those are exactly the λ whose $GL(N)$ -module restricts to $O(N)$ stably under the twist by $\det$ (Corollary 8.7). The shapes are the self-complementary ones of [AB19], where the same alphabet without the letter -1 makes the character factorize instead; the determinant of the alphabet decides which. The sufficient direction is a short corollary of complementation and we say so; the content is the converse, and we prove it for every r and every λ , modulo one rigidity theorem for products of Schur polynomials.

It is worth drawing the map, and Figure 3 is it. Adjoining r reciprocal pairs to the orbit μ_t gives a family in the two parameters (t, r) , and the zero locus is settled on three parts of it — one row, one column, and every odd column, which is half the plane.

Theorem 3.1 is $r = 1$: one pair, every t , and there we have the value and not merely its vanishing. Theorem 8.6 is $t = 2$: every number of pairs, the smallest orbit. And Corollary 8.22 is *every odd t and every r* at once: there the character vanishes if and only if a residue class is absent, so the concentric branch is a phenomenon of even t alone. The third of these is not an edge but half the plane, and it is the cheapest of the three — it drops out of the extremal argument of §8.5 with no external input at all, where the second needs Theorem 8.24. Off those three regions the locus is not settled, but it is no longer untouched. Theorem 8.35 gives a condition on the beta set — a reflection of its excess part, and one increment

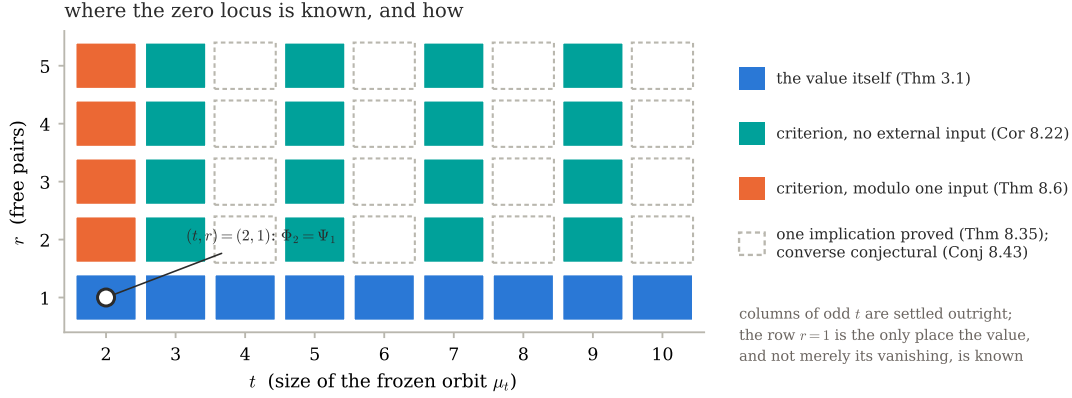


Figure 3. The two-parameter family, and what is known on it. Each cell is an alphabet $\mu_t \cup \{z_1^{\pm 1}, \dots, z_r^{\pm 1}\}$; the colour is not assigned by hand but computed from the statements themselves, one rule per line of the paper. The row $r = 1$ is the only place where the *value* is known and not merely its vanishing, which is Theorem 3.1; the columns of odd t are settled by Corollary 8.22 with no external input; the column $t = 2$ by Theorem 8.6, which uses one. On the remaining cells Theorem 8.35 proves one implication and Conjecture 8.43 is the other. The marked corner is where the two halves of the paper meet: there $\Phi_2 = \Psi_1$, and it is the only alphabet both of them see. Computed by `anc/fig_plane.py`.

hitting the centre — that forces the character to vanish for *every* t and r , by an involution pairing off every term of the expansion; Corollary 8.34 gives a necessary condition from the other side. That the two meet is Conjecture 8.43, and what separates them is a single implication, which §8.6 shows cannot come from the stratum that argument works in. The product formula of Theorem 3.1 already fails at $r = 2$, by (D1) — that no such formula exists for any $r \geq 2$ is Conjecture 10.4 and not something we prove — so past the line $r = 1$ the locus is what we can offer.

One thing the alphabet gives away deserves saying here rather than in passing. It has a representation-theoretic reading — of the kind [CK09] record as missing for their factorizations, as do [AB19] and [AK25] — and the mechanism is not ours: the alphabet is closed under inversion, hence a torus element of $O(N)$ of determinant $(-1)^{t+1}$, so for even t it sits in the non-identity component, where a character is a *twining* character in the sense of Jantzen [Jan73] and Kumar–Lusztig–Prasad [KLP09] and the folding is by the orbit Lie algebra. Remark 7.2 makes that precise and Lemma 8.9 makes it concrete — there the folded algebra is C_r and the twining character is a symplectic character on the nose. The proof of Theorem 3.1 uses none of it, which is why we state it as a reading and not as a method.

The article is organized as follows. We begin by fixing notation and recalling the two evaluations we build on in Section 2. In Section 3 we state the main theorem, identify its three integers in terms of the core and the t -quotient, describe the partitions they fail to separate, and record the vanishing criterion and the image of the resulting map. Section 4 contains the proof. In Section 5 we compare the theorem with the independence criterion of [AK25] and determine the extra family it acquires on the reciprocal locus. In Section 6 we read the theorem as an enumeration and assemble, at $t = 2$, the sign-reversing involution it asks for, and in Section 7 we try four deformations of the alphabet, each of which destroys the identity. Section 8 determines the zero locus at $t = 2$ for an arbitrary number of reciprocal

pairs — and, at no external cost, at every odd t as well — reads that criterion on the group as a determinant-twist rigidity, and gives a sufficient criterion, with a conjectural converse, for arbitrary t and r . Section 9 collects the numerical checks and Section 10 the open problems.

2. Background and notation

We follow the notation of [AK22, AK25]. All algebraic groups here are over $\mathbb{C}$; in particular $O(N)$ means $O(N, \mathbb{C})$, which is what an alphabet closed under $x \mapsto x^{-1}$ gives when its letters are allowed off the unit circle.

2.1. Partitions, beta sets, cores and quotients. A partition $\lambda = (\lambda_1, \dots, \lambda_n)$ is a weakly decreasing sequence of nonnegative integers; $\mathcal{P}_n$ denotes the set of partitions of length at most n , $|\lambda|$ the size and $\ell(\lambda)$ the length. Throughout, $t \geq 2$ is a fixed integer, ζ is a primitive t -th root of unity, $\mu_t = \{1, \zeta, \dots, \zeta^{t-1}\}$, and

$$N := t + 2.$$

For $\lambda \in \mathcal{P}_N$ the *beta set* is $\beta(\lambda) = (\beta_1, \dots, \beta_N)$ with $\beta_j = \lambda_j + N - j$, so that $\beta_1 > \beta_2 > \dots > \beta_N \geq 0$; we index by *columns* $j = 1, \dots, N$, and note that a larger β means a smaller column index. For $0 \leq i \leq t-1$ we write $n_i(\lambda)$ for the number of parts of $\beta(\lambda)$ congruent to i modulo t , so

$$\sum_{i=0}^{t-1} n_i(\lambda) = N = t + 2. \quad (3)$$

We call the vector $(n_0(\lambda), \dots, n_{t-1}(\lambda))$ the *residue profile* of λ . By (3) we have $\sum_i (n_i(\lambda) - 1) = 2$; if moreover every class is occupied, that is an excess of exactly two above one each, and only two partitions of two are available. So a profile is of exactly one of three kinds: *degenerate*, if some $n_i = 0$; *two-class*, if two entries equal 2; or *size-three*, if one entry equals 3. Everything in this paper is controlled by which of the three occurs. Let $\sigma \in S_N$ be the permutation rearranging the parts of $\beta(\lambda)$ so that they are grouped by residue class, the classes in increasing order and the values within each class in decreasing order; this is the permutation of [AK25, (2.1)]. We write $\text{core}_t(\lambda)$ and $\text{quot}_t(\lambda) = (\lambda^{(0)}, \dots, \lambda^{(t-1)})$ for the t -core and t -quotient, defined from $\beta(\lambda)$ as in [AK25, §2.1]. By [GKS90], t -cores are in bijection with vectors of the root lattice of type A_{t-1} , and the residue counts n_i are the natural coordinates.

We set $\bar{z} := z^{-1}$, put $z = e^\theta$, and abbreviate

$$f(u) := z^u - z^{-u} = 2 \sinh(u\theta), \quad V := \prod_{0 \leq k < k' \leq t-1} (\zeta^{k'} - \zeta^k).$$

For a set S of columns, b_S is the word of residues $\beta_j \bmod t$ for $j \in S$ read in increasing column order, and $\text{inv}(b_S)$ its number of inversions. We use the bialternant

$$s_\lambda(x_1, \dots, x_N) = \frac{\det(x_i^{\beta_j})_{i,j=1}^N}{\det(x_i^{N-j})_{i,j=1}^N}.$$

2.2. The two evaluations we build on. We recall the two statements the present result sits between. The first is Littlewood's evaluation, in the form of [AK25, Theorem 2.2].

Theorem 2.1 ([LR34, Theorem IX]). *Let $\lambda \in \mathcal{P}_t$. Then*

$$s_\lambda(1, \zeta, \dots, \zeta^{t-1}) = \begin{cases} (-1)^{\binom{t}{2}} \text{sgn}(\sigma) & \text{if } \text{core}_t(\lambda) = \emptyset, \\ 0 & \text{otherwise.} \end{cases}$$

The second is the independence criterion we extend, [AK25, Theorem 5.3]; we state the case we need. Let f_λ be a universal character of type s , sp or o , let Y be a set of m variables and let $x_1, \dots, x_{n-tm}$ be free. Then

$$f_\lambda(x_1, \dots, x_{n-tm}, Y, \zeta Y, \dots, \zeta^{t-1} Y) = f_\lambda(x_1, \dots, x_{n-tm}) \iff \lambda = \text{core}_t(\lambda), \quad (4)$$

for $\lambda \in \mathcal{P}_{n-tm}$. The criterion is uniform in the type — one statement for s , sp and o at once — which is what makes it quotable here: our alphabet is a GL alphabet, and the question we put to it is about a specialization and not about a group. Taking $m = 1$, $Y = (1)$ and $(x_1, x_2) = (z, \bar{z})$ gives $n = t + 2$ and our alphabet, but that substitution is precisely where the hypothesis that x_1, x_2 be free is given up. So (4) is not being applied on the reciprocal locus; it is being asked about there, which is §5.

Remark 2.2. The nearest statement in the literature adjoins two extra letters, as we do, so it is worth recording why it does not contain (1). Kumari [Kum24, Theorem 2.2] evaluates $s_\lambda(X, \zeta X, \dots, \zeta^{t-1} X, y, \zeta y, \dots, \zeta^{m-1} y)$; with $X = (1)$ and $m = 2$ the adjoined block is $\{y, \zeta y\}$, while ours is $\{z, z^{-1}\}$. The two coincide only when $z^2 = \zeta^{\pm 1}$, a finite set of points rather than a free variable. The same applies to the single extra free variable of [AK22, Theorem 2.7] and to Littlewood–Richardson’s Theorem XI.

Figure 4 carries out the whole dictionary on one shape.

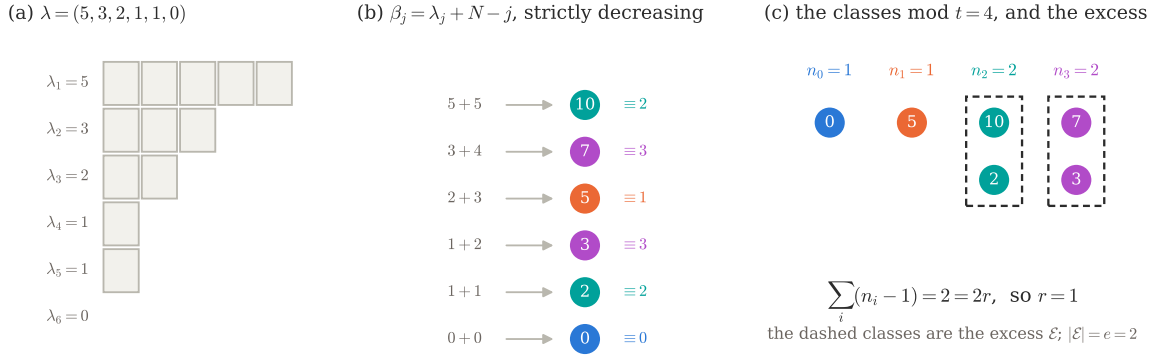


Figure 4. The dictionary the rest of the paper runs on, on $\lambda = (5, 3, 2, 1, 1, 0)$ at $t = 4$. (a) the shape, padded to N parts. (b) $\beta_j = \lambda_j + N - j$, which turns a weakly decreasing sequence into a strictly decreasing one and so into a set; each value is then coloured by its residue modulo t . (c) the residue profile $(n_0, \dots, n_{t-1})$ and the count (3): $\sum_i (n_i - 1) = 2r$ always, so when every class is occupied the classes with $n_i \geq 2$ — the excess $\mathcal{E}$, dashed — carry exactly $2r$ elements more than one each. If some class is empty the signed sum is still $2r$, but it is no longer an excess; that is the degenerate branch, and it is settled first. Everything the paper proves is a statement about that picture. Computed by `anc/fig_intro.py`.

3. The evaluation

The theorem in one sentence. The value depends on λ through its residue profile, an interval triple (d_1, d_2, d_3) and a sign, and through nothing else: an arbitrarily large partition is compressed to three integers and a sign.

Theorem 3.1. *Let $t \geq 2$ and $\lambda \in \mathcal{P}_N$, $N = t + 2$.*

- (i) *If $n_i(\lambda) = 0$ for some i , then $\Phi_t(\lambda; z) = 0$.*
- (ii) *Otherwise, by (3), either exactly two residue classes have $n_i = 2$, or exactly one has $n_i = 3$. Let $r_A \leq r_B$ be the residues involved and let*

$$A = \{a_1 > a_2\}, \quad B = \{b_1 > b_2\}$$

be, in the first case, the parts of $\beta(\lambda)$ in the classes r_A and r_B , and in the second case the two overlapping pairs $A = \{p, q\}$, $B = \{q, r\}$ of the class $\{p > q > r\}$, so that $r_A = r_B$. Set

$$d_1 = a_1 - a_2, \quad d_2 = b_1 - b_2, \quad \tilde{d}_3 = a_1 + a_2 - b_1 - b_2, \quad d_3 = |\tilde{d}_3|,$$

and call $\tilde{d}_3$ the oriented coupling. If $\tilde{d}_3 = 0$ then $\Phi_t(\lambda; z) = 0$. Otherwise, with $z = e^\theta$,

$$\Phi_t(\lambda; z) = \varepsilon_\lambda \cdot \frac{\sinh(d_1\theta/2) \sinh(d_2\theta/2) \sinh(d_3\theta/2)}{\sinh^2(t\theta/2) \sinh \theta} \quad (5)$$

where, writing j_{A_1} and j_{B_1} for the columns carrying a_1 and b_1 , S for the complement of $\{j_{A_1}, j_{B_1}\}$ in $\{1, \dots, N\}$, and b_S for the word of residues of $\beta(\lambda)$ on S read in increasing column order,

$$\varepsilon_\lambda = (-1)^{t + \binom{N+1}{2}} (-1)^{j_{A_1} + j_{B_1} + \text{inv}(b_S)} \text{sgn}(a_1 - b_1) \text{sgn}(\tilde{d}_3). \quad (6)$$

Each of the four factors is ± 1 — the last because $\tilde{d}_3 \neq 0$ is exactly the hypothesis of (ii) — so $\varepsilon_\lambda = \pm 1$. In the size-three profile $r_A = r_B$ and $\tilde{d}_3 = p - r > 0$, so that factor is $+1$.

The left-hand side is a Laurent polynomial in z . We read (5) as an identity in $\mathbb{C}(z)$; after cancellation the right-hand side is a Laurent polynomial too, and at the zeros of the denominator — $z = \pm 1$ and $z^t = 1$ — we use that continuation. We do so in §6, where the value is taken at $z = 1$.

Splitting d_3 into its size and its orientation is not bookkeeping. The sign (6) needs the orientation and the formula needs only the size, and at $\tilde{d}_3 = 0$ the orientation does not exist: a statement carrying $\text{sgn}(a_1 + a_2 - b_1 - b_2)$ and asserting that it is ± 1 is false on the concentric locus, where that argument is zero. The value there is 0 by the factor $\sinh(d_3\theta/2)$, so nothing downstream changes; what changes is that the case is now excluded before the sign is written rather than after.

We write

$$d(\lambda) := (d_1, d_2, d_3)$$

and call it the *interval triple* of λ , after the reading (2); and we call d_3 the *coupling term*, since it is the only one of the three that mixes the two distinguished classes.

It is worth giving the whole datum a name, because the theorem is better read as a statement about it than as a formula. Set

$$I_t(\lambda) := \begin{cases} 0 & \text{if the residue profile is degenerate, or } \tilde{d}_3 = 0, \\ (d_1, d_2, d_3, \varepsilon_\lambda) & \text{otherwise,} \end{cases} \quad (7)$$

and call it the *evaluation invariant* of λ . Its first branch is exactly the vanishing locus of Corollary 3.2, and it is one branch and not two because on the whole of it there is no sign to record: (6) needs $\tilde{d}_3 \neq 0$. Theorem 3.1 then says two things at once. First, Φ_t *factors through* I_t : the map $\lambda \mapsto \Phi_t(\lambda; z)$ is the composite of $\lambda \mapsto I_t(\lambda)$ with a function of the invariant alone. Second, that invariant is *complete* for this evaluation: if two partitions have the same I_t they have the same Φ_t , whatever their sizes. Completeness is what makes the compression announced above worth a name rather than an observation — it says the invariant is not a convenient summary of the dependence but the whole of it — and it turns the natural next question, what the invariant forgets, into a well-posed one. Proposition 3.10 closes the question from the other side: once the triple is read as a multiset, nothing in I_t can be dropped either. So §3.2 settles what is forgotten for the triple and Problem 10.7 is what the sign leaves. The (d_1, d_2, d_3) of Figure 6 are the values this invariant takes, drawn.

That factorisation is one pipeline, and every arrow in it loses information that never comes back:

$$\begin{aligned} \lambda &\longrightarrow \beta(\lambda) \longrightarrow (\text{residue profile, the two distinguished intervals}) \\ &\longrightarrow (d_1, d_2, d_3, \varepsilon_\lambda) \longrightarrow \Phi_t(\lambda; z). \end{aligned}$$

The intervals have to be carried alongside the profile: the profile says which classes are distinguished and nothing about where their beta numbers sit, so it does not determine the triple. Coordinate by coordinate, the invariant divides the work as follows.

datum of λ	what it controls
residue profile	the degenerate vanishing branch
the two interval lengths d_1, d_2	the first two factors
the distance between the centres, d_3	the coupling factor, and the concentric vanishing branch
the orientation $\text{sgn}(\tilde{d}_3)$	one factor of the sign
the sorting sign ε_λ	the overall sign

Neither vanishing branch is the other's: the profile cannot see the concentric one, and the triple cannot see the degenerate one, which is why the criterion of Corollary 3.2 has two clauses and §3 spends a proposition on where each of them lives.

One count runs through the paper. With t residue classes and $N = t + 2$ parts, on the branch where every class is occupied the excess is exactly two, so by (3) a profile can only be degenerate, two-class or size-three: that trichotomy is the shape of Theorem 3.1, the case split of its proof in §4, and the first branch of the zero locus in §8. Two distinguished classes are what the reciprocal pair has to couple, hence exactly one coupling term and three factors rather than two.

The count of three has a second explanation, which owes nothing to the proof. The excess two gives two distinguished classes, which are two intervals on the beta line, and an interval pair is four numbers (a_1, a_2, b_1, b_2) . The three arguments of (5) see how that pair is shaped and not where it sits: a common shift $\beta \mapsto \beta + m$ leaves d_1, d_2 and $\tilde{d}_3$ alone. So what can enter them is a linear functional killing the common translation, and those form the hyperplane $\sum \alpha_i = 0$ of the four-dimensional dual: a space of dimension exactly *three*. The

triple $d_1 = (1, -1, 0, 0)$, $d_2 = (0, 0, 1, -1)$ and $\tilde{d}_3 = (1, 1, -1, -1)$ is a basis of it. Modulo the common translation an interval pair has three independent linear coordinates, and that is where the count of three comes from. What Theorem 3.1 adds, and what the dimension count does not force, is that in this specialization those three coordinates *separate multiplicatively*: a product of one factor each, rather than some other function of three variables.

The *value* is not translation-invariant, and saying so is worth a line, because what it is instead is a constraint on the sign. A shift $\beta \mapsto \beta + m$ is $\lambda \mapsto \lambda + (m^N)$, and $s_{\lambda+(m^N)}(A) = \det(A)^m s_\lambda(A)$ for any alphabet A , so by (1)

$$\Phi_t(\lambda + (m^N); z) = (-1)^{(t+1)m} \Phi_t(\lambda; z). \quad (8)$$

For odd t the value is invariant; for even t an odd shift reverses it. The triple does not move, so the whole of that reversal is carried by ε_λ — which makes (8) a check on (6) that costs nothing and that the sign passes, over the shapes of §9. It is also the reason the slogan has to be stated about the triple and not about the value: for t even the value is not a function of the triple's translation class.

The sign (6) is a sorting sign, of the same species as the one in Theorem 2.1; Proposition 3.11 below records a shorter expression for it, in the notation of [AK25].

The right-hand side of (5) is a ratio, and the content of the theorem is that the quotient is a Laurent polynomial in z . For odd t the individual factor $\sinh(t\theta/2)$ is not a Laurent monomial, so the three factors do not separately lie in $\mathbb{Z}[z, z^{-1}]$; only their combination does. The statement is uniform in t nonetheless.

Corollary 3.2. $\Phi_t(\lambda; z) \equiv 0$ if and only if some residue class modulo t is empty, or $a_1 + a_2 = b_1 + b_2$. In particular the size-three profile never vanishes.

The vanishing is identical in z and not at a point: when it happens, λ contributes nothing at any element of the one-parameter family the reciprocal pair sweeps out. That is the form in which Prasad's question was posed in §2, and the sense in which (1) is read outside combinatorics — the frozen half of the alphabet being a holonomy of finite order, and the free pair what survives of the torus.

Corollary 3.3. Write $\chi_k(z) = (z^{k+1} - z^{-k-1})/(z - z^{-1})$. For an integer $k \geq 0$ this is the character of the $(k+1)$ -dimensional irreducible $\mathfrak{sl}_2$ -module. For half-integral k we keep the same formula, read in $u = z^{1/2}$; it is then notation and not a character, since the individual factors are not Laurent in z . Then (5) can be written with no exponential and no convention for θ at all:

$$\Phi_t(\lambda; z) = \varepsilon_\lambda \frac{\chi_{\frac{d_1}{2}-1}(z) \chi_{\frac{d_2}{2}-1}(z) \chi_{\frac{d_3}{2}-1}(z)}{\chi_{\frac{t}{2}-1}(z)^2} \quad (9)$$

For $t = 2$ the denominator is $\chi_0 = 1$ and every d_i is even, so Φ_2 is a signed product of three $\mathfrak{sl}_2$ characters outright. For odd t the indices are half-integers, which is the statement above in another form: the three factors are not separately Laurent, and only the ratio is. Each factor is read in $u = z^{1/2}$ and reverses sign under $u \mapsto -u$ exactly when its index is half-integral; since $d_1 + d_2 + d_3$ is even — it has the parity of $d_1 + d_2 + \tilde{d}_3 = 2(a_1 - b_2)$ — the ratio does not, and it is the Laurent polynomial in z of (5).

Proof. Write $f(u) = z^u - z^{-u} = 2 \sinh(u\theta)$, so that $\chi_k = f(k+1)/f(1)$. The numerator of (5) is $f(d_1/2)f(d_2/2)f(d_3/2)/8$ and its denominator is $f(t/2)^2 f(1)/8$, so (5) equals $\varepsilon_\lambda f(d_1/2)f(d_2/2)f(d_3/2)/(f(t/2)^2 f(1))$; dividing three factors of $f(1)$ out of the numerator and two out of the denominator gives (9). $\square$

3.1. The three integers come from the quotient. The following identifies (d_1, d_2, d_3) in the language of the root-of-unity line. For a partition ν put $k(\nu) = \nu_1 - \nu_2$, its $\mathfrak{sl}_2$ content.

Proposition 3.4. *In the two-class profile, with $\text{quot}_t(\lambda) = (\lambda^{(0)}, \dots, \lambda^{(t-1)})$,*

$$d_1 = t(k(\lambda^{(r_A)}) + 1), \quad d_2 = t(k(\lambda^{(r_B)}) + 1), \quad \tilde{d}_3 = t(|\lambda^{(r_A)}| - |\lambda^{(r_B)}|) + 2(r_A - r_B).$$

In the size-three profile the single distinguished component $\lambda^{(r_A)}$ has three parts and

$$d_1 = t(\lambda_1^{(r_A)} - \lambda_2^{(r_A)} + 1), \quad d_2 = t(\lambda_2^{(r_A)} - \lambda_3^{(r_A)} + 1), \quad \tilde{d}_3 = d_1 + d_2.$$

Proof. Write $a_i = t\tilde{a}_i + r_A$. The class r_A carries two beta numbers, so $\lambda^{(r_A)} = (\tilde{a}_1 - 1, \tilde{a}_2)$ by the definition of the quotient, whence $k(\lambda^{(r_A)}) = \tilde{a}_1 - \tilde{a}_2 - 1$ and $d_1 = t(\tilde{a}_1 - \tilde{a}_2)$, which is the first identity; the second is the same computation. For the third, $\tilde{a}_1 + \tilde{a}_2 = |\lambda^{(r_A)}| + 1$ and likewise for B , so $a_1 + a_2 - b_1 - b_2 = t(|\lambda^{(r_A)}| - |\lambda^{(r_B)}|) + 2(r_A - r_B)$. The size-three case is identical with $\lambda^{(r_A)} = (\tilde{p} - 2, \tilde{q} - 1, \tilde{r})$. $\square$

For $t = 2$ the third identity reads $\tilde{d}_3 = 2(|\lambda^{(0)}| - |\lambda^{(1)}| - 1)$, which already shows that $\tilde{d}_3$ is a functional on the quotient sizes and not on the residue vector, so it is not one on the root lattice of type A_{t-1} under the Garvan–Kim–Stanton correspondence [GKS90] between that lattice and the t -cores. The condition $r_B - r_A = t/2$, which governs the family of Theorem 5.2 below, is an order-two element of the relevant quotient, and this is the source of the parity phenomena throughout the paper.

Proposition 3.5 (the concentric locus). *In the two-class profile, ordered so that $r_A < r_B$,*

$$d_3 = 0 \iff t \text{ is even, } r_B - r_A = \frac{t}{2}, \quad \text{and} \quad |\lambda^{(r_A)}| = |\lambda^{(r_B)}| + 1$$

In particular, for t odd $\Phi_t(\lambda; z) = 0$ if and only if some residue class modulo t is empty: the concentric clause of Corollary 3.2 is vacuous there.

Proof. Since $a_1, a_2 \equiv r_A$ and $b_1, b_2 \equiv r_B$ modulo t , the equality $a_1 + a_2 = b_1 + b_2$ forces $2r_A \equiv 2r_B$, that is $t \mid 2(r_B - r_A)$. For t odd this gives $t \mid r_B - r_A$ and hence $r_A = r_B$, which is the size-three profile, where $d_3 = d_1 + d_2 > 0$ by Proposition 3.4. For t even and $0 \leq r_A < r_B \leq t - 1$ the only solution of $t \mid 2(r_B - r_A)$ is $r_B - r_A = t/2$. Writing $a_1 + a_2 = t(|\lambda^{(r_A)}| + 1) + 2r_A$ as in the proof of Proposition 3.4, and likewise for B , the equality becomes $t(|\lambda^{(r_A)}| - |\lambda^{(r_B)}|) = 2(r_B - r_A) = t$. $\square$

The two clauses of Corollary 3.2 therefore live in different halves of the core–quotient decomposition, and that is the answer to the question the criterion raises. The first is a condition on the residue profile, hence on the t -core: the coordinate hyperplanes $n_i = 0$ cut out of the slice $\sum_i n_i = t + 2$, which is not the Coxeter arrangement of type A_{t-1} . The second is not a condition on the core at all; it is a single hyperplane in the quotient sizes, and the admissible pairs of residues are indexed by the order-two element $r_B - r_A = t/2$ rather than by any reflection. No restatement of the vanishing criterion in terms of $\text{core}_t(\lambda)$ alone can exist, which is for Φ_t what Remark 8.8 records for Ψ_r .

3.2. What the triple forgets. Proposition 3.4 reads the triple off the quotient. Run backwards, the same reading says which partitions the triple cannot tell apart, and says it exactly. Nothing here is deep — it is the core–quotient bijection at a fixed profile — but it is what turns the compression of Theorem 3.1 from a picture into a count.

Proposition 3.6 (the two-class stratum is a lattice). *Let $t \geq 2$, $N = t+2$, and let $\mathcal{T}_t \subset \mathcal{P}_N$ be the set of partitions of two-class profile. Sending λ to its two distinguished residues together with its t -quotient is a bijection*

$$\mathcal{T}_t \xrightarrow{\sim} \coprod_{0 \leq r_A < r_B \leq t-1} \mathcal{P}_2 \times \mathcal{P}_2 \times \mathbb{Z}_{\geq 0}^{t-2}, \quad \lambda \mapsto (r_A, r_B; \lambda^{(r_A)}, \lambda^{(r_B)}; (m_i)_{i \neq r_A, r_B}), \quad (10)$$

where m_i is the single part of $\lambda^{(i)}$. Under it $\text{core}_t(\lambda)$ depends only on the pair (r_A, r_B) ; write $\kappa(r_A, r_B)$ for its size. Then

$$|\lambda| = |\text{core}_t(\lambda)| + t \left(|\lambda^{(r_A)}| + |\lambda^{(r_B)}| + \sum_i m_i \right). \quad (11)$$

Proof. A class carrying n_i beta numbers $a_1 > \dots > a_{n_i}$, written $a_j = t\tilde{a}_j + i$, contributes $\lambda^{(i)} = (\tilde{a}_1 - n_i + 1, \dots, \tilde{a}_{n_i})$, so a class of size n_i gives a partition with at most n_i parts, and every such partition arises from exactly one strictly decreasing $(\tilde{a}_j)$. In the two-class profile the sizes are 2, 2 and 1 repeated $t-2$ times, which is (10); the residues are recorded rather than forgotten because the profile is part of the datum, and they determine $\text{core}_t(\lambda)$ by the Garvan–Kim–Stanton coordinate of Remark 8.8. Then (11) is the core-quotient bijection. $\square$

Corollary 3.7 (the fibre of the triple, and its generating function). *Let (d_1, d_2, d_3) be an interval triple attained in the two-class profile, and set $k_A = d_1/t - 1$ and $k_B = d_2/t - 1$. Its fibre in $\mathcal{T}_t$ is infinite. It is the disjoint union of the branches indexed by the pairs $r_A < r_B$ and the signs $\eta = \pm 1$ for which*

$$c := \frac{1}{2} \left(\frac{\eta d_3 - 2(r_A - r_B)}{t} - k_A + k_B \right) \in \mathbb{Z}, \quad (12)$$

the two signs giving one and the same branch when $d_3 = 0$. On a branch the free parameters are an integer $v \geq \max(0, c)$ and $(m_i) \in \mathbb{Z}_{\geq 0}^{t-2}$, with $\lambda^{(r_A)} = (v + k_A, v)$ and $\lambda^{(r_B)} = (v - c + k_B, v - c)$. Consequently

$$\sum_{\lambda \text{ in the fibre}} q^{|\lambda|} = \frac{\sum_{\text{branches}} q^e}{(1 - q^{4t})(1 - q^t)^{t-2}}, \quad e = \kappa(r_A, r_B) + t(k_A + k_B - 2c) + 4t \max(0, c), \quad (13)$$

and the number of λ in the fibre with $|\lambda| \leq n$ is eventually a quasi-polynomial in n of degree $t-1$ — eventually, because the shifts q^e in the numerator put the count at 0 below the smallest of them.

Proof. By Proposition 3.4 the first two entries fix $k(\lambda^{(r_A)})$ and $k(\lambda^{(r_B)})$, so each distinguished component is determined by its size alone; writing them $(v + k_A, v)$ and $(u + k_B, u)$, the third entry reads $t((2v + k_A) - (2u + k_B)) + 2(r_A - r_B) = \eta d_3$, which is (12) with $c = v - u$; the absolute value in d_3 is what puts the two signs there, and it is vacuous at $d_3 = 0$. Substituting in (11) gives $|\lambda| = e + 4t(v - \max(0, c)) + t \sum_i m_i$, and summing the geometric series in v and in each m_i gives (13). Its denominator has $t-1$ factors, whence the degree. $\square$

Remark 3.8 (the sign is not along for the ride). Two readings, and the second is a warning. First, a fibre is infinite for every t and every attained triple, which is what Figure 7 shows and (13) explains: the columns there do not thin out because a fibre is a lattice section and not a finite accident. Second, (13) is the fibre of the *triple* and not of I_t . The $t-2$ parts m_i are invisible to (d_1, d_2, d_3) by Proposition 3.4, and it would be natural to read them as directions along which Φ_t is constant. They are not, from $t=3$ on: the sign sees them, as (6) makes possible, since ε_λ carries $\text{inv}(b_S)$, a statistic of the whole beta-word, while (10)

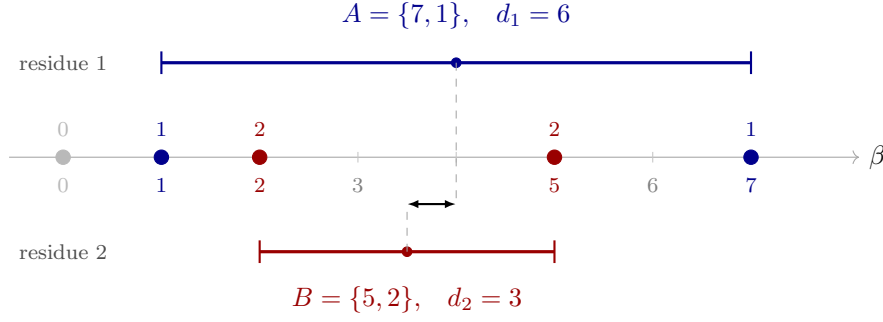


Figure 5. The mechanism for $t = 3$, $\lambda = (3, 2)$: here $N = 5$, $\beta = (7, 5, 2, 1, 0)$ with residues $(1, 2, 2, 1, 0)$ modulo 3, so no class is empty and two classes have size two. The three arguments of (5) are the two interval lengths and twice the distance between the interval centres; the short double arrow marks that distance, here $\frac{1}{2}$, so that $d_3 = 1$.

splits that word into two distinguished classes and $t - 2$ free letters. Over the ranges of §9 the sign is constant in (m_i) at fixed v in all 107 of the slots at $t = 2$, but in only 43 of 62 at $t = 3$ and 26 of 36 at $t = 4$. What the sign does on a branch is Problem 10.7.

3.3. The interval reading. Figure 5 draws (2) in an example. Corollary 3.2 reads: the character vanishes when a residue class is empty, or when the two intervals are *concentric*; and two intervals sharing an endpoint are concentric only if they coincide, which is why the size-three profile never vanishes.

Example 3.9. For $t = 3$, $\lambda = (3, 2)$ as in Figure 5, $(d_1, d_2, d_3) = (6, 3, 1)$ and

$$\Phi_3((3, 2); z) = \varepsilon \frac{\sinh(3\theta) \sinh(\theta/2)}{\sinh(3\theta/2) \sinh \theta}.$$

For $t = 3$, $\lambda = (3, 1)$ the profile is the size-three one, $(d_1, d_2, d_3) = (3, 3, 6)$ and $\Phi_3 = \varepsilon \chi_2(z)$. For $t = 4$, $\lambda = (1, 1, 1)$ we have $\beta = (6, 5, 4, 2, 1, 0)$, the class 3 is empty and $\Phi_4 = 0$.

3.4. The image of the map. Theorem 3.1 sends every partition with at most N rows to three integers, so the whole family collapses onto a sparse subset of $\mathbb{Z}^3$. By Proposition 3.4 the first two coordinates are multiples of t ; the third is not a multiple of t except in the size-three profile, where $d_3 = d_1 + d_2$. Figure 6 shows the image for $t = 3$ and $t = 4$, coloured by the number of partitions collapsing onto each point.

Figure 6 is one half of the picture: it shows which triples occur, not what a triple hides. Figure 7 is the other half. Over each cell of the (d_1, d_2) floor it stands the column of sizes $|\lambda|$ that land there, so a column is a set of partitions of different sizes on which Φ_t takes one and the same value — and the columns do not thin out as $|\lambda|$ grows, since the values are confined to a lattice and the partitions are not. The most visible one is the fibre of the empty partition: at $t = 3$ the invariant $I_3 = (3, 3, 2, +1)$ is shared by 26 partitions, of 19 different sizes between 0 and 20, so Φ_3 does not distinguish $\emptyset$ from a shape of twenty boxes.

Figure 7 makes one thing about (7) precise: I_t is complete but not minimal. The numerator of (5) is symmetric in d_1, d_2, d_3 , so the value sees only the multiset $\{d_1, d_2, d_3\}$ together with the sign, and that is a real reduction — over $t = 4$ and $|\lambda| \leq 20$ the 121 invariants take only 110 values, while at $t = 3$ in the same range the two counts agree. That reduction is the

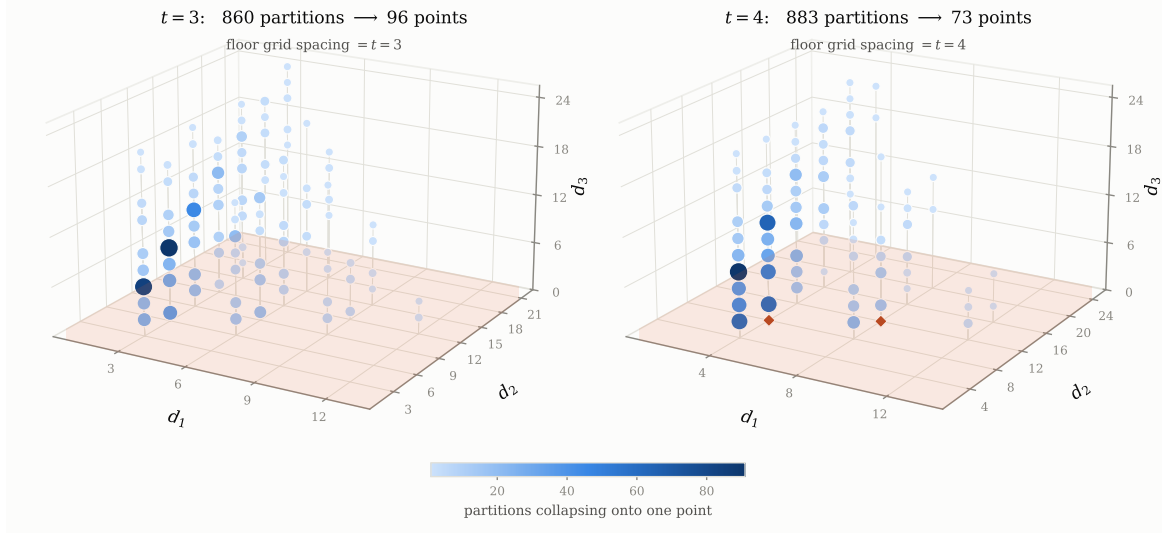


Figure 6. The image of $\lambda \mapsto (d_1, d_2, d_3)$ over all λ with $|\lambda| \leq 20$ and at most $t + 2$ rows, for $t = 3$ (left) and $t = 4$ (right), taken up to the exchange of d_1 and d_2 , under which (5) is symmetric; without that identification, and with A and B in the order $r_A \leq r_B$ of Theorem 3.1, the two panels carry 126 and 92 points rather than 96 and 73. The compression is the point of the picture: 860 partitions land on 96 triples on the left and 883 on 73 on the right, and colour and area give the number of partitions mapping to each. The floor grid spacing is t , since $d_1, d_2 \in t\mathbb{Z}$ by Proposition 3.4; the shaded plane $d_3 = 0$ is the concentric locus of Corollary 3.2, where s_λ vanishes, so those shapes are not in the image and are drawn on the plane as diamonds instead. The two panels are the two parities, which is the content of Proposition 3.5: the odd one carries no diamond, and can carry none; the even one carries two, between them the images of 32 partitions.

whole of it: nothing below the multiset and the sign is lost, and this is where the compression announced in the introduction becomes an exact statement rather than an upper bound.

Proposition 3.10 (the multiset and the sign are exactly what is seen). *Fix $t \geq 2$ and let $\lambda, \mu \in \mathcal{P}_N$ with $\Phi_t(\lambda; z) \neq 0$. Then*

$$\Phi_t(\lambda; z) = \Phi_t(\mu; z) \iff \{d_1, d_2, d_3\}(\lambda) = \{d_1, d_2, d_3\}(\mu) \text{ as multisets and } \varepsilon_\lambda = \varepsilon_\mu.$$

Proof. $(\Leftarrow)$ is (5), whose right-hand side is symmetric in the three arguments. For $(\Rightarrow)$, note first that $\Phi_t(\mu; z) = \Phi_t(\lambda; z) \neq 0$, so μ too has a non-degenerate profile and $\tilde{d}_3(\mu) \neq 0$ and both invariants are defined. Write $d = d(\lambda)$ and $e = d(\mu)$, all six entries being ≥ 1 , and put $u = z^{1/2}$, so that $2 \sinh(v\theta/2) = u^v - u^{-v}$ with integer exponents. The denominator of (5) depends on t alone, so the hypothesis is

$$\varepsilon_\lambda \prod_{i=1}^3 (u^{d_i} - u^{-d_i}) = \varepsilon_\mu \prod_{i=1}^3 (u^{e_i} - u^{-e_i}).$$

Using $u^v - u^{-v} = u^{-v}(u^{2v} - 1)$ on each factor and setting $D = \sum_i d_i$, $E = \sum_i e_i$,

$$\varepsilon_\lambda u^{-D} \prod_i (u^{2d_i} - 1) = \varepsilon_\mu u^{-E} \prod_i (u^{2e_i} - 1).$$

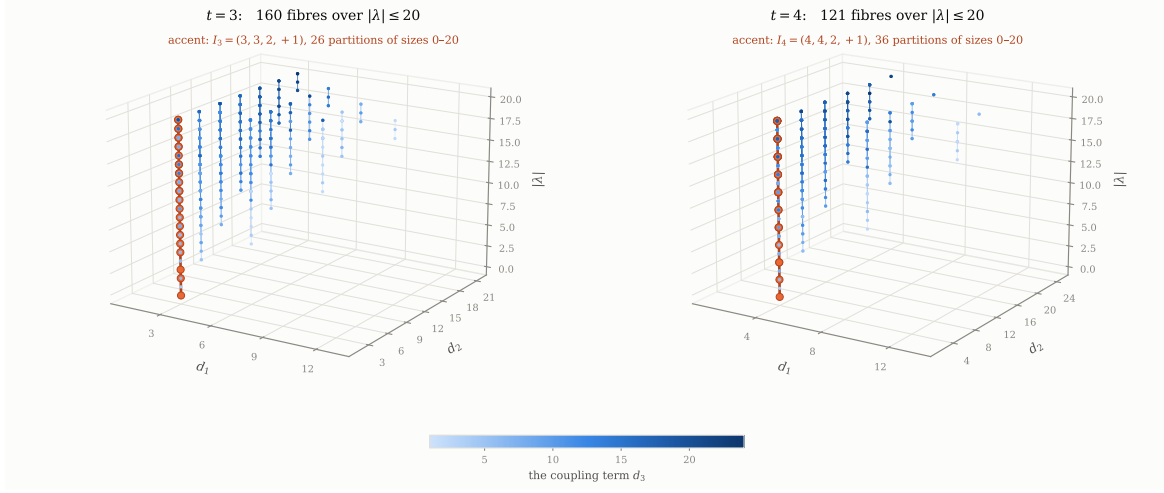


Figure 7. The fibres of the evaluation invariant (7), the complement of Figure 6: the same range $|\lambda| \leq 20$, with the coupling term moved to colour and the size $|\lambda|$ on the vertical axis. A column is one fibre, so its height is the range of sizes this alphabet cannot tell apart; the floor ticks are the lattice $t\mathbb{Z}$ of Proposition 3.4, and only the half $d_1 \leq d_2$ is occupied because the triple is taken up to the exchange, as in Figure 6. That every marker of a column carries the same value is computed and not assumed: over the 860 partitions on the left and the 883 on the right the bialternant is constant on each fibre to 10^{-37} . The accent column is the fibre of $\lambda = \emptyset$.

Each product is a polynomial with constant term $(-1)^3 = -1$, so the lowest exponent occurring is $-D$ on the left and $-E$ on the right, with coefficients $-\varepsilon_\lambda$ and $-\varepsilon_\mu$: hence $D = E$ and $\varepsilon_\lambda = \varepsilon_\mu$, and the two products agree. Now $u^{2v} - 1 = \prod_{m|2v} Q_m(u)$, where Q_m is the m -th cyclotomic polynomial — written Q and not Φ , which in this paper is the object. The Q_m are irreducible and pairwise distinct, so by unique factorization in $\mathbb{Z}[u]$ the two sides carry the same multiset of cyclotomic factors,

$$\biguplus_i \{m : m \mid 2d_i\} = \biguplus_i \{m : m \mid 2e_i\}.$$

Its largest element is $\max_i 2d_i$ on the left and $\max_i 2e_i$ on the right, so those agree; deleting one copy of the divisor set of that maximum from each side and repeating twice more gives $\{2d_1, 2d_2, 2d_3\} = \{2e_1, 2e_2, 2e_3\}$. $\square$

So the pair (multiset, sign) is a *complete separating* invariant for this evaluation off the vanishing locus — complete because the value factors through it, separating because distinct data give distinct values — and that is a theorem and not a range. We stop short of calling it *minimal* outright, since minimality is relative to a class of invariants one would have to fix first. The hypothesis $\Phi_t(\lambda; z) \neq 0$ is not removable and not a defect: on the vanishing locus every invariant collapses to one value, which is why (7) sends the whole of it to 0. We keep the ordered form there because d_3 is the coordinate the reciprocal pair contributes and the other two are not, and Corollary 3.7 describes the fibres of either, since the two differ by a symmetry of the numerator and not by anything the lattice sees. Lemma 5.1 below is the same argument run against a target that is not itself a value of Φ_t .

Proposition 3.11 (a shorter form of the sign). *Let the residue profile be non-degenerate and $\tilde{d}_3 \neq 0$, and order the two distinguished classes so that $r_A \leq r_B$. Then*

$$\varepsilon_\lambda = (-1)^{\lfloor t/2 \rfloor} \operatorname{sgn}(\sigma) (-1)^{r_A + r_B} \operatorname{sgn}(\tilde{d}_3) \quad (14)$$

where σ is the permutation of §2.

Proof. Let w be the residue word of $\beta(\lambda)$ read in increasing column order. Since β decreases with the column index, grouping the parts by class with the values decreasing inside each class is the stable sort of w , so $\operatorname{sgn}(\sigma) = (-1)^{\operatorname{inv}(w)}$. Both sides of (14) carry the factor $\operatorname{sgn}(\tilde{d}_3)$, and $\operatorname{sgn}(a_1 - b_1) = (-1)^{[a_1 < b_1]}$, so by (6) the assertion is equivalent to

$$j_{A_1} + j_{B_1} + \operatorname{inv}(w) + \operatorname{inv}(b_S) + [a_1 < b_1] \equiv \left\lfloor \frac{t}{2} \right\rfloor + t + \binom{N+1}{2} + r_A + r_B \pmod{2}. \quad (15)$$

We use one parity count. Let u be a word, p a position, $c = u_p$, and let $I(p)$ be the number of inversions of u in which p takes part; write $\nu_{<c}$ for the number of letters of u strictly below c and e_p for the number of occurrences of c strictly before p . Putting $x = \#\{j < p : u_j > c\}$, the positions before p split as $p-1 = x + e_p + \#\{j < p : u_j < c\}$ and the letters below c split as $\nu_{<c} = \#\{j < p : u_j < c\} + \#\{j > p : u_j < c\}$, whence

$$I(p) = x + \#\{j > p : u_j < c\} = 2x + e_p + \nu_{<c} - (p-1) \equiv (p-1) + \nu_{<c} + e_p \pmod{2}. \quad (16)$$

Since b_S is w with the letters at the positions j_{A_1} and j_{B_1} deleted,

$$\operatorname{inv}(w) - \operatorname{inv}(b_S) = I(j_{A_1}) + I(j_{B_1}) - [\{j_{A_1}, j_{B_1}\} \text{ is an inversion of } w],$$

and $\operatorname{inv}(w) + \operatorname{inv}(b_S) \equiv \operatorname{inv}(w) - \operatorname{inv}(b_S)$, so (16) computes the left-hand side of (15).

In the two-class profile $r_A < r_B$, and w carries r_A twice, r_B twice and every other residue once. Because β decreases with the column index, $a_1 > a_2$ puts j_{A_1} at the *first* occurrence of r_A , so $e = 0$ there, and likewise at j_{B_1} . The letters below r_A are one copy of each of $0, \dots, r_A - 1$, so $\nu_{<r_A} = r_A$; those below r_B are the same together with the second copy of r_A , so $\nu_{<r_B} = r_B + 1$. The two positions form an inversion exactly when the larger letter comes first, that is when $j_{B_1} < j_{A_1}$, that is when $a_1 < b_1$. Hence

$$\begin{aligned} \operatorname{inv}(w) - \operatorname{inv}(b_S) &\equiv (j_{A_1} - 1 + r_A) + (j_{B_1} - 1 + r_B + 1) + [a_1 < b_1] \\ &\equiv j_{A_1} + j_{B_1} + r_A + r_B + 1 + [a_1 < b_1], \end{aligned}$$

and substituting this in (15) cancels the columns and $[a_1 < b_1]$ in pairs, leaving $\lfloor t/2 \rfloor + t + \binom{N+1}{2} \equiv 1$.

In the size-three profile $r_A = r_B =: r$ and the class $\{p > q > p'\}$ sits at columns $j_1 < j_2 < j_3$ with $A = \{p, q\}$ and $B = \{q, p'\}$, so $j_{A_1} = j_1$ and $j_{B_1} = j_2$ carry the same letter r , with $e = 0$ and $e = 1$ respectively and $\nu_{<r} = r$ in both; two equal letters are not an inversion, and $a_1 > b_1$. So $\operatorname{inv}(w) - \operatorname{inv}(b_S) \equiv j_1 + j_2 + 1$, which is the displayed expression again because $r_A + r_B = 2r$ and $[a_1 < b_1] = 0$, and (15) reduces to the same congruence.

Finally $N = t + 2$, so that congruence reads $\lfloor t/2 \rfloor + t + \binom{t+3}{2} \equiv 1$: for $t = 2m$ its three terms are m , 0 and $m + 1$, and for $t = 2m + 1$ they are m , 1 and m . Both sums are odd, and the identity holds for every $t \geq 2$. $\square$

The ordering by residue is not a convenience, and it is worth saying why. Exchanging the two blocks negates the sign κ_{11} of Lemma 4.4 — through its $\operatorname{sgn}(a_1 - b_1)$ factor — and negates $\operatorname{sgn}(\tilde{d}_3)$ as well, so it leaves ε_λ , which is their product, invariant; as it must, since ε_λ is attached to λ . The right-hand side above is *not* invariant: $(-1)^{r_A + r_B}$ is symmetric, so

only its last factor changes sign. Without a rule fixing which block is A , the right-hand side is therefore not a function of λ at all. The hypothesis enters the proof at one point and one only: it is what makes $\nu_{<r_A} = r_A$ and $\nu_{<r_B} = r_B + 1$ rather than the other way round, and dropping it exchanges those two counts and breaks (15).

Read the other way round, (14) says that Φ_t is a function of a small combinatorial datum and of nothing else, and it says so for every t rather than over a range: by Theorem 3.1 the triple (d_1, d_2, d_3) determines Φ_t up to sign, and adjoining $\text{sgn}(\sigma)$, the ordered pair $r_A \leq r_B$ and the *orientation* $\text{sgn}(\tilde{d}_3)$ determines it outright. The shape of λ enters nowhere else. The orientation is not optional, and that part remains a check rather than a proof: over $t \leq 6$ and $|\lambda| \leq 14$, putting $\text{sgn}(a_1 - b_1)$ in its place leaves collisions at $t = 2$.

4. Proof of Theorem 3.1

The proof uses the Vandermonde determinant, the Laplace expansion, and one cancellation lemma. We give the steps separately so that a reader checking the ancillary scripts can localise any disagreement.

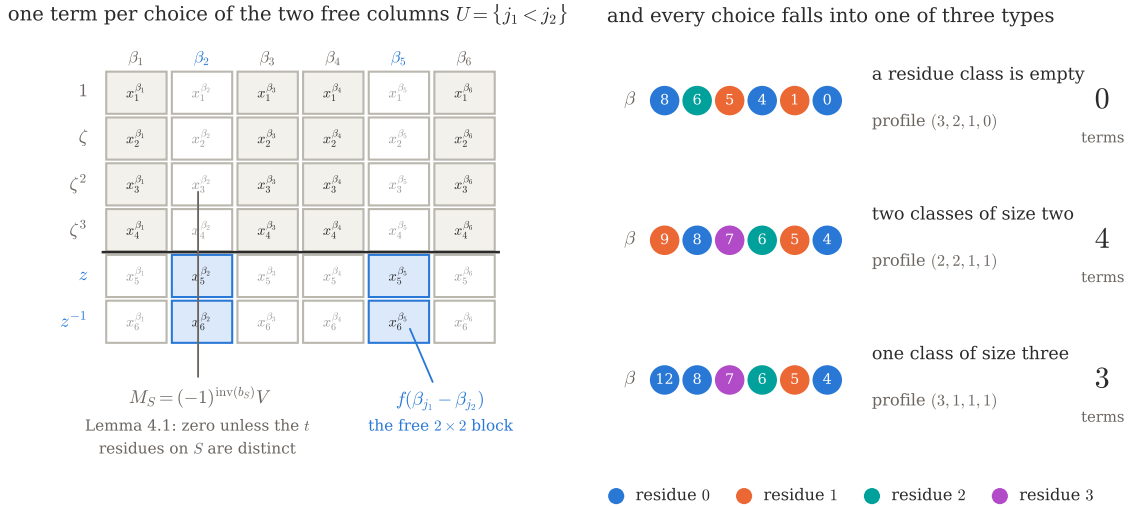


Figure 8. The architecture of the proof, drawn at $t = 4$, $N = 6$. *Left:* the determinant $\det(x_i^{\beta_j})$ expanded along the t frozen rows. Each choice of the two columns $U = \{j_1 < j_2\}$ left to the free rows z, z^{-1} leaves a $t \times t$ minor M_S on the complementary columns, which by Lemma 4.1 is $\pm V$ or zero, times the 2×2 block $f(\beta_{j_1} - \beta_{j_2})$. *Right:* by the excess-two count (3) only three residue profiles occur, one per row, each drawn on a beta set that realises it; the number at the right is how many choices of U survive, counted rather than asserted — 0, 4 and 3, which is the trichotomy the section is organised around. Colour is the residue class. Computed by `anc/fig_proof.py`.

The expansion itself is the standard move of this literature and we claim nothing for it: Albion [Alb25] applies the Laplace expansion to a determinant of this shape *according to the given block structure*, holding a first set of rows fixed, following Krattenthaler and Ciucu–Krattenthaler. What is particular here is not the expansion but what it runs into.

What the argument really does is classify rather than compute. Expanding along the t frozen rows leaves one term for each way of choosing which two columns are *not* frozen,

and the excess-two count says every such choice falls into exactly one of three combinatorial types: none survives (a residue class is empty), four survive (two classes of size two), or three do (one class of size three). Everything after that is bookkeeping inside a type, and Lemma 4.5 shows the last two types are one statement seen twice. Figure 8 is that sentence drawn.

Lemma 4.1. *Let S be a set of t columns and $M_S = \det(\zeta^{k\beta_j})_{0 \leq k < t, j \in S}$. Then $M_S = 0$ unless the residues $\{\beta_j \bmod t : j \in S\}$ are pairwise distinct, in which case $M_S = (-1)^{\text{inv}(b_S)} V$.*

Proof. Since $\zeta^{k\beta_j} = (\zeta^{\beta_j})^k$, the matrix is a Vandermonde matrix in the values $y_j = \zeta^{\beta_j}$, whose determinant is $\prod_{j < j'} (y_{j'} - y_j)$. The value y_j depends on β_j only modulo t , so two columns of S sharing a residue make the determinant vanish. If the residues are pairwise distinct then $|S| = t$ forces $\{y_j\}_{j \in S} = \mu_t$ as a set, and the Vandermonde determinant is alternating in its arguments, so it equals V times the sign of the permutation sorting b_S increasingly, which is $(-1)^{\text{inv}(b_S)}$. $\square$

Lemma 4.2. $\det(x_i^{N-j})_{i,j=1}^N = (-1)^{\binom{t}{2}} V(z^t - 1)(z^{-t} - 1)(z - z^{-1})$ for the alphabet $x = (1, \zeta, \dots, \zeta^{t-1}, z, z^{-1})$ in that order.

Proof. The determinant is $\prod_{i < i'} (x_i - x_{i'})$. Pairs inside μ_t contribute $\prod_{k < k'} (\zeta^k - \zeta^{k'}) = (-1)^{\binom{t}{2}} V$; pairs (ζ^k, z) contribute $\prod_k (\zeta^k - z) = (-1)^t (z^t - 1)$ and pairs (ζ^k, z^{-1}) contribute $(-1)^t (z^{-t} - 1)$; the last pair contributes $z - z^{-1}$. The two factors $(-1)^t$ cancel. $\square$

We compute it because it is three lines, not because it is unfamiliar: a determinant over an alphabet closed under $x \mapsto x^{-1}$, evaluated in closed product form, is exactly the subject of Krattenthaler's complement to the determinant calculus [Kra05, §5.11], whose Lemma 20 carries the Weyl denominator formulas for B_n , C_n and D_n among its special cases. What that literature does not evaluate — and what the next lemma needs — is the same determinant with a root-of-unity orbit sitting inside the alphabet.

Lemma 4.3. *Let $\mathcal{U}$ be the set of pairs $U = \{j_1 < j_2\}$ of columns whose complement S_U carries all t residues. Then*

$$\Phi_t(\lambda; z) = \frac{(-1)^{t + \binom{N+1}{2}}}{(z^t - 1)(z^{-t} - 1)(z - z^{-1})} \sum_{U \in \mathcal{U}} (-1)^{j_1 + j_2 + \text{inv}(b_{S_U})} f(\beta_{j_1} - \beta_{j_2}).$$

Proof. Expanding the numerator by Laplace along the first t rows,

$$\det(x_i^{\beta_j}) = \sum_{|S|=t} (-1)^{\binom{t+1}{2} + \sum_{j \in S} j} M_S C_{S^c},$$

where for $S^c = \{j_1 < j_2\}$ the complementary 2×2 minor is $z^{\beta_{j_1} - \beta_{j_2}} - z^{-(\beta_{j_1} - \beta_{j_2})} = f(\beta_{j_1} - \beta_{j_2})$. By Lemma 4.1 only S with pairwise distinct residues contribute. Writing $\sum_{j \in S} j = \binom{N+1}{2} - j_1 - j_2$ and dividing by Lemma 4.2, the factors V cancel and $(-1)^{\binom{t+1}{2} - \binom{t}{2}} = (-1)^t$. $\square$

If some $n_i = 0$ then no S of size t carries all residues, $\mathcal{U} = \emptyset$, and $\Phi_t = 0$: this is Theorem 3.1(i). Otherwise all $n_i \geq 1$ and by (3) either two classes have size two, in which case S_U must drop one element of each and $|\mathcal{U}| = 4$, or one class has size three, in which case S_U drops two of the three and $|\mathcal{U}| = 3$.

Lemma 4.4 (the column move). *Let A and B be distinct classes of size two, at columns $j_{A_1} < j_{A_2}$ and $j_{B_1} < j_{B_2}$. For $i, j \in \{1, 2\}$ let U_{ij} consist of the column of a_i and the column*

of b_j , and set $\kappa_{ij} = (-1)^{j_{A_i} + j_{B_j} + \text{inv}(b_{S_{U_{ij}}})} \text{sgn}(a_i - b_j)$, so that the U_{ij} term of Lemma 4.3 equals $\kappa_{ij} f(a_i - b_j)$. Then $\kappa_{ij} = \kappa_{11} (-1)^{i-1} (-1)^{j-1}$.

Proof. Fix j and compare $i = 1$ with $i = 2$. The sets $S_{U_{1j}}$ and $S_{U_{2j}}$ differ only in that the first contains the column j_{A_2} and the second the column j_{A_1} , and the letter carried is the residue r_A in both. Passing from one word to the other moves that letter across exactly the kept columns strictly between j_{A_1} and j_{A_2} , and none of those letters equals r_A because the class A has only two elements; each crossing changes the parity of inv . Hence

$$(-1)^{\text{inv}(b_{S_{U_{1j}}}) - \text{inv}(b_{S_{U_{2j}}})} = (-1)^{(j_{A_2} - j_{A_1} - 1) - [a_2 < b_j < a_1]},$$

using that β is strictly decreasing in the column index, so that j_{B_j} lies strictly between j_{A_1} and j_{A_2} exactly when $a_2 < b_j < a_1$. Multiplying by the Laplace factor $(-1)^{j_{A_1} + j_{A_2}}$,

$$\frac{\kappa_{1j}}{\kappa_{2j}} = (-1)^{2j_{A_2} - 1} (-1)^{[a_2 < b_j < a_1]} \text{sgn}(a_1 - b_j) \text{sgn}(a_2 - b_j) = -1,$$

because $\text{sgn}(a_1 - b_j) \text{sgn}(a_2 - b_j) = -1$ exactly when b_j lies between a_2 and a_1 . Exchanging the roles of A and B gives $\kappa_{i1}/\kappa_{i2} = -1$. $\square$

Lemma 4.5. For all c, p, q and all u, v ,

$$\begin{aligned} \text{(a)} \quad & \sum_{\epsilon, \eta \in \{\pm 1\}} \epsilon \eta f(c + \epsilon p + \eta q) = f(c) f(p) f(q), \\ \text{(b)} \quad & f(2u) - f(2u + 2v) + f(2v) = -f(u) f(v) f(u + v). \end{aligned}$$

Proof. For (a), $\sum_{\epsilon, \eta} \epsilon \eta z^{c + \epsilon p + \eta q} = z^c (\sum_{\epsilon} \epsilon z^{\epsilon p}) (\sum_{\eta} \eta z^{\eta q}) = z^c f(p) f(q)$, and the same computation for the negative exponents gives $z^{-c} (-f(p)) (-f(q))$; subtract. For (b), put $s = z^u$, $w = z^v$ and expand $(s - s^{-1})(w - w^{-1})(sw - s^{-1}w^{-1})$: the eight terms recombine as $(s^2 w^2 - s^{-2} w^{-2}) - (s^2 - s^{-2}) - (w^2 - w^{-2}) = f(2u + 2v) - f(2u) - f(2v)$. $\square$

Proof of Theorem 3.1. In the two-class profile, Lemmas 4.3 and 4.4 give

$$\sum_{U \in \mathcal{U}} (\cdots) = \kappa_{11} \sum_{i,j} (-1)^{i-1} (-1)^{j-1} f(a_i - b_j).$$

Writing $a_i = \alpha + \epsilon_i p$ and $b_j = \gamma + \eta_j q$ with $p = d_1/2$, $q = d_2/2$ and $\epsilon = \eta = (+1, -1)$, we have $a_i - b_j = c + \epsilon_i p - \eta_j q$ with $c = \alpha - \gamma$. Replacing η by $-\eta$ turns the sum into $-\sum_{\epsilon, \eta} \epsilon \eta f(c + \epsilon p + \eta q)$, which is $-f(c) f(p) f(q)$ by Lemma 4.5(a). Since $(z^t - 1)(z^{-t} - 1) = 2 - z^t - z^{-t} = -f(t/2)^2$ and $z - z^{-1} = f(1)$, the two minus signs cancel, and substituting $f(u) = 2 \sinh(u\theta)$ the numerical factors cancel. Here $2c = \tilde{d}_3$. If $\tilde{d}_3 = 0$ then $f(c) = 0$ and the whole sum vanishes, which is the first assertion of (ii); otherwise $\sinh(c\theta) = \text{sgn}(\tilde{d}_3) \sinh(d_3\theta/2)$, giving (5) with $\varepsilon_\lambda = (-1)^{t + \binom{N+1}{2}} \kappa_{11} \text{sgn}(\tilde{d}_3)$, which is (6) by the definition of κ_{11} in Lemma 4.4.

For the size-three profile $\{p > q > r\}$ at columns $j_1 < j_2 < j_3$ the same column-move argument applies, again because no letter strictly between two of these columns carries the class residue, and gives alternating signs $(\varsigma, -\varsigma, \varsigma)$ for the three terms $f(p - q)$, $f(p - r)$, $f(q - r)$. By Lemma 4.5(b) with $u = (p - q)/2$, $v = (q - r)/2$ the sum is $-\varsigma f(u) f(v) f(u + v)$, which is the two-class formula for $A = \{p, q\}$, $B = \{q, r\}$ with $\varsigma = \kappa_{11}$. Equivalently, the missing fourth selection of the two-class profile would drop q twice and its term carries $f(0) = 0$, so the two profiles are one statement. $\square$

All the statements of this paper were also checked numerically; the counts are collected once, in §9, rather than repeated as they arise.

5. Independence on the reciprocal locus

Theorem 5.3 of [AK25] is stated for free variables, and $(z, \bar{z})$ is not a free pair. Nothing in this section contradicts it: the reciprocal locus lies outside its hypotheses, not against them, and what we do here is determine what its criterion becomes there. Read at face value, (4) would say that for $\ell(\lambda) \leq 2$ one has $s_\lambda(z, \bar{z}, \mu_t) = s_\lambda(z, \bar{z})$ exactly on the t -cores. Theorem 3.1 evaluates the left-hand side for every λ , and comparing the two shows that the reciprocal locus carries further solutions — exactly one family. Measuring how far a criterion survives a degeneration is a way of saying how sharp it was, and the answer here is that it is sharp up to a single family.

We first isolate when Φ_t degenerates to a single character. Recall $\chi_k(z) = (z^{k+1} - z^{-k-1})/(z - z^{-1})$.

Lemma 5.1. $\Phi_t(\lambda; z) = \pm \chi_k(z)$ for some $k \geq 0$ if and only if the multiset $\{d_1, d_2, d_3\}$ contains t at least twice, and in that case the remaining entry equals $2(k+1)$.

Proof. Put $u = z^{1/2}$, so that $2 \sinh(d_i \theta/2) = u^{d_i} - u^{-d_i}$, $2 \sinh(t \theta/2) = u^t - u^{-t}$ and $2 \sinh \theta = u^2 - u^{-2}$; all exponents are integers. By (5), the asserted equality is

$$\prod_{i=1}^3 (u^{d_i} - u^{-d_i}) = \pm (u^t - u^{-t})^2 (u^{2k+2} - u^{-2k-2}).$$

Using $u^n - u^{-n} = u^{-n}(u^{2n} - 1)$ on each factor and comparing the monomial prefactors gives $d_1 + d_2 + d_3 = 2t + 2(k+1)$ together with

$$\prod_{i=1}^3 (u^{2d_i} - 1) = \pm (u^{2t} - 1)^2 (u^{4(k+1)} - 1),$$

and evaluating at $u = 0$ fixes the sign to $+$. Now $u^{2n} - 1 = \prod_{m|2n} Q_m(u)$ with Q_m the cyclotomic polynomials as in Proposition 3.10, which are irreducible and pairwise distinct, so by unique factorization in $\mathbb{Z}[u]$ the two sides have the same multiset of cyclotomic factors. On the left that multiset is $\bigsqcup_i \{m : m \mid 2d_i\}$ and on the right it is $2 \times \{m : m \mid 2t\} \sqcup \{m : m \mid 4(k+1)\}$. Its largest element is $\max_i 2d_i$ on the left and $\max(2t, 4(k+1))$ on the right; deleting the divisor set of that maximum from both sides and repeating twice more yields

$$\{2d_1, 2d_2, 2d_3\} = \{2t, 2t, 4(k+1)\}$$

as multisets, which is the assertion. The converse is the same computation read backwards. $\square$

Theorem 5.2. Let $\ell(\lambda) \leq 2$. Then

$$\Phi_t(\lambda; z) = s_\lambda(z, \bar{z})$$

— an equality, not an equality up to sign — if and only if either $\lambda = \text{core}_t(\lambda)$, or

$$t \text{ is even and } \lambda = \left(\lambda_2 + \frac{3t}{2} - 1, \lambda_2 \right) \text{ with } \frac{t}{2} \leq \lambda_2 \leq t-1. \quad (17)$$

In the second case $\{d_1, d_2, d_3\} = \{3t, t, t\}$ and $\varepsilon_\lambda = +1$, while $\text{core}_t(\lambda) = (\frac{t}{2} - 1 + j, j)$ and $\text{quot}_t(\lambda)$ has the single nonempty component (2) , in slot $j = \lambda_2 - \frac{t}{2}$.

Figure 9 is the statement computed rather than asserted: both families are plotted over a range of t , and the odd t where (17) cannot occur is drawn as the control.

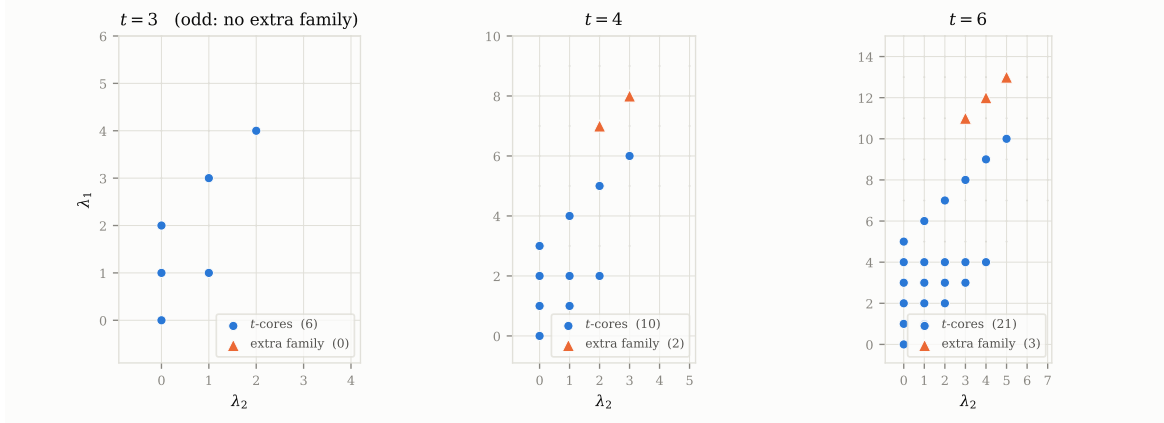


Figure 9. Theorem 5.2, computed rather than asserted. For every two-row λ in range, both sides of $\Phi_t(\lambda; z) = s_\lambda(z, \bar{z})$ were evaluated, sign included; the partitions where they agree are plotted and coloured by family. Circles are the t -cores, the solutions already described by (4); triangles are the extra family (17), which the reciprocal specialization creates. The odd case $t = 3$ is the control: by Theorem 5.2 no extra solution can exist there, and none is found.

Proof. Since $\ell(\lambda) \leq 2$ and $N = t+2$, the beta set is $\beta(\lambda) = (A'+t, B'+t, t-1, t-2, \dots, 1, 0)$ where

$$A' := \lambda_1 + 1, \quad B' := \lambda_2, \quad A' > B' \geq 0.$$

The block $\{0, 1, \dots, t-1\}$ meets every residue class exactly once, so no class is empty and the two large parts decide the profile. Write $A' = t\alpha + \rho_A$ and $B' = t\gamma + \rho_B$ with $0 \leq \rho_A, \rho_B \leq t-1$. We first determine the solutions of $\Phi_t(\lambda; z) = \pm s_\lambda(z, \bar{z})$ and settle the sign at the end. As $s_\lambda(z, \bar{z}) = \chi_{\lambda_1 - \lambda_2}(z)$, Lemma 5.1 says that this holds precisely when $\{d_1, d_2, d_3\}$ contains t twice and the third entry is $2(\lambda_1 - \lambda_2 + 1) = 2(A' - B')$.

Case $\rho_A = \rho_B =: \rho$. One class has size three, namely $\{A'+t, B'+t, \rho\}$, and by Theorem 3.1 the three arguments are $d_1 = A' - B'$, $d_2 = B' + t - \rho = t(\gamma + 1)$, $d_3 = A' + t - \rho = t(\alpha + 1)$. Here $t \mid A' - B'$, say $A' - B' = ts$ with $s \geq 1$. The pair $d_2 = d_3 = t$ forces $\alpha = \gamma = 0$, hence $A' = B'$, excluded. The pair $d_1 = d_3 = t$ forces $s = 1$ and $\alpha = 0$, hence $A' < t$ and $A' = B' + t \geq t$, excluded. The pair $d_1 = d_2 = t$ forces $s = 1$ and $\gamma = 0$, i.e. $A' = B' + t$ with $B' < t$; then $\alpha = 1$ and the third entry is $d_3 = 2t = 2(A' - B')$, as required. So this case contributes exactly the λ with $A' = B' + t$, $B' < t$.

Case $\rho_A \neq \rho_B$. Two classes have size two, namely $\{A'+t, \rho_A\}$ and $\{B'+t, \rho_B\}$, and the three arguments are

$$d_A = t(\alpha + 1), \quad d_B = t(\gamma + 1), \quad d_3 = |t(\alpha - \gamma) + 2(\rho_A - \rho_B)|.$$

If $d_A = d_B = t$ then $\alpha = \gamma = 0$, so $A', B' < t$ and $d_3 = 2|\rho_A - \rho_B| = 2(A' - B')$ automatically: every such λ is a solution. Otherwise exactly one of d_A, d_B equals t and $d_3 = t$. If $\alpha = 0$ and $\gamma \geq 1$ then $|-t\gamma + 2(\rho_A - \rho_B)| = t$ with $|\rho_A - \rho_B| \leq t-1$ forces $\gamma = 2$ and $\rho_A - \rho_B = t/2$; but then $\lambda_1 = A' - 1 = \rho_A - 1 < t$ while $\lambda_2 = B' \geq 2t$, contradicting $\lambda_1 \geq \lambda_2$. If $\gamma = 0$ and $\alpha \geq 1$ then $|t\alpha + 2(\rho_A - \rho_B)| = t$ forces, for $\alpha = 1$, $\rho_A = \rho_B$ (excluded), and for $\alpha = 2$,

$$\rho_B - \rho_A = \frac{t}{2},$$

which requires t even; $\alpha \geq 3$ gives $|2(\rho_A - \rho_B)| \geq 2t$, impossible. So $A' = 2t + \rho_A$, $B' = \rho_B = \rho_A + \frac{t}{2}$, whence

$$\lambda_1 = 2t + \rho_A - 1, \quad \lambda_2 = \rho_A + \frac{t}{2}, \quad \lambda_1 - \lambda_2 = \frac{3t}{2} - 1,$$

and $\rho_B \leq t - 1$ gives $0 \leq \rho_A \leq \frac{t}{2} - 1$, i.e. $\frac{t}{2} \leq \lambda_2 \leq t - 1$. The remaining entry is $d_A = t(\alpha + 1) = 3t = 2(\lambda_1 - \lambda_2 + 1)$, as required. This is (17).

It remains to identify the solutions collected in the first case of each analysis with the t -cores. For $\ell(\lambda) \leq 2$ the beta set in two slots is (A', B') , and λ is a t -core if and only if neither beta number can be lowered by t into an unoccupied nonnegative position, i.e. if and only if $B' < t$ and either $A' < t$ or $A' = B' + t$. These are exactly the two families found above. Finally, in case (17) the class ρ_A carries $\{2t + \rho_A + t, \rho_A\}$, so its quotient component is (2), every other component is empty, and the core is as stated.

The sign. The two families get it for different reasons. On the t -cores the equality is (4) itself, which is an equality and not an equality up to sign, so there is nothing to prove. On (17) write $t = 2m$ and $\lambda_2 = m + j$ with $0 \leq j \leq m - 1$, so that $\rho_A = j$ and $\rho_B = m + j$ and

$$\beta(\lambda) = (6m + j, 3m + j, 2m - 1, 2m - 2, \dots, 1, 0).$$

Reducing modulo $t = 2m$, the residue word read in increasing column order is

$$w = (j, m + j, 2m - 1, 2m - 2, \dots, 1, 0),$$

since $6m + j \equiv j$ and $3m + j \equiv m + j$, the latter because $j \leq m - 1$. Its inversions are of three kinds: the decreasing tail contributes $\binom{2m}{2}$; the first letter is above exactly the j tail letters $0, \dots, j - 1$; the second is above exactly the $m + j$ tail letters $0, \dots, m + j - 1$; and the first two letters are in increasing order, so they contribute none. Hence

$$\text{inv}(w) = \binom{2m}{2} + j + (m + j) = 2m^2 + 2j \equiv 0 \pmod{2}, \quad (18)$$

so $\text{sgn}(\sigma) = (-1)^{\text{inv}(w)} = +1$ by the first line of the proof of Proposition 3.11. The two distinguished classes are $r_A = j < r_B = m + j$, carrying $A = \{6m + j, j\}$ and $B = \{3m + j, m + j\}$, so $\tilde{d}_3 = (6m + 2j) - (4m + 2j) = 2m > 0$ and $\text{sgn}(\tilde{d}_3) = +1$. The remaining two factors of (14) are $(-1)^{\lfloor t/2 \rfloor} = (-1)^m$ and $(-1)^{r_A + r_B} = (-1)^{m + 2j} = (-1)^m$, whose product is $+1$. Therefore $\varepsilon_\lambda = +1$, and since $d_1 = 6m = 3t$, $d_2 = 2m = t$ and $d_3 = 2m = t$, (5) reads

$$\Phi_t(\lambda; z) = \frac{\sinh(3t\theta/2) \sinh^2(t\theta/2)}{\sinh^2(t\theta/2) \sinh \theta} = \chi_{\frac{3t}{2}-1}(z) = \chi_{\lambda_1 - \lambda_2}(z) = s_\lambda(z, \bar{z}),$$

with no sign to carry. $\square$

What (18) buys is not decoration. Stated up to sign, Theorem 5.2 says that the specialization creates a family of shapes on which the twist is invisible *to within a sign*, which is a weaker statement than the one (4) makes on the cores; with the sign settled the two families are solutions of the same equation, and “the reciprocal locus acquires exactly one further family” is literally true.

Remark 5.3. Lemma 5.1 controls the value only up to sign, which is why the sign of Theorem 5.2 needs the separate paragraph above rather than falling out of the classification.

The extra family belongs to the specialization rather than to the alphabet. The step is [AK25, Lemma 5.2], on which [AK25, Theorem 5.3] rests: it reduces the independence to the vanishing of $s_{\lambda/\mu}$ at the twisted alphabet for all $\mu \subsetneq \lambda$, and the reduction uses the linear

independence of the universal characters f_μ in the free variables. On the reciprocal locus $s_\mu(z, \bar{z}) = \chi_{\mu_1 - \mu_2}(z)$ depends only on $\mu_1 - \mu_2$, so $s_{(1)}$, $s_{(2,1)}$ and $s_{(3,2)}$ agree there and that independence is not available; Theorem 5.2 measures the difference. A control confirms the reading: for $\lambda = (3, 1)$ and $t = 2$ one has $s_\lambda(\mu_t, z, \bar{z}) = s_\lambda(z, \bar{z})$, while $s_\lambda(\mu_t, x, y) \neq s_\lambda(x, y)$ for free x, y .

So the criterion does not survive the specialization unchanged, and what it acquires is small and nameable: one family, for two-row shapes, indexed by an order-two element of the residue lattice and classified by its core and quotient. The extra solutions belong to the reciprocal locus, not to the alphabet — they exist because a specialization has removed a linear independence the original argument used, and that missing step is exactly what Theorem 5.2 measures.

6. An enumeration in which a parameter survives

Expanding s_λ over semistandard tableaux turns Theorem 3.1 into a product formula for a weighted count. Let $m_k(T)$ be the number of entries equal to k in T .

Corollary 6.1. *Expanding s_λ over semistandard tableaux and applying Theorem 3.1: for every $t \geq 2$ and every $\lambda \in \mathcal{P}_{t+2}$,*

$$\sum_{T \in \text{SSYT}(\lambda, t+2)} \zeta^{\sum_{k=1}^t (k-1)m_k(T)} z^{m_{t+1}(T) - m_{t+2}(T)} = \varepsilon_\lambda \frac{\sinh(d_1\theta/2) \sinh(d_2\theta/2) \sinh(d_3\theta/2)}{\sinh^2(t\theta/2) \sinh \theta}.$$

For $t = 2$ the weight is $(-1)^{m_2(T)}$, and if $\lambda = (c^a)$ is rectangular then $\text{SSYT}(\lambda, 4)$ is in bijection with plane partitions in an $a \times (4 - a) \times c$ box, equivalently with lozenge tilings of a hexagon. Corollary 6.1 is then a (-1) -enumeration of those objects *refined by a free parameter z* , which records the difference between the numbers of entries 3 and 4. Setting $z = 1$ recovers the plain signed count, which by (5) equals $\pm d_1 d_2 d_3 / (2t^2)$: for $a = 2$, $c = 2$ it is 4; for $a = 2$, $c = 4$ it is 9; for $a = 3$, $c = 3$ it is -6 .

Example 6.2. Taking $\lambda = (c^a)$ with $1 \leq a \leq 4$ and $t = 2$, the shape has at most four rows, the tableaux are the plane partitions in an $a \times (4 - a) \times c$ box, and the closed form gives the signed counts of Figure 10. Reading it off:

$$\begin{aligned} 1 \times 3 \times c &: \lfloor (c+2)^2/4 \rfloor, \\ 2 \times 2 \times c &: (c/2 + 1)^2 \text{ for } c \text{ even,} & 0 \text{ for } c \text{ odd,} \\ 3 \times 1 \times c &: (-1)^c \lfloor (c+2)^2/4 \rfloor, \\ 4 \times 0 \times c &: (-1)^c. \end{aligned}$$

The third row is the transpose of the first, which is why the magnitudes agree; the fourth is the degenerate box, where $s_{(c^4)}(x) = (x_1 x_2 x_3 x_4)^c$ and the alphabet has determinant -1 . We do not claim these particular counts as new — the boxes are small and the sign here is the specialization $x_2 = -1$, not the complementation sign of Kuperberg's (-1) -enumerations — and we record them only to show what the theorem produces. What does appear to be new is that each of them is the $z \mapsto 1$ value of a one-parameter refinement with the same product formula.

The perfect squares in Example 6.2 are not an accident of small cases: they are what the interval reading of §3.3 says a rectangle must produce.

Proposition 6.3. *Let $t = 2$ and $\lambda = (c^a)$ with $1 \leq a \leq 4$. Then one of d_1, d_2, d_3 equals 2, and:*

- (i) *if $a = 4$, all three equal 2 for every c and the count is $(-1)^c$; this is the full rectangle, where $s_{(c^4)}(x) = (x_1 x_2 x_3 x_4)^c$ collapses to a monomial and the alphabet contributes only its determinant;*
- (ii) *if $a \leq 3$ and c is even, the other two are both equal to $c + 2$, so the two intervals have the same length and the signed count is the perfect square $(\frac{c+2}{2})^2$;*
- (iii) *if c is odd and $a = 2$, the two intervals are concentric and the count is 0;*
- (iv) *if c is odd and $a \in \{1, 3\}$, the other two are $c + 1$ and $c + 3$, and the count is $\pm \frac{(c+1)(c+3)}{4} = \pm \lfloor \frac{(c+2)^2}{4} \rfloor$.*

Case (i) is not a degeneration of the interval triple but a collapse of the shape itself, and it is the only case in which the parity of c plays no role.

Proof. For $\lambda = (c^a)$ the beta set consists of two blocks of consecutive integers, namely $c + 3, c + 2, \dots, c + 4 - a$ and $3 - a, 2 - a, \dots, 0$; residues alternate along each block, which is what forces the coincidences. Explicitly, for $a = 1, 2, 3, 4$ the beta set is $(c + 3, 2, 1, 0)$, $(c + 3, c + 2, 1, 0)$, $(c + 3, c + 2, c + 1, 0)$ and $(c + 3, c + 2, c + 1, c)$. Splitting each by parity gives, for c even,

$\{c+3, 1\}, \{2, 0\}; \quad \{c+3, 1\}, \{c+2, 0\}; \quad \{c+3, c+1\}, \{c+2, 0\}; \quad \{c+3, c+1\}, \{c+2, c\},$
 whence (d_1, d_2, d_3) equals $(c+2, 2, c+2)$, $(c+2, c+2, 2)$, $(2, c+2, c+2)$ and $(2, 2, 2)$ respectively; in each case the product over $2t^2 = 8$ is $(\frac{c+2}{2})^2$, resp. 1. For c odd and $a = 2$ the split is $\{c+3, 0\}, \{c+2, 1\}$, so $d_3 = |c+3 + 0 - c-2 - 1| = 0$: the intervals are concentric and Corollary 3.2 applies. For c odd and $a = 1, 3$ the parity puts three beta numbers in one class, $\{c+3, 2, 0\}$ and $\{c+3, c+1, 0\}$ respectively, and the size-three profile gives $(c+1, 2, c+3)$ and $(2, c+1, c+3)$, with product over 8 equal to $\frac{(c+1)(c+3)}{4}$. For c odd and $a = 4$ the four beta numbers are again consecutive, so the split is $\{c+3, c+1\}, \{c+2, c\}$ and the triple is $(2, 2, 2)$ exactly as for c even: at full height the answer does not see the parity of c , which is why $a = 4$ is separated out. $\square$

So the square is the square of half the common interval length, the zeros are the concentric configurations, the quarter-squares are the case where the two lengths differ by 2, and at full height all three intervals coincide, the shape becomes a monomial and only a sign survives. Which pair of the three coincides rotates with a , which is why the same value appears in different columns of Figure 10.

The (-1) -enumeration literature computes such signed counts, and it is worth being exact about how, because the difference here is narrow. Kuperberg [Kup02] weights the objects by ± 1 directly. Eisenkölbl [Eis07] specializes $x_i = (-1)^i$ in a Schur function identity of the kind Stanley [Sta86] used — an identity recorded there as also being [IOTZ06, Cor. 7.3] — and gets the (-1) -enumeration of self-complementary plane partitions with an odd side. Ciucu and Krattenthaler evaluate the corresponding determinants. In each of these the alphabet is a number and nothing is left free, which is the one point of difference: here the reciprocal pair keeps a variable alive through the signed count. That is the alphabet's doing and not an argument of ours.

A refinement of a different kind is standard in the ribbon world, and it is worth saying which: the LLT polynomials of Lascoux, Leclerc and Thibon [LLT97] grade ribbon tableaux by a spin statistic. That q is an extra grading rather than a letter, and the object it grades is indexed by a tuple of shapes; the parameter here is the alphabet's own and what carries it

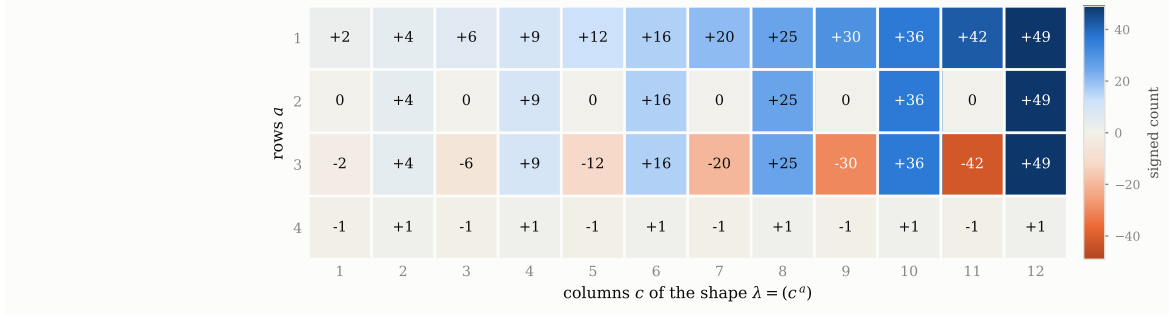


Figure 10. The signed counts of Example 6.2: the value of Corollary 6.1 at $z = 1$ and $t = 2$ for the rectangular shapes $\lambda = (c^a)$, that is the (-1) -enumeration of plane partitions in an $a \times (4 - a) \times c$ box. Every entry is the closed form $\varepsilon_\lambda d_1 d_2 d_3 / (2t^2)$, cross-checked against a direct enumeration of the tableaux. The row $a = 2$ vanishes for odd c and is a perfect square for even c .

is a single s_λ . The two refine different things, which is why we put the point narrowly: it is the reciprocal pair, and not a statistic, that keeps a variable alive through the signed count.

A signed enumeration with a product formula asks for a sign-reversing involution, and the obstruction is specific: the involution must preserve $m_{t+1} - m_{t+2}$, or the free parameter is not carried across, and neither Bender–Knuth moves nor the single-cell toggle do. The construction in [KumT] of a fixed-point-free sign-reversing involution on semistandard tableaux over a $\pm$ -paired alphabet, whose fixed set is the domino-tileable tableaux, does have the required property in its $i = 1$ instance: it touches only cells holding 1 or 2, hence preserves $z^{m_3 - m_4}$ and reverses $(-1)^{m_2}$. The triple product it would have to land on is the one of Corollary 3.3, which at $t = 2$ is a product of three $\mathfrak{sl}_2$ characters on the nose, so the missing bijection is a Clebsch–Gordan statement. That is one half. The other is the cancellation across the splitting of the alphabet, and at $t = 2$ it concerns a single one-parameter family. By (20) below,

$$s_\lambda(1, -1, z, \bar{z}) = \sum_{\ell(\nu) \leq 2} s_{\lambda/\nu}(1, -1) \chi_{\nu_1 - \nu_2}(z),$$

and an involution carrying the parameter must preserve $k = \nu_1 - \nu_2$, since that is the index of the character its term sits on. For fixed k the admissible ν are exactly $(m + k, m)$ with $m \geq 0$, so that cancellation is a statement about one word.

The size of the terms is classical. Littlewood’s theorem has a skew form — $\varphi_t s_{\lambda/\nu}$ vanishes unless λ/ν is tileable by t -ribbons, and is otherwise a ribbon sign times a product over the quotient — due to Macdonald [Mac95, p. 91], and proved there by the same Jacobi–Trudi route we take below; the case where ν is the t -core is Farahat’s. At $t = 2$ on our alphabet it gives $\sigma_m \in \{0, \pm 1\}$ at once. What is not classical is how the sign moves with m , and that is what the involution needs.

Proposition 6.4 (the runs alternate). *Let $t = 2$, let $k \geq 0$, and put $\sigma_m = s_{\lambda/(m+k, m)}(1, -1)$ for $m \geq 0$. Then*

- (i) $\sigma_m \in \{0, \pm 1\}$ — *this part is Macdonald’s, not ours;*
- (ii) *if k is odd, no two consecutive σ_m are nonzero;*
- (iii) *if σ_m and σ_{m+1} are both nonzero then $\sigma_{m+1} = -\sigma_m$.*

Consequently, pairing m with $m + 1$ inside each maximal run of consecutive nonzero σ is a sign-reversing involution, and $\sum_m \sigma_m$ is the signed count of the runs of odd length.

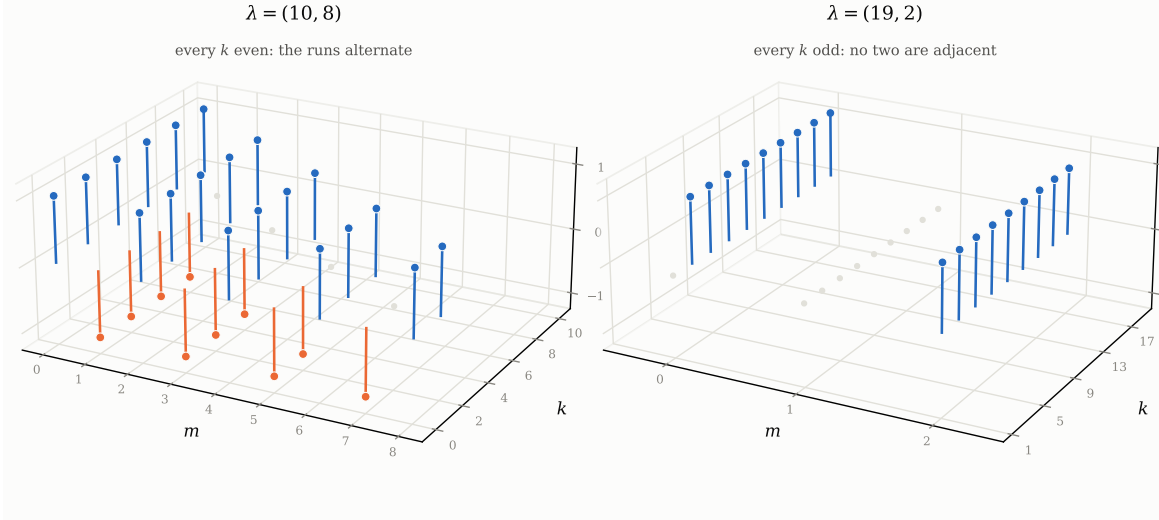


Figure 11. Proposition 6.4, drawn. For each k the admissible ν are the single family $(m+k, m)$, and $\sigma_m = s_{\lambda/(m+k, m)}(1, -1)$ is plotted against m and k ; the grey dots are the zeros. Every bar has height exactly ± 1 , which is part (i): the terms are a set, so the cancellation the enumeration asks for is a matching and not something carrying multiplicity. On the left every k is even and the colour alternates along each run, which is part (iii) and what makes $m \leftrightarrow m+1$ sign-reversing; where a run ends, a zero ends it. On the right every k is odd and no two nonzero terms are adjacent, which is part (ii): the runs have length one and there is nothing to cancel.

The three parts are visible at once in Figure 11: the bar heights are part (i), the gaps on the odd panel are part (ii), and the alternation of colour along each run is part (iii).

Proof. Write $\beta_i = \lambda_i - i$ and $c_j = \nu_j - j$, and take the Jacobi–Trudi matrix of $s_{\lambda/\nu}$ in $n \geq \ell(\lambda)$ rows. Since $h_j(1, -1) = 1$ for $j \geq 0$ even and 0 otherwise,

$$M_{ij} = [\beta_i \geq c_j] \cdot [\beta_i \equiv c_j \pmod{2}], \quad \sigma_m = \det M. \quad (19)$$

The second factor says that column j is supported on the rows of one parity class, the class of c_j ; and since $\beta_1 > \dots > \beta_n$, inside that class the first factor cuts an *initial segment*. Order the rows by parity class: M becomes block diagonal, so it is singular unless each class carries as many columns as it has rows, and inside a class it is the staircase [the rank of i in its class $\leq \ell_j$] of the segment lengths ℓ_j . That staircase is singular if two lengths agree or one is 0, and otherwise the lengths are a permutation of $1, \dots, e$ and its determinant is the sign of that permutation. This gives (i), which is [Mac95, p. 91] in this case; we re-derive it because the β -form (19) is what (ii) and (iii) rest on.

For $\nu = (m+k, m)$ we have $c_1 = m+k-1$, $c_2 = m-2$ and $c_j = -j$ for $j \geq 3$. So $c_1 - c_2 = k+1$, and passing from m to $m+1$ raises c_1 and c_2 by one and leaves every other column alone: exactly the first two columns change parity class.

If k is odd then $c_1 \equiv c_2$, so columns 1 and 2 lie in the same class and both leave it. That class's column count changes by two while its row count does not, so at most one of σ_m , σ_{m+1} can be nonzero, which is (ii).

If k is even then $c_1 \not\equiv c_2$ and the two columns exchange classes. Suppose both are nonzero. In the class of c_1 the columns are column 1 together with a set of columns $j \geq 3$ that does

not move; by (i) the lengths are a permutation of $1, \dots, e$ before and after, and the fixed columns contribute the same lengths to both, so the length column 2 brings in equals the one column 1 took away — and symmetrically in the other class. Each class therefore sees the same lengths in the same order of columns, and both block determinants are unchanged. What does change is the shuffle separating the two classes of columns: columns 1 and 2 are adjacent and exchange membership, which alters the number of inversions of that shuffle by exactly one. Hence $\sigma_{m+1} = -\sigma_m$, which is (iii). $\square$

Remark 6.5 (the sign here is the sign of Theorem 3.1). Write σ_ν for the permutation of §2 attached to a partition ν : the one sorting β by residue class, which Proposition 3.11 places inside ε_λ . The terms of Proposition 6.4 are values of that same sign. Indeed σ_m vanishes unless $\text{core}_2(\lambda) = \text{core}_2(\nu)$ and both components of the skew 2-quotient are horizontal strips, and equals $\text{sgn}(\sigma_\lambda) \text{sgn}(\sigma_\nu)$ when it does not — the identity $\text{sgn}(\sigma_\lambda) \text{sgn}(\sigma_\nu) = (-1)^{\sum_B \text{ht}(B)}$ being the ribbon sign of [Mac95, p. 91] written as a sorting sign, which is [KumT, Lemma 6.3]. Parts (ii) and (iii) are then one statement about σ_ν alone: as $\nu = (m+k, m)$ moves to $(m+k+1, m+1)$, $\text{sgn}(\sigma_\nu)$ reverses for k even and is constant for k odd. So the permutation that carries the sign of the main theorem is the permutation that makes the involution of this section alternate; the two halves of the paper are signed by one object.

The pairing is a bijection of tableaux and not only of terms, which is what carries the parameter. A semistandard tableau of shape (a, b) in the letters 3, 4 has second row 4^b and first row $3^p 4^{a-p}$ with $b \leq p \leq a$, of weight z^{2p-a-b} ; adjoining a first column $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ sends it to the tableau of shape $(a+1, b+1)$ with $p+1$ in place of p , and $2(p+1) - (a+1) - (b+1) = 2p - a - b$, so $m_3 - m_4$ is preserved exactly. Composing that with Proposition 6.4 pairs the terms of Corollary 6.1 in the way the enumeration asks for.

One half is not ours, and we say which. Proposition 6.4 disposes of the cancellation across ν ; the cancellation *inside* each $s_{\lambda/\nu}(1, -1)$ is [KumT], and the two fit together. There, at $n = 1$, the alphabet is $(x, -x)$ and the weight of a two-letter filling is $(-1)^{c_2}$, which is σ_m ; the involution [KumT, Lemma 2.18] is fixed-point-free on the fillings that are *not coverable*, so the sum reduces to the coverable ones, which [KumT, Lemma 2.16] matches with domino tableaux. What makes the reduction usable here is [KumT, Lemma 2.14]: the parity of the number of vertical dominoes does not depend on the tableau, so every survivor carries the same sign and $\sigma_m = \pm \#\{\text{coverable fillings}\}$. With part (i) that forces the count to be at most one, so the fixed set is a single filling when $\sigma_m \neq 0$ and empty otherwise — checked directly over the 1134 skew shapes with at most four rows and $|\lambda| \leq 10$, where the two counts agree without exception.

At $t = 2$ the involution the enumeration asks for therefore exists: cancel inside each ν by [KumT, Lemma 2.18], pair what is left across ν by Proposition 6.4, and carry the parameter by the column $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$. Its fixed points are one filling for each run of odd length, and Corollary 6.1 counts them. The construction is Kumari's; the assembly is ours.

7. The factorization is isolated

A factorization is only a result if it is not a specialization of a known one, and it is only a *sharp* result if the alphabet cannot be deformed. We do not prove that in general. What we do is try four deformations of (1), and all four destroy the product.

The evidence in this section is computational, and we label it as such: none of (D1)–(D4) below is a theorem, and in particular we prove no no-go statement. What we assert is that

the identity (5) is false for each deformed alphabet, which a single counterexample settles, and we give counts to indicate how far from true it is.

- (D1) *More reciprocal pairs.* For $s_\lambda(\mu_t, z_1, z_1^{-1}, \dots, z_r, z_r^{-1})$ the product law of Theorem 3.1 already fails at $r = 2$: in the computed range typical values carry one large irreducible factor, and the only factors observed to split off are binomials in the rotated variables $z_1 z_2$ and $z_1 z_2^{-1}$, that is, the couplings between the pairs' $u(1)$ charges. What does survive is the *zero locus*: at $t = 2$ it survives every r and Theorem 8.6 determines it completely, while for arbitrary t and r Theorem 8.35 gives a uniform sufficient criterion and Conjecture 8.43 is the converse. On this evidence the product looks like a rank-one accident and the vanishing like the part that generalizes; the first half of that sentence is Conjecture 10.4 and only the second is a theorem.
- (D2) *More orbit variables.* For $s_\lambda(X, \zeta X, \dots, \zeta^{t-1} X, z, z^{-1})$ the product law fails already at $|X| = n = 2$. At $t = 2, n = 2$ the value for $\lambda = (2)$ is $(x^2 z^2 + y^2 z^2 + z^4 + z^2 + 1)/z^2$, irreducible over $\mathbb{Q}$; for $\lambda = (2, 2)$ it is $(x^2 + y^2 + z^2)(x^2 z^2 + y^2 z^2 + 1)/z^2$, two factors rather than three; and for $\lambda = (3, 1)$ it is again irreducible. By contrast, with the pair deleted the same alphabet obeys [AK25, Theorem 2.3]: $s_{(2)}(x, y, -x, -y) = x^2 + y^2$ and $s_{(2,2)}(x, y, -x, -y) = (x^2 + y^2)^2$.
- (D3) *The orbit replaced by a coset.* For $\zeta' \mu_t$ with $\zeta' = e^{\pi i/t}$, the odd powers of a primitive $2t$ -th root of unity — the twist of [AK25, §3], and for t even the full set of primitive $2t$ -th roots studied in [HI24] — the identity fails, vanishing criterion included.
- (D4) *The reciprocal pair replaced by a free pair.* $s_\lambda(\mu_t, x, y)$ was irreducible in every case we computed. We make no claim beyond that range.

Counts for (D3) and (D4), over $t = 2, \dots, 6$ and $|\lambda| \leq 10$: the identity holds in all 600 cases for (1), in 383 of 600 for (D3), and in 200 of 600 for (D4). The test is therefore capable of failing, which is the point of quoting the first row.

One reading of the proof accounts for all four. Splitting the alphabet and applying Theorem 2.1 to the frozen half gives, for any free alphabet B ,

$$s_\lambda(\mu_t \cup B) = \sum_{\nu} s_{\lambda/\nu}(\mu_t) s_{\nu}(B) = \sum_{\nu} \varepsilon_{\nu} s_{\nu}(B), \quad \varepsilon_{\nu} \in \{0, \pm 1\}, \quad (20)$$

the coefficients being ribbon signs: that the skew evaluation $s_{\lambda/\nu}(\mu_t)$ is 0 or a ribbon sign is the skew form of Theorem 2.1, due to Macdonald [Mac95, p. 91]. Now $B = (z, z^{-1})$ has exactly two letters and they are reciprocal, so $s_{\nu}(B) = \chi_{\nu_1 - \nu_2}(z)$ is a *single* $\mathfrak{sl}_2$ character depending on one difference, and the short signed sum collapses. Step one requires the frozen half to be a full orbit, since Theorem 2.1 needs the rows to meet every residue class exactly; a coset does not, which is (D3). Step two requires the free half to be exactly one reciprocal pair: a free pair has $\det = xy$ and couples to both central charges, which is (D4), and enlarging the free half — by more pairs, or by more orbit variables, which enlarges ν — makes $s_{\nu}(B)$ cease to be a single character and the sum cease to collapse, which is (D1) and (D2). *The four failures are one condition seen from four sides.*

Remark 7.1. Equivalently: the reciprocal pair contributes exactly *one* coupling, the cross term d_3 , because $\det = 1$ prevents it from seeing the two central charges separately. This is why there are three factors and not two, and Ayer–Kumari's factorizations, which are products over quotient components with no coupling at all, are the case where that contribution is absent.

Remark 7.2 (a representation-theoretic reading). The alphabet (1) is closed under inversion, hence a torus element of $O(N)$, and its determinant is $(-1)^{t+1}$. For even t it therefore lies in the non-identity component, where a character is a *twining* character; the identity computing such characters by folding is due to Jantzen [Jan73] and, in the form closest to this one, to Kumar–Lusztig–Prasad [KLP09], the relevant folded algebra being the orbit Lie algebra. The passage from their statement to ours is not a single step, and it is worth naming the two: restricting the GL_N -representation to $O(N)$ is a branching problem, and once it is decomposed into irreducible constituents the twining formula applies to each of them on the non-identity component; Clifford theory is what governs the $SO(N)/O(N)$ passage inside that, not the branching. Our restriction is reducible — for a general λ it is a $\mathbb{Z}$ -combination of symplectic characters and not one of them, as the last remark of §8 records — so what one reads off is a sum of twining characters rather than a single one. We record this because Ciucu–Krattenthaler [CK09] wrote that they could not offer a representation-theoretic interpretation of their factorizations, and [AB19] and [AK25] record that it is still missing; the reading here is of that kind. The mechanism is entirely theirs, and the proof above does not use it. Lemma 8.9 makes the reading concrete for the alphabet (21), where the folded algebra is C_r and the twining character is a symplectic character on the nose.

8. The zero locus survives every r

This section is past the factorization, and reads differently from the seven before it. Those prove statements about an identity; this one asks what is left when the identity is gone, and the answer is a locus rather than a formula. It has two halves. The first fixes $t = 2$ and proves both directions for every r and every λ , the converse resting on one external input; that is Theorem 8.6, and the route to it is what takes the section its length. The second lets t go free again and asks the same question of the whole family: there we prove one implication for every t and every r — Theorem 8.35, by an involution that pairs off every term of the expansion — record what the other direction forces, and state as Conjecture 8.43 that the two conditions are the same one. The change of register is the subject changing, not the standard.

By (D1) the product of Theorem 3.1 does not survive a second reciprocal pair. Its *zero locus* does, and for every r it has a closed form. Write

$$\Psi_r(\lambda) := s_\lambda(1, -1, z_1, \bar{z}_1, \dots, z_r, \bar{z}_r), \quad N = 2r + 2, \quad (21)$$

so that Ψ_1 is (1) at $t = 2$. Throughout this section λ is padded to exactly N parts, $\beta = \beta(\lambda)$ is its beta set, and E and O are its even and odd entries. Call λ *self-complementary of width* w if $\lambda_i + \lambda_{N+1-i} = w$ for every i , in which case $w = \lambda_1 + \lambda_N$. The two conditions the section is about are

$$\boxed{\begin{array}{cc} \underbrace{|E| \in \{0, N\}}_{\text{branch (a)}} & \underbrace{\lambda_i + \lambda_{N+1-i} = w \text{ for all } i, \ w \text{ odd}}_{\text{branch (b)}} \end{array}} \quad (22)$$

Branch (a) says the beta set has constant parity; branch (b) says λ is self-complementary of odd width, the width being $w = \lambda_1 + \lambda_N$.

We state the four claims separately, because they are not proved to the same extent and a single biconditional would hide that.

Theorem 8.1 (the two branches). *For every $r \geq 1$ and every $\lambda \in \mathcal{P}_N$: if branch (a) or branch (b) holds, then $\Psi_r(\lambda) = 0$.*

Corollary 8.2. *If λ is self-complementary of odd width w then $|\lambda| = w(r+1)$. In particular branch (b) occurs only for $|\lambda| \equiv r+1 \pmod{2(r+1)}$, and a size outside that progression needs no further test.*

Proof. Summing $\lambda_i + \lambda_{N+1-i} = w$ over i counts $|\lambda|$ twice, so $2|\lambda| = wN = w(2r+2)$. $\square$

The progression is visible in Figure 12, and it is worth recording that the analogous statement for the whole locus is false: branch (a) is not confined to it, which is why the condition has to be read off branch (b) alone.

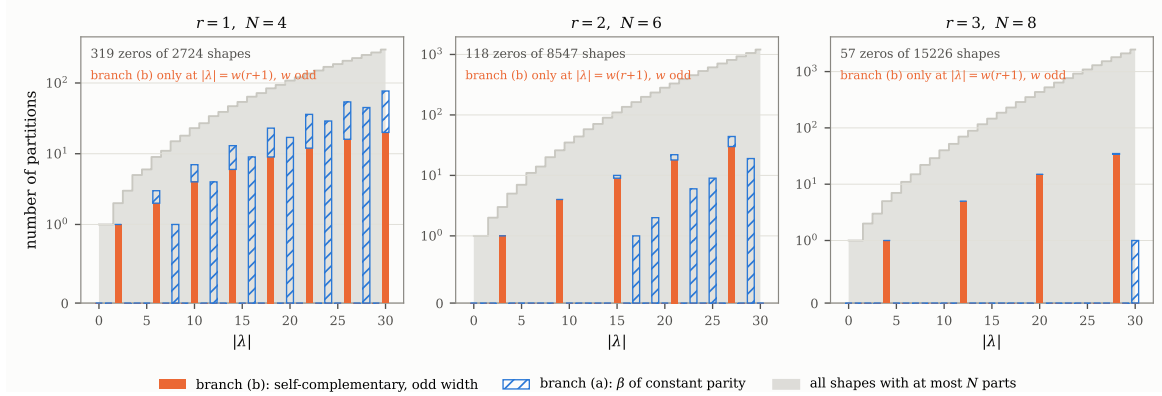


Figure 12. The zero locus of Theorem 8.6 — the biconditional, of which Theorem 8.1 is the sufficient half — counted rather than plotted. For each size $|\lambda|$ the bars give the number of partitions with at most N parts that vanish, split by branch; the shaded step is the total number of shapes of that size, on the same axis, so that the rarity of the locus is visible and not asserted. The sizes carrying branch (b) are exactly the multiples $w(r+1)$ with w odd, by Corollary 8.2; branch (a) occupies other sizes and needs the beta set to miss one parity entirely, which is why it is absent at small $|\lambda|$ once N grows.

Theorem 8.3 (one pair). *For $r = 1$ the converse holds as well: $\Psi_1(\lambda) = 0$ only if (a) or (b).*

Theorem 8.4 (inside Littlewood's range). *For every r and every λ with $\ell(\lambda) \leq N/2$, the converse holds. Equivalently, on that range $\Psi_r(\lambda) = 0$ if and only if $\lambda = (k^{N/2})$ with k odd.*

Remark 8.5 (rectangles are where the general arguments stop). It is worth noticing that both places where a general argument in this paper fails are rectangles, at two different heights and for two different reasons. At height $N/2$ an odd rectangle is self-complementary of odd width, so it vanishes; by Theorem 8.4 these are the only shapes in Littlewood's range that do, and in the proof they are precisely the shapes for which the associate witness cannot be built. At full height the object does not vanish but degenerates the other way: $s_{(c^N)}(x) = (x_1 \cdots x_N)^c$ is a monomial, so on any alphabet A the value is $\det(A)^c$ and nothing of the shape survives — which at $t = 2$ is case (i) of Proposition 6.3, the one case there in which the parity of c plays no part. Everywhere between the two heights the generic argument applies unchanged. The rectangles are not a nuisance to be handled case by case; they are the boundary of the shape, and each of the two extremes breaks a different hypothesis — the first the existence of a witness, the second the non-degeneracy of the interval triple.

Theorem 8.6 (the criterion in general). *For every r and every $\lambda \in \mathcal{P}_N$, $\Psi_r(\lambda) = 0$ if and only if (a) or (b).*

Proved at the end of §8, modulo Theorem 8.24.

Theorems 8.1–8.4 are proved in §§8.1–8.3, and Theorem 8.6 — the criterion for every r — at the end of this section, by a route independent of all three. It is stated apart from them for two reasons. It is the only statement in the paper that rests on an external input, Theorem 8.24; and the two routes to it are not the same route. Littlewood’s reduces it, in §8.4, to one existence statement, and that reduction is not completed: the range $\ell(\lambda) > N/2$ is exactly where the restriction rule stops applying, and while the values there are settled by Lemma 8.9, the statement about the coefficients is not — the certificate we proposed for it fails, by Proposition 8.11. The extremal argument of §§8.5 and 8.6 does not pass through that range at all: it works in the degree filtration of the numerator and never meets a Littlewood coefficient.

Read on the group rather than on the alphabet, the same theorem describes a rigidity property, and we state it separately because it is the form in which a representation theorist would want it. The translation is not ours and is not new: that a representation of $O(N)$ is det-stable exactly when its character vanishes off the identity component is elementary, and that this is the condition $m_\nu = m_{\nu^*}$ on Littlewood’s associates [Lit50] is classical. The corollary therefore claims exactly what Theorem 8.6 claims and nothing besides.

Corollary 8.7 (determinant-twist rigidity). *Let $N = 2r + 2$, let $\lambda \in \mathcal{P}_N$ and let V_λ be the irreducible polynomial $GL(N)$ -module of highest weight λ . Then*

$$V_\lambda|_{O(N)} \cong V_\lambda|_{O(N)} \otimes \det$$

if and only if $\beta(\lambda)$ has constant parity, or λ is self-complementary of odd width. Equivalently: (22) classifies exactly the irreducible polynomial $GL(N)$ -modules whose restriction to $O(N)$ is stable under the character of the component group.

Proof. Over characteristic 0 a finite-dimensional $O(N)$ -module is semisimple and is determined by its character, so for $W = V_\lambda|_{O(N)}$ one has $W \cong W \otimes \det$ if and only if $\chi_W = \chi_{W \otimes \det}$. Those two characters agree on the identity component automatically, and on the other one $\chi_{W \otimes \det} = -\chi_W$; so the condition is that χ_W vanish on $O(N)^-$. Every semisimple element of $O(N)^-$ is conjugate to one with spectrum $(1, -1, z_1^{\pm 1}, \dots, z_r^{\pm 1})$, which is the alphabet of (21); and since the semisimple elements are dense in $O(N)^-$ and χ_W is regular, vanishing on that torus is vanishing on the whole component. The condition is therefore $\Psi_r(\lambda) \equiv 0$, and Theorem 8.6 is the criterion. The dictionary between the two readings is Littlewood’s associate expansion, recalled in §8.3. $\square$

The credit splits exactly as it does for Theorem 8.6, and it is worth repeating here because the group language can make a known statement look new. That the two families (22) are det-stable is not ours: at the endpoint of the family the first is Karmakar’s [Kar24, Theorem 4.1(A)], and the second is complementation over the index family of Ayer and Behrend [AB19], as (23) records. What Theorem 8.6 adds, and all it adds, is that there are no others.

Remark 8.8 (what the core sees, and what it does not). Branch (a) is a condition on the core, and a simple one: at a fixed number N of beta numbers the residue profile determines $\text{core}_t(\lambda)$ — it is the Garvan–Kim–Stanton coordinate [GKS90] — so every condition on the profile is one on the core. It is not, however, the vanishing of the core but its opposite

extreme: at $t = 2$ branch (a) says

$$\text{core}_2(\lambda) = (N-1, N-2, \dots, 1) \quad \text{or} \quad \text{core}_2(\lambda) = (N, N-1, \dots, 1),$$

the two largest 2-cores that N beta numbers admit, whereas $\text{core}_2(\lambda) = \emptyset$ is the balanced profile. Branch (b) moves neither the profile nor the core, and is therefore invisible to it. Consequently core_2 does *not* determine whether Ψ_r vanishes, and the smallest witness is the empty core itself: among the shapes with $\text{core}_2(\lambda) = \emptyset$ at $r = 1$,

$$(1, 1), (3, 3), (2, 2, 1, 1), \dots \quad \text{vanish, while} \quad \emptyset, (2), (1^4), \dots \quad \text{do not,}$$

and the same split occurs at $r = 2$ and $r = 3$, there also in the classes with core (1) , $(2, 1)$ and $(3, 2, 1)$: over the ranges of §9 five core classes carry both behaviours, and the profile determines the core with no clash over 260 profiles at $t \leq 5$. No restatement of Theorem 8.1 in terms of $\text{core}_2(\lambda)$ can exist: the criterion needs the shape and not only its core. Read against Theorem 2.1, where the empty core is exactly what decides, the two criteria disagree in both directions — $(1, 1)$ has empty 2-core and $\Psi_1 = 0$, while (1) has non-empty 2-core and $\Psi_1 \neq 0$.

The index family in (b) is not new: it is exactly the family Ayyer and Behrend single out [AB19, remark after Thm. 3], together with the parity obstruction they record there — a partition cannot be self-complementary in an $a \times b$ rectangle with a and b both odd, and here $a = N$ is even, so $b = w$ odd is the admissible case. What their theorems say about those shapes is that the character *factorizes*; the alphabet there carries the fixed point $+1$ and has determinant $+1$. Ours carries $+1$ and -1 and has determinant -1 .

On the *same* family of shapes, the alphabet of determinant $+1$ makes the character factorize and the alphabet of determinant -1 makes it vanish. One letter separates a factorization from an annihilation, and the determinant of the alphabet is what decides; (23) below is where the separation happens.

8.1. Branch (a): the squared alphabet. This branch is Karmakar's, at the endpoint. [Kar24, Theorem 4.1(A)] evaluates a $\text{GL}(n, \mathbb{C})$ character at $C_{n-k, k} = (1, \dots, 1, -1, \dots, -1)$ and gives $\chi_\lambda(C_{n-k, k}) = 0$ when $\#\eta_0(\lambda) > n - k$, where the $\eta_i(\lambda) = \{a \in \lambda + \rho : a \equiv i \pmod{2}\}$ is exactly the partition of $\beta(\lambda)$ into E and O used here; at $k = 1$, after the normalization by the determinant character, that is branch (a). The one-line argument below is included because it is the one the rest of the section reuses, not because it is new.

Proof of Theorem 8.1, branch (a). Let $A = \{1, -1, z_1, \bar{z}_1, \dots\}$. If $\beta = 2\gamma$ then $\det(a^{\beta_j})_{a \in A} = \det((a^2)^{\gamma_j})_{a \in A}$, and if $\beta = 2\gamma + 1$ the same holds after extracting $\prod_{a \in A} a$. Either way the bialternant factors through the squared alphabet $A^2 = \{1, 1, z_1^2, \bar{z}_1^2, \dots\}$, in which the letter 1 occurs twice because $1^2 = (-1)^2$. Two rows coincide. $\square$

8.2. Branch (b): a determinant twist, and self-duality.

Proof of Theorem 8.1, branch (b). Let $\hat{\lambda}$ be the complement of λ in the $w \times N$ box, reversed. In beta sets $\beta(\hat{\lambda}) = c - \beta(\lambda)$ read backwards, with $c = w + N - 1$, and $\hat{\lambda}$ is the highest weight of $V_\lambda^* \otimes \det^w$ as a $\text{GL}(N)$ -module. If $\lambda = \hat{\lambda}$ then $V_\lambda \cong V_\lambda^* \otimes \det^w$. Restrict to $O(N)$: every representation of $O(N)$ is self-dual, so $V_\lambda|_{O(N)} \cong V_\lambda^*|_{O(N)} \cong V_\lambda|_{O(N)}$, whence

$$V_\lambda|_{O(N)} \cong V_\lambda|_{O(N)} \otimes \det^w.$$

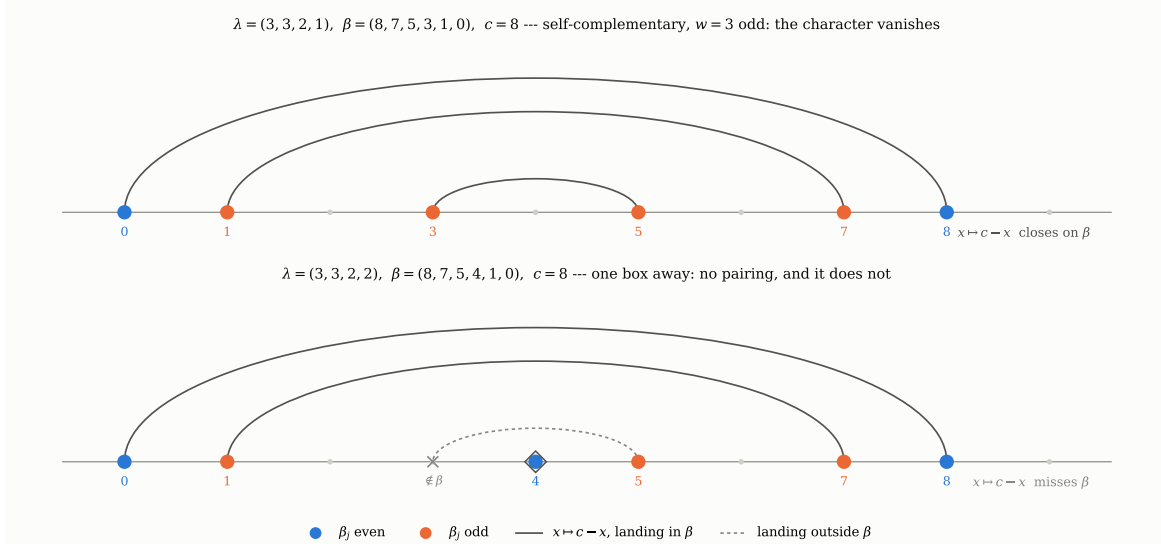


Figure 13. Why self-complementarity kills the character. Expanding along the two frozen rows leaves terms indexed by pairs (e, o) with $e \in E$, $o \in O$, and the reflection $x \mapsto c - x$ acts on them. Above, a self-complementary shape of odd width: every arc lands in β and joins entries of equal parity, because c is even. Below, the same shape with one box moved: the arc $5 \mapsto 3$ leaves β , so there is no pairing and nothing cancels. The diamond marks the fixed point of the reflection, which cannot occur inside an opposite-parity pair — which is why the involution is free.

The alphabet (21) is a torus element of the $\det = -1$ component, so for w odd the twist is by $\det$ itself and the character vanishes there. For w even $\det^w$ is trivial and the argument gives nothing, which is why the parity of w is not decoration. $\square$

Equivalently, and this is the computation rather than the representation, the standard complementation identity gives

$$s_{\hat{\lambda}}(A) = \left(\prod_{a \in A} a \right)^c (-1)^{\binom{N}{2}} (-1)^r s_{\lambda}(A) = (-1)^c (-1) s_{\lambda}(A), \quad (23)$$

using that A is closed under inversion — A^{-1} is A with the r reciprocal pairs transposed, contributing $(-1)^r$ — and that $\prod_{a \in A} a = -1$. With c even this reads $s_{\hat{\lambda}} = -s_{\lambda}$, and a self-complementary shape is annihilated. Branch (b) of Theorem 8.1 is therefore a two-line corollary of a textbook identity over a known index family, and we claim it as nothing more. Figure 13 draws the pairing the identity performs, and next to it the same shape with one box moved, where the pairing breaks; the content of the section is the converse.

8.3. The converse.

Proof of Theorem 8.3. Put $c_* = z + \bar{z}$ and let S_n be defined by $S_0 = 0$, $S_1 = 1$, $S_{n+1} = c_* S_n - S_{n-1}$, so $\deg_{c_*} S_n = n - 1$ and $S_{-n} = -S_n$; the value $S_{-1} = -1$ is used whenever $\beta_N = 0$. Since $\Phi_A = (x^2 - 1)(x^2 - c_* x + 1)$, a polynomial P supported on β is divisible by Φ_A exactly when $P(1) = P(-1) = 0$ and $P \equiv 0$ modulo $x^2 - c_* x + 1$. Splitting $P(\pm 1)$ by parity and reducing $x^n \equiv S_n x - S_{n-1}$, admissibility becomes the singularity of the 4×4 matrix with rows $[\beta_j \text{ even}]$, $[\beta_j \text{ odd}]$, S_{β_j} , S_{β_j-1} . Expanding along the two parity rows and

using d’Ocagne’s identity [Doc] for this recurrence,

$$S_b S_{b'-1} - S_{b'} S_{b-1} = -S_{b-b'}, \quad (24)$$

each 2×2 block collapses to a *single* S , indexed by the difference of the two surviving β ’s. Distinct S_n have distinct degrees in c_* and are therefore linearly independent, so the signed sum vanishes only by exact pairing of equal indices with opposite signs. The case analysis is finite: $|E| \in \{0, 4\}$ is branch (a); $|E| \in \{1, 3\}$ leaves three terms whose indices are the three pairwise differences of a 3-set and hence always distinct, so no pairing exists and the value is nonzero; and for $|E| = 2$ there are four terms and three perfect matchings, and one checks the six parity patterns directly. For $E = \{\beta_1, \beta_4\}$ and for $E = \{\beta_2, \beta_3\}$ exactly one of the three matchings is available and it imposes the single condition $\beta_1 + \beta_4 = \beta_2 + \beta_3$, which is branch (b); for the four remaining patterns every one of the three matchings forces two distinct entries of β to coincide, so none is available and the value is nonzero. $\square$

Identity (24) is classical. The collapse it produces is an accident of rank one: two letters admit a single gap, so $s_{(m,n)}(z, \bar{z}) = S_{m-n+1}$ depends on that gap alone. There is no rank- r analogue. One would be a closed form for s_μ at a reciprocal alphabet for *every* μ ; [CK09] supply that for rectangular shapes and [AB19] for self-complementary ones, and nobody in general.

For general r we use associates, in Littlewood’s sense [Lit50]. On the $\det = -1$ coset $[\nu^*] = [\nu] \otimes \det$ gives $\Psi_r = \sum_\nu (m_\nu - m_{\nu^*}) \chi_{[\nu]}$, so $\Psi_r = 0$ if and only if $m_\nu(\lambda) = m_{\nu^*}(\lambda)$ for every ν ; equivalently, $V_\lambda|_{O(N)}$ is $\det$ -invariant. That equivalence is the dictionary Corollary 8.7 is read through, and it is what turns the criterion into a statement about modules rather than about an alphabet. A witness is therefore a single ν with $m_\nu \neq m_{\nu^*}$, and it can be constructed.

Karmakar’s [Kar24, Theorem 4.1] determines the value at the endpoint $z_i = 1$, which is an element of order two: case (A) is branch (a), case (B) is a nonzero dimension, and case (C) leaves an alternating sum of products of dimensions with no vanishing criterion attached. Branch (b) lives in that case (C), so the theorem below is not a consequence of it. Nor does case (B) help more as r grows: at $k = 1$ it asks that $N - 1$ of the N beta numbers share a parity, and the smallest shape doing so is the staircase $(2r - 1, 2r - 2, \dots, 1)$, of size $r(2r - 1)$, so on a fixed range of $|\lambda|$ that nonvanishing case is visible only for small r — the result is sharpest exactly where ours is easiest. What connects the endpoint to the curve at all is rigidity, and rigidity is a statement about our object rather than about the order-two element: over the ranges of §9 it holds on $\ell(\lambda) \leq N/2$ without exception and fails outside, which is Remark 8.12.

Proof of Theorem 8.4. Branch (a) cannot occur: with $\ell(\lambda) \leq N/2$ at least two trailing parts vanish, so β contains two consecutive integers. In Littlewood’s range $m_\nu(\lambda) = \sum_{\beta' \text{ even}} c_{\nu\beta'}^\lambda$, and the term $\beta' = \emptyset$ gives $m_\mu \geq 1$ for every $\mu \subseteq \lambda$. Take μ as follows. If $\ell(\lambda) < N/2$, take $\mu = \lambda$: then $\ell(\lambda^*) = N - \ell(\lambda) > \ell(\lambda)$, so $\lambda^* \not\subseteq \lambda$ and $m_{\lambda^*} = 0 \neq m_\lambda$. If $\ell(\lambda) = N/2$ and $\lambda_{N/2}$ is even, take $\mu = (\lambda_1, \dots, \lambda_{N/2-1})$; then $\beta' = (\lambda_{N/2})$ is even, λ/μ is a horizontal strip and $m_\mu \geq 1$ while $\ell(\mu) < N/2$, so the previous computation applies to μ . If $\ell(\lambda) = N/2$ and λ is not a rectangle, let $j \leq N/2 - 1$ be largest with $\lambda_j > \lambda_{j+1}$ and remove both the last row and one box from row j ; then $\beta' = (\lambda_{N/2} + 1)$ is even and $\lambda_j \geq \lambda_{N/2} + 1$ keeps λ/μ a horizontal strip. An even rectangle $\lambda = (k^{N/2})$ has already been covered by the second construction, so the only case left is k odd, where neither construction is available — and that is branch (b). $\square$

8.4. Outside Littlewood’s range. For $\ell(\lambda) > N/2$ the restriction rule acquires non-standard labels, and the modification rules that reduce them are King’s [Kin71]; Koike and Terada [KT87] record that the type- D specialization sends a universal character to 0 or to $\pm$ one other, and Fauser, Jarvis, King and Wybourne observe that beyond the two classical cases “*the general theory, having infinitely many cases, needs a not yet available formalism*” [FJKW05, §4]. Extensions of Littlewood’s rule to all parameters do exist for the two classical cases — Newell’s modification rules [New51], Sundaram [Sun90] in the symplectic case, Gavarini’s Brauer-algebra proof [Gav99], and Enright and Willenbring [EW04], whose Theorem 4 gives the multiplicity as an alternating sum over the Weyl group which in favourable cases collapses to a difference of two Littlewood coefficients [EW04, §7.11]. That observation of [FJKW05] is from 2005, and it is worth recording that the gap it names has since been filled on the side we need: Frohmader [Fro25] gives combinatorial branching rules for $GL_n \downarrow O_n$ and $GL_{2n} \downarrow Sp_{2n}$ that generalize Littlewood’s restriction rules and are, in its words, “*valid for all parameters, i.e. there are no stable range constraints*”; the multiplicity becomes a sum of generalized Littlewood–Richardson coefficients $c_{\mu\nu}^\lambda(O_n)$ read off flag conditions on ordinary Littlewood–Richardson tableaux, with no modification rule anywhere. We do not use it below, and say why in a moment; but it is the natural instrument for the one question this section leaves open, and we return to it in Problem 10.5.

The route taken here is the dual one: keep the coefficients universal, which they are in every range, and move the whole range-dependence into the values, where one lemma disposes of it. On this alphabet no formalism for the labels is needed at all, and the reason is that the values are not type- D values.

Adding a letter c to an alphabet acts on the ring of symmetric functions by $p_k \mapsto p_k + c^k$; write ι_c for that homomorphism. From the generating function of the complete homogeneous symmetric functions, $h_m(W, c) - c h_{m-1}(W, c) = h_m(W)$ for every m .

Lemma 8.9 (the alphabet is symplectic). $\iota_{-1}\iota_1 o_\nu = sp_\nu$ for every partition ν , as an identity between Koike–Terada universal characters in Λ . In particular, for the alphabet (21),

$$o_\nu(A) = sp_\nu(z_1, \dots, z_r) \quad (25)$$

with no hypothesis on $\ell(\nu)$ and with no sign. For $\ell(\nu) > r$ the right-hand side is that universal character specialized at r variables, not an irreducible $Sp(2r)$ character; it is the modification rules below that fold it onto one.

Proof. Both steps are Ayer and Kumari’s, and both are the same manipulation: the recurrence turns the column operations $C_j \rightarrow C_j - c C_{j-1}$, applied for $j = n, \dots, 2$ in that order, into a change of character type, and operations of that shape leave a determinant unchanged. At $c = 1$ they carry the defining determinant of o_ν evaluated at $(W, 1)$ to that of the odd orthogonal universal character so_ν over W , which is Koike and Terada’s dictionary in the form $so_\nu(X) = o_\nu(X, \bar{X}, 1, 0, 0, \dots)$ [AK25, (2.13)]. At $c = -1$, where the recurrence is [AK25, Proposition 2.1] and the operations read $C_j \rightarrow C_j + C_{j-1}$, they carry so_ν to sp_ν , which is [AK25, Lemma 3.2]. Composing gives the first statement, and evaluating it on the reciprocal alphabet $(z_1, \dots, z_r)$ gives (25). $\square$

Since $\nu'_1 = \ell(\nu)$, the trichotomy on ν'_1 against $N/2 = r + 1$ is a trichotomy on the length of ν against the rank of $Sp(2r)$, and the three facts we need are then the classical symplectic modification rules — one of the two cases [FJKW05] exempts — read through (25): $o_\nu(A) = 0$ when $\ell(\nu) = r + 1$, the vanishing of a symplectic character one row past its rank; $o_\nu(A) = \pm o_{\nu^*}(A)$ when $\ell(\nu) > r + 1$, King’s rule folding the label back onto a dominant one; and the

same for non-standard ν , which by Lemma 8.10 are exactly the tall ones. The expansion $s_\lambda = \sum_\nu c_\nu o_\nu$, with $c_\nu = \sum_{\beta' \text{ even}} c_{\nu\beta'}^\lambda$, is an identity of symmetric functions and holds in every range; specializing it at (21) therefore lands every term on $\{o_\mu(A) : \ell(\mu) \leq r\}$, which by (25) is the set of irreducible characters of $Sp(2r)$ and so is linearly independent. Figure 14 sorts the labels into the four classes this paragraph describes and computes $o_\nu(A)$ on each, so that the reduction can be seen to land where it is claimed to. Hence

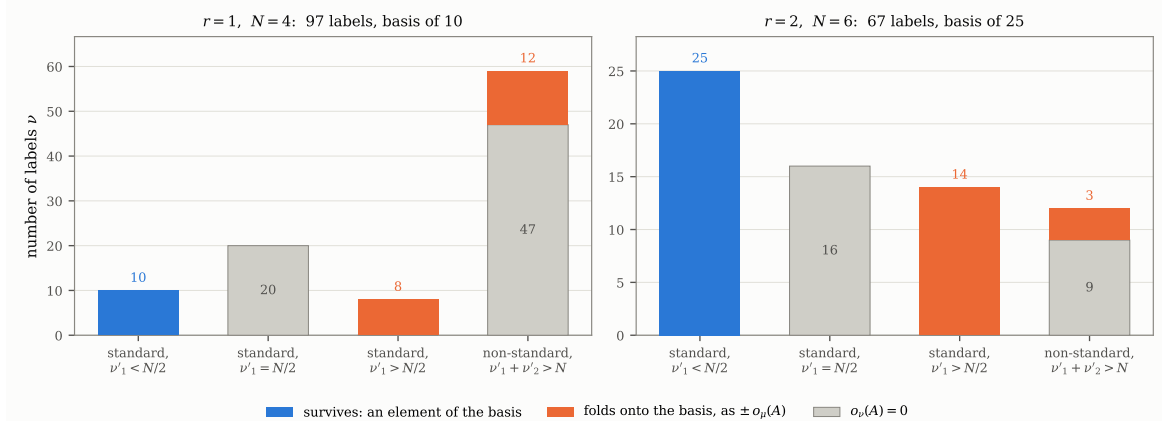


Figure 14. The reduction of Lemma 8.9, drawn. Each label ν occurring in $s_\lambda = \sum_\nu c_\nu o_\nu$ is placed in one of four classes and its value $o_\nu(A)$ computed. Self-associate labels vanish, every one of them; labels standard with $\nu'_1 > N/2$ fold onto the basis as $\pm o_\mu(A)$; non-standard labels either vanish or fold onto a single basis element with coefficient ± 1 . By (25) these three are the symplectic modification rules in disguise, the classes being cut by $\ell(\nu)$ against the rank r ; the figure was computed before that was noticed, and is left as drawn. Everything lands on the independent basis $\{o_\mu(A) : \ell(\mu) \leq r\}$, which is what turns the vanishing into the finite condition (26). The bar heights are label counts over the range each panel was computed on, which is smaller than the range of the corresponding row of §9; the classification, not the count, is what the picture is about.

$$\Psi_r(\lambda) = \sum_{\ell(\mu) \leq r} C_\mu(\lambda) o_\mu(A), \quad \Psi_r(\lambda) = 0 \iff C_\mu(\lambda) = 0 \text{ for all } \mu, \quad (26)$$

where $C_\mu = c_\mu \pm c_{\mu^*} + \sum \pm c_\nu$ over the non-standard ν reducing to μ . The reduction is carried out in two steps rather than one: most non-standard labels pair with an associate inside the range, while 21 of them — 18 at $r = 1$ and 3 at $r = 2$ — have no associate in range and are reduced separately, each landing in the span of the standard values with integer coefficients.

We had computed the three facts before locating them, and record the computation because it is what the figure below draws: `sec8_derivation.sage` checks (25) directly for all $|\nu| \leq 8$, confirms that the labels of length $r + 1$ vanish and that longer ones fold onto the basis with a sign, and verifies the independence as a rank computation, at $r = 1, 2, 3$. A control that had to fire: $\iota_1 o_\nu$ and $\iota_{-1} o_\nu$ each agree with sp_ν only for $\nu = \emptyset$, so both letters are needed and (25) is a statement about the reciprocal pair, not about either letter alone. `sec8_formulas.sage` checks the proof rather than the statement — the recurrence at four values of c , the two defining determinants against an independent implementation, and the column operations entry by entry. That last check earned its keep: an earlier draft of the

proof wrote $C_j \rightarrow C_j + C_{j-1}$ at $c = 1$, the sign [AK25, Lemma 3.2] uses for its own step at $c = -1$.

The non-standard labels, which are the whole obstruction here, cannot be small:

Lemma 8.10 (non-standard labels are large). *Every non-standard ν occurring in the expansion satisfies $|\nu| \geq N + 1 = 2r + 3$. Consequently, if $|\lambda| \leq 2r + 2$ then no non-standard label is contained in λ , the last sum in C_μ is empty, and $C_\mu(\lambda) = c_\mu(\lambda) \pm c_{\mu^*}(\lambda)$ — with no hypothesis on $\ell(\lambda)$, hence also outside Littlewood’s range.*

Proof. Since $c_{\nu\beta'}^\lambda = 0$ unless $\nu \subseteq \lambda$, only labels with $\ell(\nu) \leq \ell(\lambda) \leq N$ occur. Non-standard means $\nu'_1 + \nu'_2 > N$. Were ν a single column we would have $\nu'_2 = 0$ and hence $\nu'_1 = \ell(\nu) > N$, which is excluded; so $\nu'_2 \geq 1$ and $|\nu| \geq \nu'_1 + \nu'_2 > N = 2r + 2$. A non-standard ν can therefore contribute only when $|\lambda| \geq |\nu| \geq 2r + 3$. $\square$

Since $\ell(\lambda) > N/2$ forces $|\lambda| \geq r + 2$, Lemma 8.10 settles the band $r + 2 \leq |\lambda| \leq 2r + 2$ of the unstable range outright: there Littlewood’s rule applies verbatim and the converse follows by the argument of Theorem 8.4. What this route still has to cross is $|\lambda| > 2r + 2$.

Two exact facts bound the competitors. First $\ell(\mu^*) = N - \ell(\mu)$, so $\ell(\mu) \leq N - 1 - \ell(\lambda)$ forces $\mu^* \not\subseteq \lambda$ and $c_{\mu^*} = 0$. Second, a non-standard ν has $\nu'_1 + \nu'_2 > N$ with $\nu'_2 \leq \nu'_1$, hence $2\ell(\nu) > N$ and $\ell(\nu) \geq r + 2$; in particular no non-standard label fits inside λ in Littlewood’s range. Call μ *isolating* for λ if exactly one of the labels feeding C_μ is contained in λ ; then $C_\mu = \pm c_\nu \neq 0$ and $\Psi_r(\lambda) \neq 0$. What the route asks for there is one existence statement — open as an existence statement, though not as a theorem, since Theorem 8.6 does not go through it.

Proposition 8.11 (the isolating witness does not always exist). *At $r = 2$ the shape $\lambda = (5, 5, 2, 1)$ has $\Psi_2(\lambda) \neq 0$ and is of neither type (a) nor type (b), and no μ is isolating for it: every μ whose C_μ has a nonzero contributor inside λ has exactly two, namely μ and its associate μ^* .*

We had proposed the isolating witness as the route to Theorem 8.6, and it does not reach. The reason is legible in the isolation lemma: it needs $\ell(\mu) \leq N - 1 - \ell(\lambda)$, which at $\ell(\lambda) = 4$ and $N = 6$ leaves only one-row μ , and a shape as wide as $(5, 5, 2, 1)$ contains every associate. Over $r = 2$ and $|\lambda| \leq 14$ there are eleven such shapes, and none at all below $|\lambda| = 13$, which is why the certificate looks sound on small ranges. Theorem 8.6 itself is untouched — it asserts that the C_μ vanish, not that a certificate exists — but this route to it is closed. The one that reaches it, §8.6, does not go through Littlewood’s range at all, which is why the failure here costs the explanation and not the theorem.

The mechanism does reach a great deal: at $r = 2$ and $|\lambda| \leq 14$, 227 of the 238 non-vanishing unstable shapes carry a proved isolating witness. It also reaches further than fact (ii) allows on its own — the true minimum of $\ell(\nu)$ over non-standard ν with $o_\nu(A) \neq 0$ is 5 there, against the $r + 2 = 4$ the bound gives. What defeats it is width rather than height: the basis in (26) grows with r and the competing labels are constrained to be tall, so we had read isolation as getting easier with r , and along the height axis it does; the residue lives where λ is wide enough to swallow the associates. That reading predicts where the residue is *not*: at $r = 3$ the isolation lemma allows $\ell(\mu) \leq 7 - \ell(\lambda)$, so a shape must be wider than at $r = 2$ before it can swallow every associate, and the same sizes should come out clean. They do — over $r = 3$ and $|\lambda| \leq 13$ all 149 non-vanishing unstable shapes carry a proved witness and the residue is empty — which is a control on the explanation rather than a step toward Theorem 8.6. At $r = 1$ the question does not arise, since that case is settled independently by Theorem 8.3.

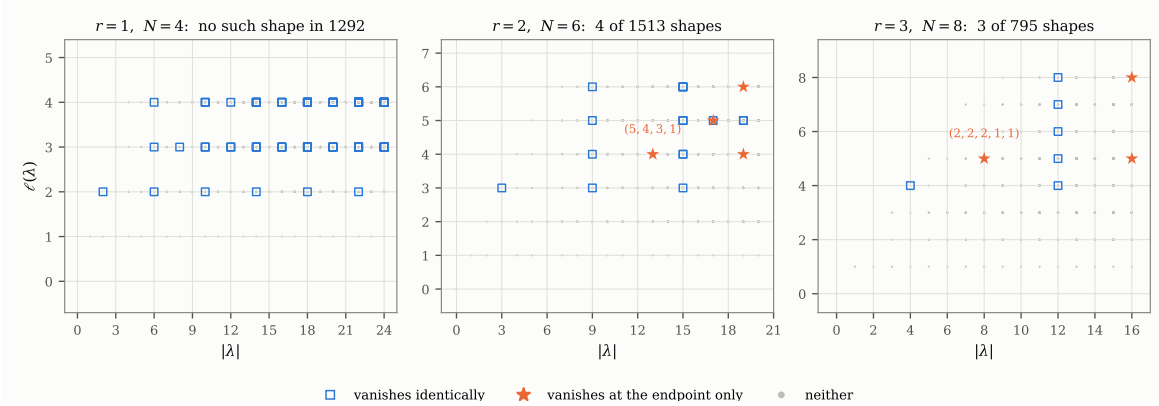


Figure 15. Where one reciprocal pair stops being typical. Each partition with at most N parts is a point at $(|\lambda|, \ell(\lambda))$; the class drawn as a star is the one that vanishes at the endpoint $z_i = 1$ without vanishing identically, and it is the whole content of the picture; each panel names its smallest member and Remark 8.12 lists the rest. The panels cover $|\lambda| \leq 24, 20$ and 16 respectively. At $r = 1$ the class is empty over all 1292 shapes in range, because the object there is $\pm$ a genuine character and its value at the identity is a dimension. From $r = 2$ on properly virtual examples occur and the class is not empty — not that every shape is one: Table 2 counts both. See Remark 8.12.

The tool one reaches for first does not fit. Saturation, in the honeycomb form of Knutson and Tao [KT99], decides when a single Littlewood–Richardson coefficient is positive, and every competitor feeding C_μ is positive as soon as its label sits inside λ . What an isolating μ asks for is not positivity but *isolation*: that exactly one of them does. The positivity is what makes the question hard rather than what answers it, and we know of no tool aimed at the second question.

Remark 8.12 (the rank-one accident, and where the two ranges part). At $r = 1$ the object is $\pm$ a genuine character, so its value at the identity is a dimension and $\Psi_1(\lambda) \equiv 0$ if and only if $\Psi_1(\lambda)|_{z=1} = 0$. At $r \geq 2$ the folded expansion *can* be properly virtual — not that it always is, and Table 2 counts both columns — and one shape where it suffices to break that equivalence: $\lambda = (5, 4, 3, 1)$ at $r = 2$ has $\Psi_2(\lambda) \neq 0$ and $\Psi_2(\lambda)|_{z_1=z_2=1} = 0$. Figure 19 shows the two regimes side by side, on a pair of shapes differing in a single box. Over the ranges of Figure 15 the class of such shapes is empty at $r = 1$ across 1292 partitions; at $r = 2$ it consists of the four shapes $(5, 4, 3, 1)$, $(5, 5, 4, 2, 1)$, $(10, 5, 3, 1)$ and $(6, 5, 4, 2, 1, 1)$, and at $r = 3$ of the three shapes $(2, 2, 2, 1, 1)$, $(5, 5, 4, 1, 1)$ and $(3, 3, 3, 2, 2, 1, 1, 1)$; it is thin, but it is not empty, and one counterexample is all the equivalence can survive. The surviving implication, exact vanishing $\Rightarrow$ vanishing at the endpoint, is what makes the endpoint a valid sieve and no more. This is why results computed at elements of order two — Karmakar [Kar24] along the whole family of them, Eisenkölbl [Eis07] at the balanced one $x_i = (-1)^i$ — bear on a point and not on the curve, and why Theorem 8.1 is not a corollary of them for $r \geq 2$.

It is worth putting a number on what the endpoint sieve loses, since the qualitative statement is easy to misread as a small correction. Over the ranges of Figure 15 the test $\Psi_r(\lambda)|_{z_i=1} = 0$ flags 143 shapes at $r = 1$, 22 at $r = 2$ and 9 at $r = 3$; of those, 0, 4 and 3 respectively do not vanish identically. So the sieve is exact at one pair, and the fraction of its verdicts that are spurious grows with the number of pairs — 0%, 18%, 33% over these

ranges. An endpoint computation is a filter that never misses a genuine zero, and past one pair it is nothing more than that. The rank-one case at $t = 2$, where the sieve happens to be exact because the object is a genuine character, was treated separately in [Mar26]; the alphabet (1) arose there, in a Wilson-line computation on an orbifold, which is why -1 appears as a letter rather than as a choice.

The reason the two ranges of $\ell(\lambda)$ behave so differently is worth separating from the mechanism that proves it. Littlewood's rule is a statement about which labels ν can occur, and $\ell(\lambda) \leq N/2$ is exactly the condition under which the non-standard ones cannot fit inside λ at all. Lemma 8.10 says that the obstruction is governed by $|\lambda|$ as well as by $\ell(\lambda)$, so the honest picture is not two ranges but a region: *this route* reaches the converse wherever either $\ell(\lambda) \leq N/2$ or $|\lambda| \leq 2r + 2$, and outside their union it does not — the isolating witness covers most of what is left but provably not all of it, by Proposition 8.11. The converse is not open there. Theorem 8.6 settles it for every r and every λ by the extremal argument of §§8.5–8.6, which never enters this range at all. What the region marks is where Littlewood's reduction stops explaining, not where the theorem stops holding.

8.5. The criterion at $t = 2$: the extremal argument. Throughout this subsection and the next two, $t \geq 2$ and $r \geq 1$ are arbitrary and

$$N := t + 2r, \quad \Phi_{t,r}(\lambda; z) := s_\lambda(\mu_t, z_1, \bar{z}_1, \dots, z_r, \bar{z}_r),$$

so that $\Phi_{t,1}$ is (1) and $\Phi_{2,r} = \Psi_r$. The beta set, the residue counts n_i and the column indexing are those of §2; $\lambda \in \mathcal{P}_N$ and $\beta_1 > \dots > \beta_N \geq 0$. Write

$$\mathcal{E} = \{i : n_i(\lambda) \geq 2\}, \quad e = |\mathcal{E}|, \quad \mathcal{S} = \{\beta_j : \beta_j \equiv i \pmod{t} \text{ for some } i \in \mathcal{E}\}$$

for the *excess classes* and the *excess values*. Since the $t - e$ remaining classes are singletons, (3) becomes

$$\sum_{i \in \mathcal{E}} (n_i(\lambda) - 1) = 2r, \quad |\mathcal{S}| = 2r + e, \quad 1 \leq e \leq 2r. \quad (27)$$

We assume throughout the *occupancy hypothesis* $n_i \geq 1$ for every i ; by Lemma 4.1 its failure gives $\Phi_{t,r} \equiv 0$ outright, which is branch (a).

Lemma 8.13 (the Laplace decomposition, general r). *Let $\mathcal{T}$ be the set of t -subsets S of columns carrying all t residues, and for $S \in \mathcal{T}$ let $A(S^c)$ be the minor of $\det(x_i^{\beta_j})$ on the $2r$ rows carrying $z_1, \bar{z}_1, \dots, z_r, \bar{z}_r$ and the columns S^c . Then*

$$\det(x_i^{\beta_j})_{i,j=1}^N = (-1)^{\binom{t+1}{2}} V \sum_{S \in \mathcal{T}} w(S) A(S^c), \quad w(S) = (-1)^{\sum_{j \in S} j + \text{inv}(b_S)}. \quad (28)$$

Proof. Laplace along the first t rows gives

$$\det(x_i^{\beta_j}) = \sum_{|S|=t} (-1)^{\binom{t+1}{2} + \sum_{j \in S} j} M_S A(S^c).$$

By Lemma 4.1, $M_S = 0$ unless the residues b_S are pairwise distinct — that is, unless $S \in \mathcal{T}$ — and then $M_S = (-1)^{\text{inv}(b_S)} V$. $\square$

At $r = 1$ the complementary minor is the 2×2 determinant $f(\beta_{j_1} - \beta_{j_2})$ and (28) is Lemma 4.3; the content below is what replaces that 2×2 minor when $r > 1$. Only the relative signs $w(S)$ matter, and since a member of $\mathcal{T}$ is determined by which excess value it keeps in each excess class — the singleton classes are kept by every S — we index the terms by that choice, written g , and set $T_g = \mathcal{S} \setminus g$, a set of $2r$ values listed decreasingly.

One question is worth settling here, because everything below moves g about and the sign has to be carried along: what happens to w when the choice in a single class changes. At $r = 1$ that is Lemma 4.4, where a class has two elements and there is nothing between them. In general there is.

Lemma 8.14 (the column move, any class size). *Let g and g' agree outside one class and choose there the values $u > v$ respectively. Then*

$$\frac{w(g')}{w(g)} = (-1)^{1+B+M}, \quad (29)$$

where $B = \#\{\beta_j : v < \beta_j < u\}$ and M is the number of frozen values lying strictly between v and u — the t values a transversal keeps, one in each residue class, so the e values of g together with the $t - e$ singleton classes. It may be counted in either g or g' , which agree strictly between v and u .

Proof. The two frozen sets differ by replacing the column of u with that of v . Since β is strictly decreasing, that column index increases by $B + 1$, which is the change in $\sum_{j \in S} j$. In the word b_S the letter carried by that column travels past exactly the M other frozen letters lying strictly between, and each crossing changes inv by one; the letter itself is unchanged, and the $B - M$ intervening columns that are *not* frozen contribute no letter at all. So $\text{inv}(b_S)$ changes by M modulo 2, and (29) follows from the definition of w in Lemma 8.13. $\square$

Lemma 4.4 is the case $B = M = 0$ of this, read twice. The bookkeeping is the one Albion [Alb23, Alb25] and Ayer–Behrend [AB19] carry out whenever they move or reverse a block of columns and collect the sign, and we follow it because it is the accounting this kind of determinant asks for; we set it out in this form only because §8.6 moves a single column many times over and the count then has to be exact rather than absorbed into a constant. The two terms must both be there: dropping either gives a rule that is right about half the time, and $B - M$ is exactly the number of intervening columns that no transversal freezes — which is why a class of size two, with nothing between its elements, hides the distinction. The singleton classes must be counted in M as well, and that is not a formality: reading M over the choices g alone already fails at $t = 3$, $r = 1$ on $\beta = (4, 3, 2, 1, 0)$, where moving the class of 1 from $u = 4$ to $v = 1$ has $B = 2$ and one intervening frozen value that lies in a singleton class, so the two readings differ by a sign.

The degree filtration. For a monomial $\prod_j z_j^{c_j}$ we use the *total degree* $\sum_j c_j$. Every monomial of $A(S^c)$ has the form $\prod_j z_j^{u - u'}$ for a matching of T_g into ordered pairs assigned to the variables, so writing $T = (u_1 > \dots > u_{2r})$,

$$\deg(T) := \sum_{a \leq r} u_a - \sum_{a > r} u_a \quad (30)$$

is the largest total degree occurring in A , attained exactly by the matchings joining the top half $H = (u_1, \dots, u_r)$ to the bottom half $L = (u_{r+1}, \dots, u_{2r})$. Writing $a_X(z) = \det(z_j^{X_i})_{r \times r}$ and $P(T) = a_H(z) a_L(z^{-1})$, which is nonzero, being a product of two $GL(r)$ alternants, the component of maximal total degree of A is

$$[A]_{\deg(T)} = (-1)^{\binom{r}{2}} P(T). \quad (31)$$

Indeed in maximal degree each variable receives one row of H with a positive exponent and one row of L with a negative one, the two assignments being independent, so the sum factorises into the two determinants; the sign is that of the shuffle sending the rows of H

to the columns $z_1, \dots, z_r$ and those of L to the columns $z_1^{-1}, \dots, z_r^{-1}$, which is $(-1)^{\binom{r}{2}}$. Under the grading by $\sum_j |c_j|$ the maximal component splits into 2^r orientation sectors — one for each subset of the variables that is inverted — with disjoint monomials, permuted by $z_j \mapsto z_j^{-1}$ with a uniform sign, so they vanish together; all statements below use the total degree.

Lemma 8.15 (reflection). *$P(c - T) = P(T)$ for every $c \in \mathbb{Z}$ and every T .*

Proof. The top half of $c - T$ is $c - L$ listed decreasingly, so $a_{c-L}(z) = (\prod_j z_j^c)(-1)^{\binom{r}{2}} a_L(z^{-1})$, and symmetrically for the bottom half. The two powers of $\prod_j z_j^c$ cancel and $(-1)^{r(r-1)} = +1$. $\square$

Let G be the set of g maximising $\deg(T_g)$ and D_1 that maximum. Since the numerator $\det(x_i^{\beta_j})$ and $\Phi_{t,r}$ differ by the nonzero factor $\det(x_i^{N-j})$, we have $\Phi_{t,r} \equiv 0$ if and only if the numerator vanishes, and *every degree and stratum below refers to the numerator*; no comparison between its homogeneous parts and those of the quotient is made. By (28),

$$[\det(x_i^{\beta_j})]_{D_1} = (-1)^{\binom{t+1}{2} + \binom{r}{2}} V \sum_{g \in G} w(g) P(T_g), \quad (32)$$

the second sign coming from (31). Only the relative signs $w(g)$ matter below, so the global constant is recorded once and not carried.

Remark 8.16 (a warning about that global sign). It is tempting to drop $(-1)^{\binom{r}{2}}$ on the ground that nothing below depends on it, and we record it for the opposite reason: an identity that is stated is an identity that gets checked, and this one is *false* without it as soon as $r \geq 2$. The sign is not decoration. It is the signature of the shuffle in (31) — $+1$ for $r = 1$ and $r \equiv 0, 1 \pmod{4}$, -1 otherwise — and it is invisible at $r = 1$, which is exactly the rank at which one first writes the formula down.

The extremal set has at most two elements. For $i \in \mathcal{E}$ list the excess values of the class decreasingly, $c_{i,1} > c_{i,2} > \dots > c_{i,n_i}$ with $n_i = n_i(\lambda)$, and define its *increments*

$$\Delta_i(k) := c_{i,k} + c_{i,k+1}, \quad 1 \leq k \leq n_i - 1. \quad (33)$$

By (27) there are exactly $2r$ of them in all; write $\Delta_{(1)} \geq \dots \geq \Delta_{(2r)}$ for the list in decreasing order and put $\tau := \Delta_{(r)}$.

Theorem 8.17 (the extremal set). *Assume $n_i \geq 1$ for every i . Then*

- (a) $g \in G$ if and only if the multiset of increments it selects, in the sense of Step 2 below, consists of all increments $> \tau$ together with enough increments equal to τ ;
- (b) $|G| \leq 2$, and $|G| = 2$ if and only if $\Delta_{(r)} = \Delta_{(r+1)}$, in which case t is even, the two tied increments belong to classes i and $i + t/2$ — both necessarily in $\mathcal{E}$ — and the two maximisers differ exactly in those two classes;
- (c) if t is odd then $|G| = 1$.

Proof. Step 1 (reformulation). For any $2r$ -set T we have

$$\deg(T) = \max_{A \subseteq T, |A|=r} (2\Sigma A - \Sigma T),$$

the maximum being at the top half. With $T = \mathcal{S} \setminus g$ this reads $\deg(\mathcal{S} \setminus g) = \max_A [2\Sigma A + \Sigma g] - \Sigma \mathcal{S}$, so we maximise $\Theta(A, g) = 2\Sigma A + \Sigma g$ over disjoint pairs with $|A| = r$ and g a choice of one excess value in each excess class.

Step 2 (shape of an optimum). Let (A, g) be optimal and let $x > y$ lie in the same class i . If $y \in A$ and $x = g_i$, exchanging them leaves g admissible and changes Θ by $x - y > 0$; if $y = g_i$ and x lies in neither A nor g , the same exchange gives $x - y > 0$. Both are impossible, so within each class, read decreasingly, the elements of A come first, then g_i , then the unchosen ones. Hence an optimum is parametrised by one integer per class, $k_i = |A \cap \text{class } i| \in \{0, \dots, n_i - 1\}$ with $\sum_i k_i = r$, via $A \cap \text{class } i = \{c_{i,1}, \dots, c_{i,k_i}\}$ and $g_i = c_{i,k_i+1}$. The map $(k_i) \mapsto g$ is injective and for fixed g the optimal A is the top half of T_g , so optimal pairs project bijectively onto G .

Step 3 (separability). With that parametrisation $\Theta = \sum_{i \in \mathcal{E}} f_i(k_i)$, where $f_i(k) = 2(c_{i,1} + \dots + c_{i,k}) + c_{i,k+1}$: the objective splits over the classes and the only coupling is $\sum_i k_i = r$.

Step 4 (the exchange). $f_i(k) - f_i(k-1) = \Delta_i(k)$, which is strictly decreasing in k because the $c_{i,k}$ are, so

$$\Theta = \sum_{i \in \mathcal{E}} f_i(0) + \Sigma, \quad \Sigma = \sum_{i \in \mathcal{E}} \sum_{k \leq k_i} \Delta_i(k),$$

and maximising Θ is maximising Σ . A feasible (k_i) selects, in each class, a *prefix* of that class's increment list, with r increments in all; call such a selection *admissible*. Now take any r increments of maximal total, admissible or not. That selection *is* admissible: if it contained $\Delta_i(k)$ without $\Delta_i(k-1)$, exchanging the two would raise the total, since $\Delta_i(k-1) > \Delta_i(k)$ — and the exchange is legitimate because $\Delta_i(k-1)$ was not already selected. And no admissible selection can beat the total of the r largest increments, being itself a selection of r of them. Hence the maximisers of Θ are exactly the admissible selections whose multiset of increments is a multiset of r largest ones, which is the assertion of (a). (This is the marginal-allocation greedy for a separable concave objective under one cardinality constraint; see [FG86, IK88]. We have given the argument because it is four lines, so that the only external input to Theorem 8.6 is Theorem 8.24.)

Step 5 (residues). $c_{i,k} \equiv c_{i,k+1} \equiv i$, hence

$$\Delta_i(k) \equiv 2i \pmod{t}. \quad (34)$$

Within a class the increments are pairwise distinct by Step 4. Across classes, $\Delta_i(k) = \Delta_{i'}(k')$ forces $2i \equiv 2i'$, that is $i' = i$ or — only if t is even — $i' = i + t/2$. Therefore *no value is shared by more than two increments*, and when it is, the two come from classes i and $i + t/2$.

Step 6. The optima are the selections containing every increment $> \tau$ together with s of those equal to τ , with $s \geq 1$. At most two increments equal τ , so either both fit ($|G| = 1$) or exactly one does ($|G| = 2$), the latter precisely when $\Delta_{(r)} = \Delta_{(r+1)}$. For t odd, $x \mapsto 2x$ is injective on $\mathbb{Z}/t$, so no tie is possible and $|G| = 1$. $\square$

Remark 8.18. The content of Theorem 8.17 is (34) together with the fact that $x \mapsto 2x$ on $\mathbb{Z}/t$ has kernel of order $\gcd(2, t)$: the multiplicity of an extremal value is controlled by the 2-torsion of the residue group, which is why the parity of t governs everything here and in §3. Steps 1–4 use only that the β_j are distinct integers; neither $\beta_j \geq 0$ nor λ being a partition nor $N = t + 2r$ enters beyond (27). Figure 16 is the theorem carried out on one shape, with the tie and the prefix form of Step 2 both visible.

Remark 8.19 (where this shape of argument comes from). Steps 1–4 are an old story told in a new alphabet. Maximising a separable concave objective under one cardinality constraint is the marginal-allocation problem of resource allocation, solved by the greedy in [FG86, IK88]; we gave the exchange in four lines only so that the reader need not leave the paper. Step 6 is also familiar, from a different direction: in tropical linear algebra the initial

form of a determinant is the signed sum over the *optimal* assignments, and the matrix is called tropically singular exactly when two or more optima attain the optimum [RGST05]. Read that way, $|G| = 2$ is a tie at the outer, *transversal* level: by (32) the component of the numerator in top degree is the signed sum of the two optimal transversal contributions, and the whole question of §8.6 is whether they cancel. We put it no more strongly than that. Tropical singularity of the full determinant is a statement about assignments, and inside a single $A(T_g)$ many assignments already attain the maximum — precisely the ones producing the two alternants of (31) — which is why $[A]_{\deg(T)}$ is a nonzero product rather than a cancellation. The tie that matters here is between the T_g , not between assignments. What is *not* standard, and is the reason the parity of t governs everything, is (34): it is a congruence, not an inequality, and it is what bounds the number of optima by two.

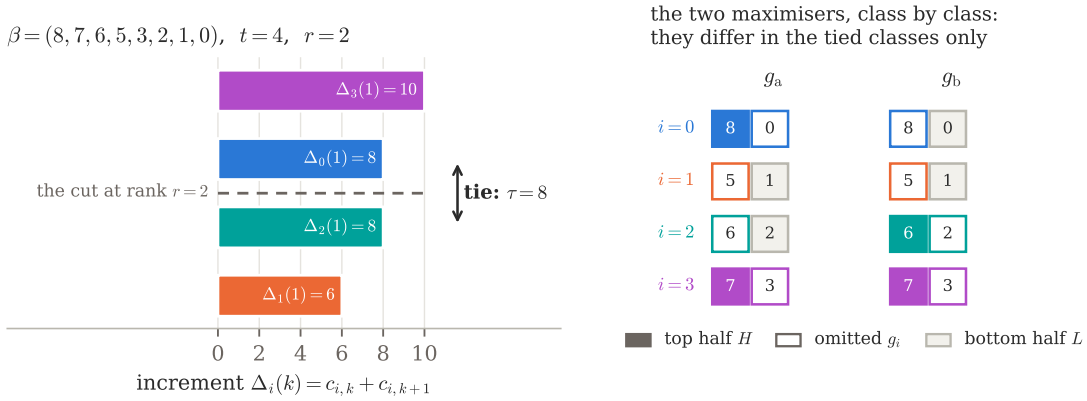


Figure 16. Theorem 8.17 computed on one shape rather than stated. For $\beta = (8, 7, 6, 5, 3, 2, 1, 0)$ at $t = 4$, $r = 2$ the four excess classes are $\{8, 0\}$, $\{5, 1\}$, $\{6, 2\}$, $\{7, 3\}$ and the four increments are $\Delta_3(1) = 10$, $\Delta_0(1) = 8$, $\Delta_2(1) = 8$, $\Delta_1(1) = 6$. A maximiser takes the $r = 2$ largest; the second and third are equal, so $\tau = \Delta_{(2)} = \Delta_{(3)} = 8$ and there are exactly two of them. The tied pair belongs to the classes 0 and $2 = 0 + t/2$, which is (34) and not a coincidence: no value is shared by increments of classes other than i and $i + t/2$. On the right the two maximisers are drawn class by class in the prefix form of Step 2 — reading a class decreasingly, first the elements of the top half, then the omitted g_i , then those of the bottom half — which makes visible that they differ in the tied classes and nowhere else. Colour encodes the class, not the rank. Computed by `anc/fig_v2.py`.

Corollary 8.20 (a unique maximiser). *If $|G| = 1$ then $\Phi_{t,r} \neq 0$.*

Proof. By (32) the component of the numerator in degree D_1 is $\pm V P(T_g) \neq 0$ for the unique g . A Laurent polynomial with a nonzero homogeneous component is nonzero, so the numerator is nonzero and hence so is $\Phi_{t,r}$. $\square$

Corollary 8.21 (the middle block). *Write $\mathcal{S} = \{s_1 > \dots > s_n\}$, $n = 2r + e$. If the block $\{s_{r+1}, \dots, s_{r+e}\}$ contains exactly one element of each excess class, then $\Phi_{t,r} \neq 0$.*

Proof. $\deg(\mathcal{S} \setminus g) \leq (s_1 + \dots + s_r) - (s_{n-r+1} + \dots + s_n)$, since the top r elements of $\mathcal{S} \setminus g$ are dominated termwise by $s_1, \dots, s_r$ and its bottom r dominate $s_{n-r+1}, \dots, s_n$. Equality

forces $\mathcal{S} \setminus g$ to contain both extreme blocks, that is $g \subseteq \{s_{r+1}, \dots, s_{r+e}\}$, and $|g| = e$ forces equality of the two sets. So the bound is attained by exactly one term and Corollary 8.20 applies. $\square$

Corollary 8.22 (odd t). *Let t be odd and $r \geq 1$. Then $\Phi_{t,r}(\lambda; z) \equiv 0$ if and only if some residue class modulo t is absent from $\beta(\lambda)$.*

Proof. If some $n_i = 0$ then $\mathcal{T} = \emptyset$ in Lemma 8.13 and the numerator vanishes. Conversely, if all $n_i \geq 1$ then $|G| = 1$ by Theorem 8.17(c) and Corollary 8.20 applies. $\square$

Remark 8.23. For $r = 1$ Corollary 8.22 is Proposition 3.5, proved there by showing that the concentric clause is vacuous for odd t — an argument internal to the two-class profile that a single reciprocal pair forces. What is new is that the statement survives for every $r \geq 1$, and by a route that does not mention the profile at all: the congruence (34). At $r = 0$ it is false, since Littlewood’s rule gives $s_\lambda(\mu_t) = 0$ whenever the t -core is nonempty, odd t included; and the observation that $\prod \mu_t = +1$ for odd t does not by itself deliver it, because the alphabet is a subtorus of dimension r with the remaining coordinates frozen.

8.6. The criterion at $t = 2$: rigidity and the reflection. Corollary 8.20 settles every shape with a unique maximiser. What follows is the other half: when $|G| = 2$ the two terms of (32) can only cancel if the two transversals are reflections of one another, and that rigidity comes from a single external input.

Theorem 8.24 (Purbhoo–van Willigenburg [PvW08, Thm. 2.5]). *For partitions with at most n parts, writing $\lambda^- := \lambda/(\lambda_n)^n$,*

$$s_\lambda(x_1, \dots, x_n) s_\mu(x_1, \dots, x_n) = s_\nu(x_1, \dots, x_n) s_\rho(x_1, \dots, x_n)$$

if and only if $\lambda_n + \mu_n = \nu_n + \rho_n$ and $\{\lambda^-, \mu^-\} = \{\nu^-, \rho^-\}$ as multisets.

Remark 8.25. Theorem 3 of [Raj04] gives the statement for n factors, but under the hypothesis that all the highest weights be nonzero. Theorem 8.24 needs no such hypothesis, and that is why it is the one used here: the degenerate case, one factor trivial, is exactly what Lemma 8.26 produces whenever α^* or $\tilde{\alpha}$ is empty. Neither statement can be shortcut by unique factorisation, since Schur polynomials are not irreducible: $s_{(3)}(x, y) = (x+y)(x^2+y^2)$.

The dictionary. Write $T = (u_1 > \dots > u_{2r})$ with halves $H = (h_1 > \dots > h_r)$ and $L = (l_1 > \dots > l_r)$, let $\delta = (r-1, r-2, \dots, 0)$, so that $a_\delta(z) = \det(z_j^{r-i})$ is the Vandermonde determinant in the notation of §8.5, and put

$$\alpha = H - \delta, \quad \tilde{\alpha} = \alpha - (\alpha_r)^r, \quad L^* = (l_1 - l_r > \dots > l_1 - l_1), \quad \alpha^* = L^* - \delta.$$

Lemma 8.26 (the dictionary). $\alpha_r^* = 0$ and

$$P(T) = (-1)^{\binom{r}{2}} \left(\prod_j z_j \right)^{h_r - l_1} a_\delta(z)^2 s_{\tilde{\alpha}}(z) s_{\alpha^*}(z). \quad (35)$$

In particular P is unchanged by $T \mapsto T + m$ and, by Lemma 8.15, by $T \mapsto c - T$.

Proof. $a_H(z) = s_\alpha(z) a_\delta(z)$ is the bialternant formula. For the second factor, multiplying column j of $\det(z_j^{-l_i})$ by $z_j^{l_1}$ gives $(\prod_j z_j^{l_1}) a_L(z^{-1}) = \det(z_j^{l_1 - l_i})$, whose row exponents increase with i ; reversing the r rows costs $(-1)^{\binom{r}{2}}$ and produces $a_{L^*}(z) = s_{\alpha^*}(z) a_\delta(z)$. Finally $s_\alpha = (\prod_j z_j)^{\alpha_r} s_{\tilde{\alpha}}$ and $\alpha_r = h_r$. $\square$

Proposition 8.27 (the orbit of P). *Let $T, \tilde{T}$ be $2r$ -sets of distinct integers with $P(T) = \pm P(\tilde{T})$. Then $\tilde{T} = T + m$ for some $m \in \mathbb{Z}$, or $\tilde{T} = c - T$ for some $c \in \mathbb{Z}$.*

Proof. Two remarks first. By (35) both sides carry the same nonzero factor $(-1)^{\binom{r}{2}} a_\delta(z)^2$; cancel it before comparing anything. What is left on each side is a monomial times $s_{\tilde{\alpha}} s_{\alpha^*}$, a nonzero polynomial with nonnegative coefficients, so the sign is $+$. The order matters: a_δ^2 does *not* have nonnegative coefficients, and reading the sign off before cancelling it proves nothing. And the monomial must be reabsorbed: with $d = h_r - l_1 \geq 1$, which holds because T is strictly decreasing, $(\prod_j z_j)^d s_{\tilde{\alpha}} = s_{\tilde{\alpha} + (d^r)}$, and it is this d that plays the role of $\lambda_n + \mu_n$ in Theorem 8.24, α^* being already normalised. Cancelling a_δ^2 and applying Theorem 8.24 with $n = r$ to $(\lambda, \mu) = (\tilde{\alpha} + (d^r), \alpha^*)$ and to the corresponding pair (ν, ρ) for $\tilde{T}$, the criterion says that the integer d agrees and that $\{\tilde{\alpha}, \alpha^*\}$ agrees as a multiset. Agreement in the same order says that $H, \tilde{H}$ and $L, \tilde{L}$ are translates, and the equality of the two integers d forces the same translation on both halves, that is $\tilde{T} = T + m$. Agreement after swapping says that H is, up to translation, the reverse of $\tilde{L}$ and L that of $\tilde{H}$, that is $\tilde{T} = c - T$. $\square$

The two maximisers reflect one another. From here to the end of the subsection assume $|G| = 2$ and write $G = \{g_a, g_b\}$, $T_a = \mathcal{S} \setminus g_a$, $T_b = \mathcal{S} \setminus g_b$, with halves $T_a = H_a \sqcup L_a$ and $T_b = H_b \sqcup L_b$, and

$$\mathcal{K} = T_a \cap T_b.$$

By Theorem 8.17(b) the two transversals differ exactly in the tied classes i and $i' = i + t/2$, say

$$g_a \ni p_1 = c_{i,k+1}, \quad q_1 = c_{i',k'}, \quad g_b \ni p_2 = c_{i,k}, \quad q_2 = c_{i',k'+1}, \quad (36)$$

so that $p_1 + p_2 = q_1 + q_2 = \tau$ by (33): the tie value is the sum of each swapped pair.

Proposition 8.28 (translation forces reflection). *If $T_b = T_a + m$ with $m \neq 0$, then $T_b = \tau - T_a$.*

Proof. $|T_a \cap (T_a + m)| = |\mathcal{K}| = 2r - 2$. Put an edge $x \rightarrow x + m$ whenever both x and $x + m$ lie in T_a ; since $m \neq 0$ each vertex has at most one outgoing and one incoming edge, so the graph is a disjoint union of directed paths. It has $2r$ vertices and, by the displayed cardinality, exactly $2r - 2$ edges, hence exactly two components. So T_a is a union of exactly two maximal arithmetic progressions of step m , say $T_a = \{p, p + m, \dots, p + (u - 1)m\} \cup \{q, \dots, q + (v - 1)m\}$ with $u + v = 2r$. Then $T_a \setminus T_b = \{p, q\}$ and $T_b \setminus T_a = \{p + um, q + vm\}$. The tie says that $x \mapsto \tau - x$ carries the first pair onto the second, and there are two matchings. If $\tau - p = p + um$ and $\tau - q = q + vm$ then $\tau - T_a = \{p + m, \dots, p + um\} \cup \{q + m, \dots, q + vm\} = T_b$. If $\tau - p = q + vm$ and $\tau - q = p + um$ then subtracting gives $um = vm$, hence $u = v$, and again $\tau - T_a = T_b$. $\square$

Proposition 8.29 (the centre is the tie value). *If $T_b = c - T_a$ then $c = \tau$.*

Proof. $x \mapsto c - x$ is an involution carrying T_a to T_b , hence T_b to T_a , hence $\mathcal{K}$ to $\mathcal{K}$; therefore it carries $T_a \setminus \mathcal{K}$ onto $T_b \setminus \mathcal{K}$. By (36) these two sets are $T_a \setminus \mathcal{K} = \{p_2, q_2\}$ and $T_b \setminus \mathcal{K} = \{p_1, q_1\}$, since T_a omits exactly p_1, q_1 and T_b omits exactly p_2, q_2 . Either $c - p_2 = p_1$, and then $c - q_2 = q_1$, whence $c = p_1 + p_2 = \tau$; or $c - p_2 = q_1$ and $c - q_2 = p_1$, and adding gives $2c = p_1 + p_2 + q_1 + q_2 = 2\tau$, again $c = \tau$. In the second case moreover $c = p_2 + q_1 \equiv i + i' \equiv 2i + t/2$ while $\tau \equiv 2i$ by (34), so $c = \tau$ would force $t/2 \equiv 0 \pmod{t}$, impossible for $t \geq 2$: the crossed matching does not occur. $\square$

Corollary 8.30 (the reflection). *If $[\det(x_i^{\beta_j})]_{D_1} = 0$ then $|G| = 2$ and $T_b = \tau - T_a$. Consequently $\tau - \mathcal{K} = \mathcal{K}$ and, since $x \mapsto \tau - x$ also fixes $\{p_1, p_2\}$ and $\{q_1, q_2\}$ setwise,*

$$\tau - (\mathcal{K} \cup \{p_1, p_2, q_1, q_2\}) = \mathcal{K} \cup \{p_1, p_2, q_1, q_2\}. \quad (37)$$

Proof. $|G| = 1$ is excluded by Corollary 8.20. With $|G| = 2$, (32) and $w(g) = \pm 1$ give $P(T_a) = \pm P(T_b)$; apply Proposition 8.27 and then Propositions 8.28 and 8.29. The identity $\tau - \mathcal{K} = \mathcal{K}$ is the first line of the proof of Proposition 8.29, and $p_1 + p_2 = q_1 + q_2 = \tau$ is the tie. $\square$

Remark 8.31 (three constants that are easy to confuse). $\tau = \Delta_{(r)}$ is the tie value, a datum of the greedy with nothing to do with symmetry; c is a centre of reflection, produced by rigidity with nothing to do with residues; $C = \min \mathcal{S} + \max \mathcal{S}$ is a statistic of the beta set, and it is the one condition (ii) uses. That $c = \tau$ (Proposition 8.29) and $C = \tau$ (Corollary 8.33) are theorems, not notation, and both need the vanishing hypothesis: writing C where τ belongs to proves less than it appears to. We made that slip; hence the three symbols.

The centre of the beta set. The two maximisers agree outside the tied classes, so their common omitted part

$$g_{\text{com}} := g_a \cap g_b, \quad \mathcal{S} = \mathcal{K} \sqcup g_{\text{com}} \sqcup \{p_1, p_2, q_1, q_2\} \quad (38)$$

is empty exactly when $e = 2$. By (37) the set $\mathcal{S} \setminus g_{\text{com}}$ is stable under $x \mapsto \tau - x$, but g_{com} need not be, so $\mathcal{S}$ itself need not be. That is more than is needed: only the two *extremes* of $\mathcal{S}$ have to avoid g_{com} , and they do. Figure 17 draws both the reflection and the shape that has $|G| = 2$ without it, which is what the next proposition must exclude.

Proposition 8.32 (the extremes avoid the common part). *Suppose $[\det(x_i^{\beta_j})]_{D_1} = 0$. Then $\max \mathcal{S} \notin g_{\text{com}}$ and $\min \mathcal{S} \notin g_{\text{com}}$.*

Proof. By Corollary 8.30, $T_b = \tau - T_a$. Since $x \mapsto \tau - x$ reverses order, it carries the bottom half of T_a to the top half of T_b and back:

$$H_b = \tau - L_a, \quad L_b = \tau - H_a. \quad (39)$$

Recall from Step 2 of the proof of Theorem 8.17 that a maximiser is described by the integers k_i : inside class i , read decreasingly, the first k_i excess values lie in the top half, the next one is the omitted g_i , and the remaining ones lie in the bottom half. Recall also from Theorem 8.17(a) that a maximiser selects every increment $> \tau$, that the increments of a class are selected in prefix order, and that the only increments equal to τ are the two tied ones, which lie in the two tied classes.

The maximum. Put $m = \max \mathcal{S}$ and let i be its class, so $m = c_{i,1}$ and $n_i \geq 2$; write $c = c_{i,2}$. Then $m \in g_a \cap g_b$ if and only if $k_i = 0$ in both maximisers. Assume it. Then class i is not tied: a maximiser taking the tied increment of a tied class has $k_i \geq 1$ there. Hence no increment of class i is $\geq \tau$, and in particular

$$\Delta_i(1) = m + c < \tau.$$

On the other hand $k_i = 0$ places c in the bottom half of both maximisers, so $c \in L_b$, and (39) gives $\tau - c \in H_a \subseteq \mathcal{S}$, whence $\tau - c \leq \max \mathcal{S} = m$, that is $\tau \leq m + c$. The two displayed inequalities contradict each other.

The minimum is the mirror image. Put $m^- = \min \mathcal{S}$, let i^- be its class, $n = n_{i^-} \geq 2$ and $c' = c_{i^-,n-1}$, so that $m^- = c_{i^-,n}$. Then $m^- \in g_a \cap g_b$ if and only if $k_{i^-} = n - 1$ in both, that is if and only if both maximisers select *every* increment of class i^- ; again class i^- cannot be tied, since the maximiser that does not take the tied increment of a tied class omits one there. Hence every increment of class i^- exceeds τ , in particular $\Delta_{i^-}(n-1) = m^- + c' > \tau$. And $k_{i^-} = n - 1$ places c' in the top half of both, so $c' \in H_b$, and (39) gives $\tau - c' \in L_a \subseteq \mathcal{S}$, whence $\tau - c' \geq \min \mathcal{S} = m^-$, that is $\tau \geq m^- + c'$. Contradiction. $\square$

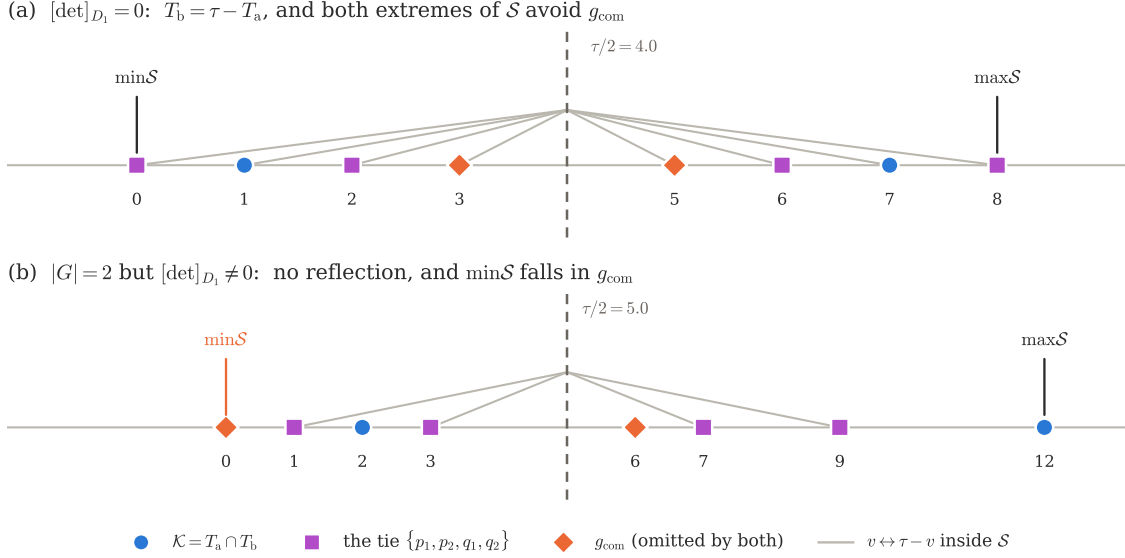


Figure 17. Corollary 8.30 and Proposition 8.32 on two shapes, both with $t = 4$, $r = 2$ and $|G| = 2$. Each panel draws $\mathcal{S}$ on a line, coloured by which part of (38) the value belongs to; an arc joins v to $\tau - v$ whenever both lie in $\mathcal{S}$, and the dashed line is the centre $\tau/2$. (a) $\beta = (8, 7, 6, 5, 3, 2, 1, 0)$: $\tau = 8$, $\mathcal{K} = \{1, 7\}$, $g_{\text{com}} = \{3, 5\}$, tie $\{0, 2, 6, 8\}$. Here $T_b = \tau - T_a$, so $\mathcal{S} \setminus g_{\text{com}}$ is symmetric about $\tau/2$, and both extremes of $\mathcal{S}$ avoid g_{com} , so $C = 0 + 8 = \tau$. Here $g_{\text{com}} = \{3, 5\}$ turns out to be symmetric as well — the arc joining its two members is drawn like any other — which is what Conjecture 8.44 asserts in general and what §8.6 does not prove: the reflection is established on $\mathcal{S} \setminus g_{\text{com}}$ and observed on the rest. (b) $\beta = (12, 9, 7, 6, 3, 2, 1, 0)$: $\tau = 10$ and $|G| = 2$ again, but the two maximisers are not reflections of one another; $\min \mathcal{S} = 0$ falls in g_{com} , and $C = 0 + 12 = 12 \neq \tau$. Panel (b) is the reason Proposition 8.32 cannot be proved from $|G| = 2$ alone. Computed by `anc/fig-v2.py`.

Corollary 8.33 (the centre is the tie value). *If $[\det(x_i^{\beta_j})]_{D_1} = 0$ then $C := \min \mathcal{S} + \max \mathcal{S} = \tau$. Moreover t is even and the two tied classes are exactly the two solutions of $2\kappa \equiv C \pmod{t}$, both lying in $\mathcal{E}$.*

Proof. By (37) the set $\mathcal{S} \setminus g_{\text{com}}$ is stable under $x \mapsto \tau - x$, so its largest and smallest elements add up to τ ; by Proposition 8.32 the two extremes of $\mathcal{S}$ lie in that set, hence they are its largest and smallest elements, and $C = \tau$. Theorem 8.17(b) gives t even and the two tied classes $i, i + t/2$, both in $\mathcal{E}$; by (34) $\tau \equiv 2i \pmod{t}$, so $2\kappa \equiv \tau = C$ has exactly the two solutions $\kappa = i$ and $\kappa = i + t/2$. $\square$

Corollary 8.34 (a necessary condition, for every even t and every r). *Let $t \geq 2$ and $r \geq 1$, and suppose $\Phi_{t,r}(\lambda; z) \equiv 0$ with every $n_i \geq 1$. Then t is even, $|G| = 2$, $\min \mathcal{S} + \max \mathcal{S} = \tau$, and $\mathcal{S} \setminus g_{\text{com}}$ is symmetric about $\tau/2$.*

Proof. $\Phi_{t,r}$ and the numerator differ by a nonzero factor, so the numerator vanishes identically and in particular $[\det(x_i^{\beta_j})]_{D_1} = 0$. Apply Corollaries 8.30 and 8.33 and (37). $\square$

This is the part of §§8.5–8.6 that survives outside $t = 2$. What it does *not* give is the step that closes $t = 2$, where $e = 2$ forces $g_{\text{com}} = \emptyset$ and the symmetry of $\mathcal{S} \setminus g_{\text{com}}$ becomes the

symmetry of β itself. For $t \geq 4$ the common part g_{com} can be non-empty, and the argument above says nothing about it: the reflection is established on $\mathcal{S} \setminus g_{\text{com}}$ only, so it constrains β without determining it. That g_{com} is symmetric too is what the measurements show and what Conjecture 8.44 states; it is not proved here.

The condition is sufficient, for every t and every r . A necessary condition invites its converse, and half of it can be had outright. The reflection that Corollary 8.34 produces from the top stratum turns out, when it holds on all of $\mathcal{S}$, to be a symmetry of *every* term of the Laplace sum and not only of the two maximisers; and an involution with no fixed point cancels what it pairs. Write

$$C := \min \mathcal{S} + \max \mathcal{S}, \quad \text{and write } x \mapsto C - x \text{ for the reflection it centres.}$$

Theorem 8.35 (the sufficient direction, all t and r). *Let $t \geq 2$, $r \geq 1$, $N = t + 2r$, and let every residue class be occupied. Suppose*

$$C - \mathcal{S} = \mathcal{S} \quad \text{and} \quad \Delta_i(k) = C \text{ for some } i \in \mathcal{E}, \quad k. \quad (40)$$

Then $\Phi_{t,r}(\lambda; z) \equiv 0$.

The two clauses do two different jobs, and it is worth saying which before the proof: the first makes the reflection act on the transversals, and the second makes that action free. Figure 18 is the whole mechanism on one shape, with a shape that satisfies the first clause and not the second beside it.

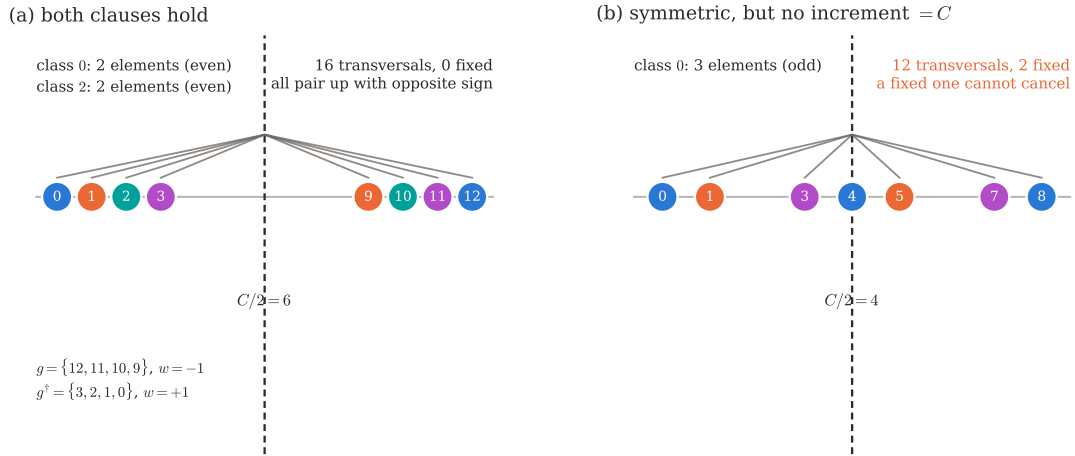


Figure 18. Why the sum cancels, and what the second clause is for. Each panel draws $\mathcal{S}$ on a line, coloured by residue class, with an arc joining v to $C - v$ and the dashed axis at $C/2$. (a) $\beta = (12, 11, 10, 9, 3, 2, 1, 0)$ at $t = 4$, $r = 2$: both classes fixed by the reflection have *even* size, so neither contains $C/2$; the 16 transversals pair off with no fixed point, and every pair carries opposite signs, so every term of (28) is cancelled by another. One such pair is printed with its two signs, computed and not asserted. (b) a shape with $C - \mathcal{S} = \mathcal{S}$ but no increment equal to C , found by search rather than chosen: there the fixed class has *odd* size, so it contains $C/2$, and two transversals are their own reflections. A term paired with itself cannot cancel, and the sum survives. That is the whole of what the second clause buys. Computed by `anc/fig_involution2.py`.

Lemma 8.36 (the reflected transversal). *Assume $C - \mathcal{S} = \mathcal{S}$. Then $x \mapsto C - x$ carries the class i onto the class $i^* := C - i \bmod t$, both in $\mathcal{E}$ or neither; so*

$$g^\dagger := (C - (g \cap \mathcal{S})) \cup (g \setminus \mathcal{S})$$

is again a transversal, $g \mapsto g^\dagger$ is an involution on them, and $T_{g^\dagger} = C - T_g$.

Proof. Every element of the class i is $\equiv i$, so $x \mapsto C - x$ sends it into the class i^* ; since $C - \mathcal{S} = \mathcal{S}$ and the map is injective, it carries the class i onto the class i^* whenever $i \in \mathcal{E}$, which forces $i^* \in \mathcal{E}$ and $n_i = n_{i^*}$. The singleton classes are not touched: their single element is frozen in every transversal. Finally $T_{g^\dagger} = \mathcal{S} \setminus g^\dagger = C - (\mathcal{S} \setminus g) = C - T_g$. $\square$

Lemma 8.37 (the reflected minor). *$A(C - T) = A(T)$ for every $2r$ -set T and every C .*

Proof. Extracting x_i^C from each of the $2r$ rows multiplies the determinant by $\prod_j z_j^C z_j^{-C} = 1$ and leaves the matrix of T with x_i replaced by x_i^{-1} , which is the matrix of T with the two rows of each reciprocal pair exchanged: a sign $(-1)^r$. And $C - T$ listed decreasingly reverses the column order of T : a sign $(-1)^{\binom{2r}{2}} = (-1)^r$, since $2r - 1$ is odd. The two cancel. $\square$

The two moves in that proof are not ours, and it is worth saying whose they are, because they are the right instrument and we chose them for that reason. Scaling each row by a power so as to replace x_i by $\bar{x}_i$, and then reversing a block of columns to collect $(-1)^{n(n-1)/2}$, is the manoeuvre Ayyer and Behrend use on the bialternant of a self-complementary shape [AB19, §3]; they describe their proofs as relying only on the determinant expressions “and the application of standard determinant operations”, and this is one of those operations. Albion performs the same reversal on the root-of-unity side [Alb23, Alb25], and at the level of characters the same fact is the invariance of sp_λ under $x_i \mapsto \bar{x}_i$, recorded in [AB19] and put to work in [AF20].

Its usefulness here is worth spelling out, since it is what makes the whole section short. A reflected minor could be attacked by expanding it, and then the singleton classes and the reflected ones would have to be tracked term by term — which is how we first tried, and it does not end. The Ayyer–Behrend manoeuvre instead disposes of the reflection in *two* sign computations that do not look at the entries at all, and it is robust in exactly the way we need: it never uses that the whole alphabet is being reflected, so it survives being applied to a *part* of the beta set, with the rest held fixed. That robustness is the reason Lemma 8.37 is four lines instead of a section, and it belongs to the tool rather than to us. It is also a small piece of evidence for the thesis of §2: the two lines this paper sits between share more machinery than their statements suggest.

The next statement is the combinatorial heart of the argument, and it needs only the first clause of (40). Reflecting a beta set *entirely* is classical — it is the complementation identity (23), and the determinant steps behind it are the standard ones just cited. What is asked here is a reflection of the excess part alone, with the singleton classes held fixed and interleaved among the reflected values; the two involutions it produces, one of β and one of $\mathbb{Z}/t$, are what the sign turns out to depend on.

Lemma 8.38 (the partial-reflection sign formula). *Assume $C - \mathcal{S} = \mathcal{S}$, and put*

$$\nu = \#\{x \in \mathcal{S} : 2x = C\}, \quad \eta = \#\{i \in \mathcal{E} : 2i \equiv C \bmod t\}.$$

Then for every transversal g

$$\frac{w(g^\dagger)}{w(g)} = (-1)^{e - \frac{\nu + \eta}{2}}. \quad (41)$$

In particular the ratio does not depend on g , nor on r , nor on N .

Corollary 8.39. *Under $C - \mathcal{S} = \mathcal{S}$, the ratio (41) equals -1 if and only if $\eta = 2$.*

Proof. An involution has as many fixed points as its set has elements, modulo two, so $\nu \equiv |\mathcal{S}|$ and $\eta \equiv e$; and $|\mathcal{S}| = 2r + e$ by (27), so $\nu \equiv e$ as well. If e is odd then $\nu = \eta = 1$ and the exponent is $e - 1$, even. If e is even then $\nu = 0$ and η is 0 or 2, giving the exponent e or $e - 1$; the first is even and the second odd. $\square$

That independence is the part that looks unlikely and is the point: the singleton classes sit at arbitrary positions among the reflected values, and moving a transversal moves them past one another — but each crossing they contribute to the column sum is undone by the crossing they contribute to the residue word, so they leave no trace in the ratio.

Lemma 8.40 (the reflected sign). *Assume (40). Then $\eta = 2$, the number e is even, $\nu = 0$, and $w(g^\dagger) = -w(g)$ for every transversal g .*

Proof of Lemma 8.38. Let $\mathcal{R}$ be $x \mapsto C - x$ on $\mathcal{S}$ and the identity on $\beta \setminus \mathcal{S}$, and let $\mathcal{R}_t$ be $i \mapsto i^*$ on $\mathcal{E}$ and the identity elsewhere; both are involutions, of β and of $\mathbb{Z}/t$ respectively, with ν and η fixed points, so $\text{sgn}(\mathcal{R}) = (-1)^{(|\mathcal{S}|-\nu)/2}$ and $\text{sgn}(\mathcal{R}_t) = (-1)^{(e-\eta)/2}$. Order the columns of a transversal g by residue and the remaining ones by decreasing β ; the sign of the resulting permutation of $\{1, \dots, N\}$ is $w(g)$ up to a factor depending on t alone. Applying $\mathcal{R}$ to that one-line notation contributes $\text{sgn}(\mathcal{R})$; restoring the residue order of the first block contributes $\text{sgn}(\mathcal{R}_t)$; and $\mathcal{R}$ reverses T_g entirely, so restoring its decreasing order contributes $(-1)^{\binom{2r}{2}}$. Multiplying the three gives (41), in which no term refers to g . $\square$

Proof of Lemma 8.40. Counting fixed points of an involution gives $|\mathcal{S}| \equiv \nu$ and $e \equiv \eta \pmod{2}$, and $|\mathcal{S}| = 2r + e$ by (27), so $e \equiv \nu$. Now the second clause of (40) says that two adjacent elements of some class i_0 sum to C ; they are exchanged by the reflection, so $i_0 = i_0^*$, and writing the class decreasingly the exchange sends its k -th entry to its $(n_{i_0} + 1 - k)$ -th, whence $n_{i_0} = 2k$ is even. The reflection $x \mapsto C - x$ has at most one fixed point, namely $C/2$; since this class is stable under it and has even cardinality, its elements pair off and it contains none, so $C/2 \notin \text{class } i_0$.

Before speaking of a second solution we must know there is one, and that is where the parity of t is decided rather than assumed. Suppose t were odd. Then 2 is invertible modulo t , so $i \mapsto i^*$ has i_0 as its only fixed residue; the remaining classes of $\mathcal{E}$ pair off, so e is odd, and $|\mathcal{S}| = 2r + e$ is odd too. An involution on a set of odd size has a fixed point, so $\nu = 1$ and $C/2 \in \mathcal{S}$; its residue κ satisfies $2\kappa \equiv C$, hence $\kappa = i_0$, putting $C/2$ in the class i_0 — which we have just shown it misses. So t is even, and $2i \equiv C$ has exactly the two solutions i_0 and $i_1 = i_0 + t/2$.

If $\nu = 1$ then $C/2 \in \mathcal{S}$, and its residue is i_0 or i_1 ; not i_0 , so $i_1 \in \mathcal{E}$. If $\nu = 0$ then e is even, and $\mathcal{E} = \{i_0\} \sqcup \{\text{pairs}\}$ would make e odd unless $i_1 \in \mathcal{E}$. Either way $\eta = 2$, hence e is even and $\nu = 0$, and Corollary 8.39 gives the sign at once. Read off (41) directly, the exponent is $e - \frac{0+2}{2} = e - 1 \equiv 1 \pmod{2}$. $\square$

Corollary 8.41 (the hypothesis, read on the residues). *Assume $C - \mathcal{S} = \mathcal{S}$. Then some $\Delta_i(k) = C$ if and only if t is even and both solutions of $2i \equiv C \pmod{t}$ are excess classes. So (40) says: the excess part of the beta set is symmetric about $C/2$, and the two residue classes that the reflection fixes are both excess.*

Proof. One direction is the first paragraph of the proof of Lemma 8.40. For the other, if both fixed classes lie in $\mathcal{E}$ then $\eta = 2$, so e is even by $e \equiv \eta$, so $|\mathcal{S}| = 2r + e$ is even and $\nu = 0$:

no element of $\mathcal{S}$ is fixed. A fixed class is therefore reflection-stable with no fixed point, hence of even size $2k$, and its k -th and $(k+1)$ -st entries are exchanged, so $\Delta_i(k) = C$. $\square$

Read that way the hypothesis is the same shape as the conclusion of Corollary 8.33, which is what makes the two halves of the section look at one another: there the two tied classes *turn out* to be the two solutions of $2\kappa \equiv C$; here we *ask* that they be excess.

Proof of Theorem 8.35. By Lemma 8.36, $g \mapsto g^\dagger$ is an involution on transversals with $T_{g^\dagger} = C - T_g$. It has no fixed point: a fixed transversal would need $C - g_i = g_i$, that is $g_i = C/2$, in every class with $i = i^*$, and the class i_0 of the proof above is such a class and misses $C/2$. By Lemmas 8.37 and 8.40,

$$w(g)A(T_g) + w(g^\dagger)A(T_{g^\dagger}) = w(g)A(T_g) - w(g)A(C - T_g) = 0,$$

so the sum of (28) cancels in pairs and the numerator vanishes identically. $\square$

Remark 8.42 (what this is, and what it is not). Three things are worth separating. First, at $t = 2$ the hypothesis (40) is branch (b): there $e = 2$ forces $\mathcal{S} = \beta$, so $C - \mathcal{S} = \mathcal{S}$ is self-complementarity, and the increment clause is the parity of the width — the two statements agree on every shape in the ranges of §9. So Theorem 8.35 contains that half of Theorem 8.6, and proves it again by a different mechanism. Second, for $t \geq 4$ it is *not* complementation: there $\mathcal{S} \subsetneq \beta$, λ is not self-complementary, and (23) says nothing. Third, the involution is global — it pairs every term of (28), not just the two maximisers, which is why it survives where the extremal argument only produces a necessary condition.

Conjecture 8.43 (the criterion, in general). *Conversely, if $\Phi_{t,r}(\lambda; z) \equiv 0$ and every residue class is occupied, then (40) holds. Equivalently, for every t and r ,*

$$\Phi_{t,r}(\lambda; z) \equiv 0 \iff \text{some class is empty, or } C - \mathcal{S} = \mathcal{S} \text{ and some } \Delta_i(k) = C.$$

Under the vanishing hypothesis, most of what (40) asks is already known, and saying which turns the conjecture into a single sentence about a single set. Corollary 8.33 gives $C = \tau$, so the tied increments equal C and the second clause holds; and (37) gives $C - (\mathcal{S} \setminus g_{\text{com}}) = \mathcal{S} \setminus g_{\text{com}}$. Since $\mathcal{S} = (\mathcal{S} \setminus g_{\text{com}}) \sqcup g_{\text{com}}$, all that is left is:

Conjecture 8.44 (the common part is symmetric). *Under occupancy, if $\Phi_{t,r}(\lambda; z) \equiv 0$ then $C - g_{\text{com}} = g_{\text{com}}$.*

Conjectures 8.43 and 8.44 are the same statement, and the second is the useful form: it names the $e - 2$ elements the argument has to reach, and no others. It also says at once why $t = 2$ closes and $t \geq 4$ does not. At $t = 2$ a tie is between the classes i and $i + 1$, so both of them are excess and $e = 2$ — not because $e = 2$ always holds there, since (27) allows $e = 1$ and it is common, but because $|G| = 2$ forces it, and g_{com} is only defined when $|G| = 2$. Hence $g_{\text{com}} = \emptyset$ and there is nothing left to be symmetric. The theorem there is not easier; there is simply no common part.

We state these as conjectures and not as theorems because only the direction proved above is proved. The evidence is in §9: over 190 443 shapes in twenty-four configurations, $t \leq 12$ and $r \leq 4$, the criterion has no false positive and no false negative; 43 010 of them are redone by an evaluation sharing no code with this section, and a further sweep of 12 937 shapes, with 131 zeros, is redone in Sage, which shares neither code nor library with either; on the ranges where a theorem already decides it agrees with Theorem 3.1 at $r = 1$ and with Theorem 8.6 at $t = 2$ without a single disagreement, over 9913 and 29 508 shapes; and a decoy that reads the condition on β instead of on $\mathcal{S}$ — that is, self-complementarity of the shape rather

than of its excess part — fails in every configuration of both sweeps. What is missing is one implication: Corollary 8.34 gives the reflection on $\mathcal{S} \setminus g_{\text{com}}$, and the top stratum cannot give more — there are shapes sharing τ , $\mathcal{S} \setminus g_{\text{com}}$ and the tied classes, of which some vanish and some do not, so the missing equation is not a refinement of §8.6 but a statement about a lower stratum.

Remark 8.45 (why not the obvious argument). One tries first to exchange inside a *single* maximiser: if $\max \mathcal{S}$ is omitted, re-choose its class and hope the degree rises. That cannot work. A maximiser may omit $\max \mathcal{S}$ — precisely when $\max \mathcal{S}$ is one of the tied values p_1, p_2 , so that one maximiser omits it and the other keeps it — so no argument reading one transversal at a time concludes. Proposition 8.32 is a statement about the *pair*, which is why the proof runs on (39) and on nothing else. Table 1 records both the steps of that proof and what happens to its conclusion when the hypothesis is dropped.

Table 1. The machinery of §§8.5–8.6, measured. The population is every shape with $|G| = 2$ over $(t, r) \in \{4, 6, 8, 10\} \times \{2, 3\}$ within the widths swept. The last row is the one that matters for Proposition 8.32: drop the hypothesis $[\det]_{D_1} = 0$ and the conclusion fails, so it is not carried for free.

statement	population	result	script
greedy parametrisation = brute force	4049 shapes	0 fail	a1_proof
A is the top half of T_g (Step 2)	same	0 fail	a1_proof
$T_b = \tau - T_a$	2522 shapes	0 fail	a1_proof
$v \in L_b \Rightarrow \tau - v \in H_a$	3034 values	0 fail	a1_proof
the extremes avoid g_{com}	2522 shapes	0 fail	a1_proof
the displayed identities of these subsections	11 identities	0 fail	v2.formulas
the same, with $[\det]_{D_1} = 0$ dropped	24718 shapes	414 fail	a1_proof

Remark 8.46. Corollary 8.30 is doing real work in Proposition 8.32, and $|G| = 2$ alone does not suffice. For $t = 4$, $r = 2$ and $\beta = (200, 199, 198, 197, 194, 193, 191, 4)$ the excess classes are $\{200, 4\}$, $\{197, 193\}$, $\{198, 194\}$, $\{199, 191\}$ and the increments are 392, 390, 390, 204, so $\tau = 390$, the tie is between the classes 1 and 3, and $|G| = 2$. But $\Delta_0(1) = 204 < \tau$, so $k_0 = 0$ in both maximisers and $\max \mathcal{S} = 200$ is omitted by both: the conclusion of Proposition 8.32 fails. Here $T_a = (198, 197, 191, 4)$ and $T_b = (199, 198, 193, 4)$ are not reflections of one another — $h_r - l_1$ is 6 for the first and 5 for the second — and accordingly the component of degree D_1 does not vanish.

The criterion at $t = 2$.

Proof of Theorem 8.6. The implication from (a) or (b) to $\Psi_r(\lambda) = 0$ is Theorem 8.1. For the converse, assume $\Psi_r(\lambda) = \Phi_{2,r}(\lambda; z) \equiv 0$ and that no residue class modulo 2 is absent from $\beta(\lambda)$, since otherwise branch (a) holds and there is nothing to prove. Then the numerator vanishes identically, so in particular its component of degree D_1 vanishes and $|G| = 2$ by Corollary 8.20.

By Theorem 8.17(b) there is a tie at rank r between classes i and $i + t/2 = i + 1$; at $t = 2$ these are the two classes, and both lie in $\mathcal{E}$, so $e = 2$. (With $e = 1$ all increments lie in a single class and are pairwise distinct by Step 4, so no tie is possible.) Since $e = 2$, neither class is a singleton and $\mathcal{S} = \{\beta_1, \dots, \beta_N\}$ is the whole beta set.

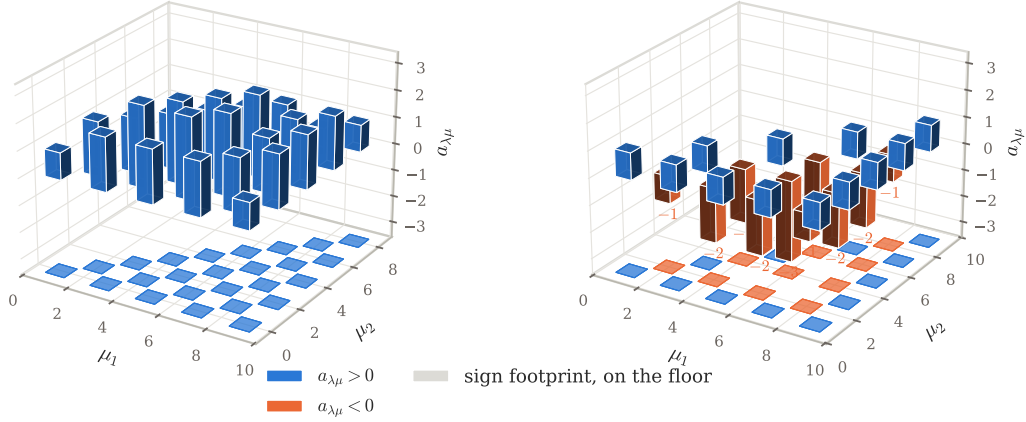
(a) $\lambda = (9, 8, 8, 6, 0, 0)$: a genuine character, all $a_{\lambda\mu} > 0$ (b) $\lambda = (9, 9, 8, 6, 0, 0)$: properly virtual

Figure 19. Why one reciprocal pair is not typical, at $r = 2$: the expansion $\Psi_r(\lambda) = \sum_{\mu} a_{\lambda\mu} sp_{\mu}(z)$ over the dominant weights of $Sp(4)$. The two shapes differ in a single box. For $\lambda = (9, 8, 8, 6, 0, 0)$ every $a_{\lambda\mu}$ is positive: the object is a genuine character, as it is for *every* λ when $r = 1$, where Theorem 3.1 writes it as a product of three $\mathfrak{sl}_2$ characters. For $\lambda = (9, 9, 8, 6, 0, 0)$ the expansion has 12 positive and 10 negative coefficients, the most negative being -3 : it is properly virtual, which is the phenomenon Remark 8.12 records and the reason an endpoint computation stops being an exact test past one pair. The two panels share the vertical scale, and the flat squares on the floor repeat the sign of each coefficient, because a coefficient of -1 is invisible beside one of $+3$. Computed by `anc/fig_v2.py`.

By Corollary 8.33, $C = \min \mathcal{S} + \max \mathcal{S} = \beta_N + \beta_1 = \tau$, and since both classes change, $g_{\text{com}} = \emptyset$ in (38), so (37) reads $\tau - \mathcal{S} = \mathcal{S}$: the beta set is symmetric about C , that is

$$\beta_j + \beta_{N+1-j} = C \quad \text{for every } j.$$

Substituting $\beta_j = \lambda_j + N - j$ turns this into $\lambda_j + \lambda_{N+1-j} = C - N + 1 =: w$ for every j , which is self-complementarity of width $w = \lambda_1 + \lambda_N$. Finally, Corollary 8.33 says that the two solutions of $2\kappa \equiv C \pmod{2}$ are the two classes, which holds exactly when C is even; and $C = w + N - 1$ with $N = 2r + 2$ even, so C even is exactly w odd. Thus λ is self-complementary of odd width: branch (b). $\square$

Remark 8.47. At $t = 2$ the congruence (34) is vacuous — every increment is even — and $|G| \leq 2$ holds for the weaker reason that only two classes exist. What is *not* vacuous at $t = 2$ is that τ is even, and it is used twice: it forbids the crossed matching in Proposition 8.29, and it *is* the parity that becomes branch (b).

Remark 8.48 (*genuine* does not mean *irreducible*). In Remark 8.12, $\pm$ a *genuine character* means a *nonnegative* combination of irreducibles, not a single one. It is not a single one: already $s_{(2)}(1, -1, z, z^{-1}) = z^2 + 2 + z^{-2} = sp_{(2)} + sp_{(0)}$. At $t = 2$, $r = 1$ Theorem 3.1 gives a product of three $\mathfrak{sl}_2$ characters, and a product of characters is a character — that is the whole claim. From $r = 2$ negative coefficients appear and no representation exists, which is what *properly virtual* means; Table 2 counts both and Figure 19 draws one of each.

Table 2. Genuine against properly virtual, over all λ with λ_1 up to the bound shown. The expansion is $\Psi_r(\lambda) = \sum_{\mu} a_{\lambda\mu} sp_{\mu}(z)$; *one sign* means every $a_{\lambda\mu}$ has the same sign, so that $\pm\Psi_r$ is a character, and *mixed* means it is properly virtual. At $r = 1$ the mixed column is empty, as Theorem 3.1 forces; it is not empty for any $r \geq 2$ we can reach. Computed by `anc/fig_v2.py`.

	shapes	$\Psi_r = 0$	one sign	mixed
$r = 1, \lambda_1 \leq 14$	3060	352	2708	0
$r = 2, \lambda_1 \leq 8$	3003	78	2531	394
$r = 3, \lambda_1 \leq 5$	1287	27	1008	252
$r = 4, \lambda_1 \leq 2$	66	2	58	6

Remark 8.49 (where the alphabet sits). For $r = 1$ the alphabet $(1, -1, z, \bar{z})$ of (21) is literally the eigenvalue list of a matrix in the non-identity component $O(4)^-$, tabulated as such by Hanany and Kalveks [HK16, Table 7], who also record the rank drop $[\text{vec}]_{O(2n)^-} \cong [\text{vec}]_{O(2)^-} \oplus [\text{fund}]_{C_{n-1}}$ for the *vector* representation. So Ψ_r is a $GL(2r+2)$ character evaluated on that component, and the folding $D_{r+1} \rightarrow C_r$ is the reason a symplectic group appears at all; we make no use of this beyond the placement, since for a general λ the restriction is a $\mathbb{Z}$ -combination of symplectic characters and not a single one.

9. Verification

Every statement above was checked numerically, by three implementations along different code paths: one in exact Laurent arithmetic, one a from-scratch bialternant evaluation written from the statements alone, and one in Sage, which shares no line of code and no library with the other two. The ranges and counts are collected here once.

statement	range	result	status
Thm 3.1, sign included	$t \leq 7, \lambda \leq 14, \dots, 10$	959 exact, 476 zeros, 0 fail	proved
the same, an implementation written from the statement	$t \leq 6, \lambda \leq 12, \dots, 9$	749 shapes, 0 fail; sign decoy fails 35/133	proved
Lemma 4.1	$t \leq 6$	454 nonzero minors	proved
Lemma 4.4	$t \leq 7$	724/724	proved
Proposition 3.4	$t \leq 7, \lambda \leq 19$	2970/2970	proved
Prop. 3.5 (concentric locus)	$t \leq 9, \lambda \leq 16$	1331/1331; 0 at odd t , 43 at even	proved
Prop. 3.11 (short form of the sign)	$t \leq 8, \lambda \leq 14$	1496/1496; proof steps 0 fail	proved
the shift law (8) and its sign	$t \leq 7, \lambda \leq 12$	826 shapes, 0 fail	proved
Prop. 3.10 (the multiset separates)	$d_i \leq 60; t \leq 5, \lambda \leq 16$	75640 invariants, 0 collisions	proved
the same, both directions, from the statement	$t \leq 5, \lambda \leq 14, \dots, 11$	676 shapes, 0 collisions; decoy 104	proved
Prop. 3.6 (the lattice)	$t \leq 5, \lambda \leq 14, 12, 10, 10$	84, 35, 20, 14 fibres, both ways	proved
Cor. 3.7 (generating function)	same range	153/153; wrong denominator 101	proved
Remark 3.8 (the sign sees the m_i)	$t \leq 4$	107/107, 43/62, 26/36	verified
Lemma 5.1	$t \leq 7, \lambda \leq 16$	2113/2113	proved
Thm 5.2	$t \leq 12, \lambda_1 \leq 33$	119 + 21, no others	proved
the sign is +1, (18)	$t \leq 14$ even; cores $t \leq 10$	28 extra + 219 cores, all +1	proved
Proposition 6.3	$t = 2, r \leq 4, c \leq 60$	240/240	proved

continues on the next page

Verification, continued

statement	range	result	status
Cor. 6.1 vs direct enumeration	$t \leq 5$	14/14; boxes 16/16	proved
Prop. 6.4 (runs alternate)	$t = 2, \lambda \leq 14, \leq 4$ rows	4823 minors; 217 pairs, 0 fail	proved
(D3), (D4) of §7	$t \leq 6, \lambda \leq 10$	600 / 383 / 200 of 600	verified
Thm 8.6, the criterion, every r	—	proof, §8.6	proved
the same, read on $O(N)$, Cor. 8.7	—	restatement, §8.3	proved
Thm 8.24, the external input	—	[PvW08]	external
Theorem 8.17, $ G \leq 2$	$t \leq 10, r \leq 3$	greedy vs. brute force, 0/4049	proved
Prop. 8.32, the extremes	$t \leq 10, r \leq 3$	2522 shapes, 0 fail	proved
the hypothesis is needed	same	414 shapes fail without it	proved
the displayed identities, §§8.5–8.6	$t = 2, r \leq 3$	11 identities, 0 fail	verified
Lemma 8.14, the column move, any class	$t \leq 6, r \leq 3$	285 600/285 600; one-count decoy 149 652	proved
Thm 8.35, the sufficient direction	$t \leq 8, r \leq 4$	438 shapes, 0 fail; (41) 438/438	proved
Conj. 8.43, both directions	$t \leq 12, r \leq 4$	190 443 shapes, 24 configurations, 0 fail	verified
the same, independent evaluation	$t \leq 8, r \leq 4$	43 010 shapes, 409 zeros, 0 fail	verified
the same again, in Sage	$t \leq 8, r \leq 3$	12 937 shapes, 131 zeros, 0 fail	verified
against Theorems 3.1 and 8.6	$r = 1; t = 2$	9913 and 29 508 shapes, 0 disagreements	proved
the decoy reading β instead of $\mathcal{S}$	same	fails in 24 of 24 and 14 of 14	proved
Thm 8.6, both directions, direct check	$r \leq 3, \lambda \leq 28, 26, 22$	9961 shapes, 318 zeros, 0 fail	proved
branch (a), Thm. 8.1	$r \leq 3, \lambda \leq 15$	16/16	proved
branch (b), Thm. 8.1	$r \leq 3$	24/24	proved
even-width control for (b)	$r \leq 3$	56/56 nonzero	proved
converse at $r = 1$, Thm. 8.3	$\beta_1 \leq 25$	14950 beta sets, 0 fail	proved
converse, $\ell(\lambda) \leq N/2$, Thm. 8.4	$r \leq 3$	372/372 witnesses	proved
the isolating witness, Prop. 8.11	$r = 2, \lambda \leq 14$	227 witnesses, 11 residue	proved
the same at $r = 3$	$r = 3, \lambda \leq 13$	149 witnesses, 0 residue	proved
Identity (23)	$r \leq 2$	474/474	proved
Identity (24), and $\deg_{c_*} S_n = n - 1$	$b \leq 31$	496/496; 39/39	proved
Lemma 8.9, the §8.4 reduction	$r \leq 3, \nu \leq 8$	identity 0 fail; folds 0 fail	proved
the reduction carried out	$r \leq 2$	236 labels, 0 unresolved	proved
(26) against the exact object	$r \leq 2$, both ranges	184 shapes, 0 fail	verified

The two shaded blocks are §8, and they are shaded because they are the only place in the paper where a row's status is not obvious from its statement. The first block is the machinery and what it reaches: Theorem 8.6 closes $t = 2$ for every r , and the one row in *external* is the external input it rests on; below it the extremal steps, then Theorem 8.35, which reaches every t and every r and needs no external input at all; and last Conjecture 8.43, in *verified* because it is the one statement here that is measured and not proved. Its two sub-rows are what makes the measurement worth quoting: the criterion is compared against the two theorems that already decide their own ranges, and a decoy that reads the condition on β instead of on $\mathcal{S}$ is run alongside and fails everywhere. The second block is the same locus checked directly against the definitions, branch by branch, which is a different code path and not a restatement. The rows below both blocks are the identities and the reduction the section rests on. In the column, *proved* means a proof is given here or cited, and *verified* means the statement is checked exhaustively over the stated range and not proved.

The (D3), (D4) row is the one that matters most in the first group: the identity holds in all 600 cases for (1) and fails in 217 and 400 of them for the two deformed alphabets, so the test is capable of failing. In the second group the even-width control plays the same role for Theorem 8.1: 56 self-complementary shapes of even width, none of which vanishes, so the parity hypothesis is doing work rather than decorating. A separate audit rebuilds all of the second group from the definitions in a single pass, independently of the scripts that produced them: 3414 checks, 0 failures. A second one takes the *displayed formulas* rather than the counts — Theorem 2.1, the interval reading (2), Lemmas 4.2 and 4.5, the splitting (20), the complementation (23), d’Ocagne (24) and the scalar caveat (43) — and evaluates both sides at numeric points from the definitions, using neither the closed forms of this paper nor the scripts behind the table: 3297 evaluations over nine displayed formulas, 0 failures. The scripts are ancillary to this paper, and so is the saved output of each one, so every count in the table above can be located in an archived run rather than taken on trust.

10. Open problems

Figure 20 places what follows against what the paper proves and what it borrows; the problems below are the boxes it leaves open.

Problem 10.1 (the other classical types). Everything above is type A . What do $sp_\lambda(\mu_t, z, \bar{z})$, $so_\lambda(\mu_t, z, \bar{z})$ and $o_\lambda(\mu_t, z, \bar{z})$ do? The root-of-unity factorizations were carried from type A to B , C , D by a uniform argument in [AK22] and to the universal characters in [Alb23]; the frozen rows of the bialternant are the same in every type, so Lemma 4.1 and the excess-two count should survive verbatim, while the collapse, Lemmas 4.4 and 4.5, would need re-deriving once per type. The alphabet (1) is already a torus element of $O(t+2)$, so the other types are not the same function at a new point but new objects, and choosing the right ones is a matter of taste inside a programme we did not build. We have measured, and report it as a measurement only. Expanding the Koike–Terada universal characters in the Schur basis and evaluating term by term with Theorem 3.1, over $|\lambda| \leq 12$ for $t \leq 3$ and $|\lambda| \leq 10$ for $t = 4, 5$, the fraction of $\ell(\lambda) \geq 2$ shapes on which the value vanishes is markedly larger for o_λ on the non-identity component than on the identity one — 83.8% at $t = 2$ against 41.8% at $t = 3$, and 58.8% at $t = 4$ against 42.1% at $t = 5$ — with sp_λ in between and the Schur case lowest. The effect is real and it tracks the component, which is what one would expect of an associate cancellation; but it is quantitative and we found no law covering the whole table. A first attempt at one, that o_λ vanishes for all $\ell(\lambda) \geq 2$ when t is even, is false: it was read off a sample and the systematic sweep refutes it.

The $t = 2$ column of that comparison is the one case which does have a law, and it is Lemma 8.9: there the alphabet is (21) at $r = 1$, so o_ν equals $sp_\nu(z)$, a *rank-one* symplectic character. That vanishes identically for $\ell(\nu) = 2$ — all 36 such shapes in range, with no exception — and folds for $\ell(\nu) > 2$, which is why the $t = 2$ figure is the outlier. What remains without a law is odd t , where the alphabet is not of the form (21) and the lemma does not apply.

The sp_λ column at $t = 2$, on the other hand, has an answer, and it explains the asymmetry of the table rather than adding a row to it. Adjoining the same two letters to a universal *symplectic* character does not collapse it. Over $|\nu| \leq 8$,

$$sp_\nu(W, 1, -1) = \sum_{\gamma \subseteq \nu} c_{\nu/\gamma} sp_\gamma(W), \quad (42)$$

three routes to the zero locus, and what each one reaches

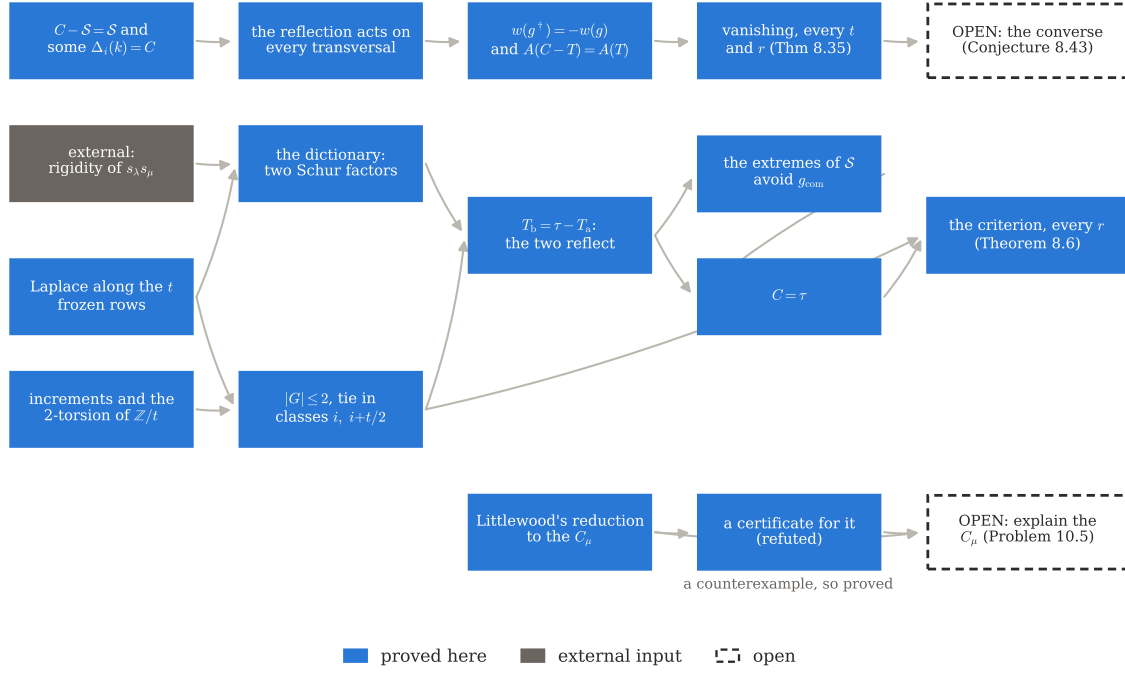


Figure 20. Three routes to the zero locus, and what each one reaches. Boxes are statements and arrows are dependencies; the colour is the status column of §9 — filled for what is proved here, grey for the single external input, dashed for what is open. The *middle* path is the extremal argument of §§8.5 and 8.6, which closes the criterion at $t = 2$ for every r . The *lower* one is Littlewood’s reduction, whose certificate is refuted by Proposition 8.11 — a counterexample, hence itself a proof — so that it no longer bears on the criterion but on the separate question of explaining the C_μ , which is Problem 10.5. The *upper* one starts from a hypothesis rather than from the expansion, and is the only one that reaches every t and every r : there the reflection pairs off all the terms at once, not just the two maximisers. The three meet at the one implication that is missing, which is what makes Conjecture 8.43 a conjecture. Statements are named rather than numbered, so that the figure cannot drift when the numbering does. Computed by `anc/fig_proof.py`.

where the coefficient depends on ν and γ only through the skew shape ν/γ — 247 coefficients over 74 distinct skew shapes, with no clash — and takes only the values 0 and ± 1 , vanishing whenever $|\nu/\gamma|$ is odd. That is the shape a branching rule for universal symplectic functions produces, and such rules are available [JLW24]. Read against it, Lemma 8.9 says that for o_ν the corresponding coefficients are concentrated at $\gamma = \nu$: the two letters collapse the orthogonal character to a single symplectic one, and leave the symplectic character spread over the whole interval below ν . The asymmetry of the measurement is that, and not an artefact of the ranges. What we have not done is identify $c_{\nu/\gamma}$, which (42) makes a question about skew universal symplectic characters at $(1, -1)$ and nothing more.

The third column resists the same route, and for a structural reason rather than an accidental one. What answers the other two is that adjoining a letter is the ring map

$p_k \mapsto p_k + c^k$, so ι_1 and ι_{-1} act on Λ and compose. The identity that would bring so into the picture, $so_\nu(X) = o_\nu(X, \bar{X}, 1, 0, \dots)$ of [AK25, (2.13)], relates the two on a *doubled* alphabet instead, and a doubling is not an adjunction: the ι 's do not reach it. So the so column is not a gap of effort but of mechanism, and we leave it stated as such.

Problem 10.2 (a Verschiebung that sees a reciprocal pair). The operator φ_t has nothing to say about a letter outside a ζ -orbit, which is the structural reason Theorem 3.1 is not a corollary of anything in that line, and the reason our proof descends to the alternant. Is there an operator on universal characters, or a factorization of φ_t through a larger algebra, acting on alphabets of the form (union of ζ -orbits) $\cup$ (reciprocal pairs)? By §7 such an operator would decide (D1) and (D2) at once. The discussion there suggests a quantitative form.

What makes the question well posed rather than vague is that each half already has a formalism, and they are not the same one. The ζ -orbits are the province of φ_t , in Albion's formulation [Alb23]. Alphabets closed under inversion are the province of the universal symplectic and orthogonal functions, which have vertex operator realizations from which their branching rules follow [JLW24]. Our alphabet is one of each, and Lemma 8.9 is the one place where the two meet: there the ζ -orbit is $\{1, -1\}$, small enough that adjoining it is a pair of letter maps rather than an operator. We know of nothing that acts on both halves at once, and the question is whether the two formalisms compose.

Proposition 10.3 (the rank is the wrong invariant). *Let $A = \mu_t \cup B$ with B closed under inversion and disjoint from μ_t , and let $\rho(B)$ be the rank of the subgroup of $\mathbb{C}^\times$ generated by B . Then $\rho(B) = 1$ does not imply that $s_\lambda(A)$ is a product of characters. At $t = 2$ and $B = \{z, z^{-1}, z^2, z^{-2}\}$, which has $\rho(B) = 1$, the value at $\lambda = (2)$ has a root off the unit circle, and so has no such factorization; the same happens for $B = \{z, z^{-1}, z^3, z^{-3}\}$.*

Proof. A product of characters $\chi_k(z) = (z^{k+1} - z^{-k-1})/(z - z^{-1})$ has all of its roots on the unit circle, so it is enough to exhibit one root off it. For $B = \{z, z^{-1}, z^2, z^{-2}\}$ and $\lambda = (2)$ the value is $z^{-4} + z^{-3} + z^{-2} + z^{-1} + 3 + z + z^2 + z^3 + z^4$. The Laurent polynomial is invariant under $z \mapsto z^{-1}$, so writing $x = z + z^{-1}$ and using $z^k + z^{-k} \in \mathbb{Z}[x]$ it becomes

$$x^4 + x^3 - 3x^2 - 2x + 3 = (x - 1)(x^3 + 2x^2 - x - 3).$$

The cubic has discriminant $18 \cdot 2 \cdot (-1)(-3) - 4 \cdot 2^3(-3) + 2^2(-1)^2 - 4(-1)^3 - 27 \cdot (-3)^2 = -31 < 0$, so two of its roots are not real. If $|z| = 1$ then $x = z + z^{-1} = 2\cos\theta$ is real; hence for a non-real root x_0 , neither root of $z^2 - x_0z + 1 = 0$ lies on the unit circle, and both are roots of the value. So the factorization is impossible, and the statement is exact rather than numerical.

Over $|\lambda| \leq 8$ the same phenomenon occurs for 17 of the 62 nonvanishing shapes, and for $B = \{z, z^{-1}, z^3, z^{-3}\}$ for 18; on the same range $B = \{z, z^{-1}\}$ has every root on the circle without exception, as Theorem 3.1 requires. $\square$

What the rank does not see is how many free directions B has. A single reciprocal pair is one direction; $\{z, z^{-1}, z^2, z^{-2}\}$ generates a group of rank one and is two, and the two see each other. So the invariant to conjecture on is not $\rho(B)$ but the number of independent letters, and with that reading the statement becomes the one §7 already makes by other means:

Conjecture 10.4 (one free direction). *Let B be closed under inversion and disjoint from μ_t , as in Proposition 10.3. Then $s_\lambda(\mu_t \cup B)$ is zero or a product of characters, for every λ , if*

and only if B is a single reciprocal pair. (The alternative is not decoration: by Corollary 3.2 the value does vanish for some λ already at one pair, and 0 is not a product of characters.)

The evidence for it is Theorem 3.1 in one direction and (D1)–(D4) together with Proposition 10.3 in the other: enlarging B destroys the product whether the enlargement raises the rank or not. We have no proof and no proposed mechanism, and this is precisely the statement a φ_t -style argument ought to produce.

That hypothesis is what the conjecture is stated under, and it is also what a general form would have to drop. Inversion-closure of B is sufficient in the evidence above, but it is not necessary for a product. Write $w = -iy$. Then the second block of the alphabet $\{1, -1, y, -y^{-1}\}$ is not closed under inversion, yet

$$s_\lambda(1, -1, y, -y^{-1}) = i^{|\lambda|} s_\lambda(-i, i, w, w^{-1}), \quad (43)$$

and the alphabet on the right is. The hypothesis a projectively invariant form would carry is therefore not inversion-closure of B but that A be a scalar multiple of an inversion-closed set. A coset $\zeta\mu_t$ is itself inversion-closed exactly when $\zeta^2 = \pm 1$, which is where (D3) does and does not break.

Problem 10.5 (the instrument for the second pair). Section 8 reduces the vanishing at $r \geq 2$ to a statement about the coefficients C_μ of (26), which are signed sums of Littlewood coefficients. That statement is no longer what is missing: Theorem 8.6 settles the vanishing, by the extremal argument of §8.6, which never meets a C_μ . What the reduction asks for now is not a proof but an *explanation* — why those signed sums cancel, in Littlewood’s own terms — and there is still no certificate, the one we proposed failing by Proposition 8.11. A problem that was a gap is now a question about a second proof, which is a better problem and a smaller one. Lemma 8.9 settles the other half of the reduction — what the values $o_\nu(A)$ are — and settles it symplectically: the two readings of the alphabet agree because they are the same reading, the orbit Lie algebra of D_{r+1} being C_r rather than the fixed subalgebra B_r . What it does not touch is the coefficients. Kumari and Stokke [KS26] have given Murnaghan–Nakayama rules for symplectic, orthogonal and orthosymplectic Schur functions; by Lemma 8.9 the relevant one here is the symplectic [KS26, Theorem 3.4], not the even orthogonal [KS26, Theorem 3.10]. Does the recursion it provides compute C_μ ? Their rule carries a third sum beyond the adding and removing of border strips, and it is the one they single out as complicated; [KS26, Corollary 3.5] shows it drops out once $\mu_n + 1 \geq r$, so it is a shallow-shape phenomenon. An earlier form of this problem asked whether that sum is what accounts for the failure of Littlewood’s rule outside its range; Lemma 8.9 answers that it is not, since the failure is carried by the symplectic modification rules, which act on the labels and not inside the recursion. The coefficients are what is left, and whether that recursion computes them is what we have not attempted.

There is a second instrument, and it is the one we would try first. Frohmader’s branching rules [Fro25] compute $\text{mult}(\pi_{O_n}^\nu, \pi_{GL_n}^\lambda)$ as $\sum_\mu c_{\mu\nu}^\lambda(O_n)$ over partitions μ with even rows, with no stable-range constraint and no modification rule. Our C_μ of (26) are signed sums of exactly the multiplicities those rules compute, assembled by the associate involution $\nu \mapsto \nu^*$. So the question becomes concrete: is $C_\mu(\lambda) = 0$, on the two branches, an identity between generalized Littlewood–Richardson coefficients that the flag conditions make visible? That is a question about tableaux, which is the form in which we would like to see it answered.

Problem 10.6 (the extra locus, in general). Does Theorem 5.2 have an analogue for sp , so and o ? More generally, for which specializations of the free variables does the criterion

(4) acquire extra solutions, and can the extra locus always be read off the kernel of the specialization acting on the span of the universal characters, as Remark 5.3 suggests? If each type acquires a family indexed by an order-two element of the residue lattice, the principle is real.

The question needs the free part pinned down, and for a computed reason: the extra family is a phenomenon of the *minimal* alphabet. Adjoin a second reciprocal pair and it is gone — over $t = 3, 4, 5, 6$ and $|\lambda| \leq 14$ the locus is then exactly the t -cores, for type s no less than for sp and o , while at one pair type s has the two extras of Theorem 5.2 at $t = 4$ and one at $t = 6$. (We test the second pair at $z_2 = z_1^2$, a specialization, which can only create coincidences and not remove them, so a count of zero there is a count of zero.) An analogue for another type must therefore be sought at *that type's* minimal free part, one letter of each kind; comparing types across free parts of different sizes measures the free part instead, which is what our first attempt did.

The kernel is the right thing to look at, but its *presence* does not settle the locus; its size does. Two free parts of the same size can both collapse the independence and behave quite differently. On the reciprocal locus $s_\mu(z, \bar{z}) = \chi_{\mu_1 - \mu_2}(z)$ retains one grading, and the extra locus is the single family of Theorem 5.2. Replace the pair by the ζ_2 -orbit $(z, -z)$, where $s_\mu(z, -z) = z^{|\mu|} s_\mu(1, -1)$ retains only $|\mu|$ and a sign, and over $\ell(\lambda) \leq 2$ the criterion acquires 28 further solutions at $t = 2$ and 186 at $t = 8$, in no family we can name. The control is the free pair (z, w) , which has no kernel and returns the t -cores and nothing else at every $t \leq 8$ — that is (4) itself, so the control is their theorem rather than a measurement of ours. So the reciprocal pair is the collapsing specialization of *smallest* kernel, and that, rather than the mere existence of a kernel, is what makes its extra locus one family.

Problem 10.7 (the fibres, refined by the sign). Corollary 3.7 describes the fibre of the interval triple, and only that. The invariant is $I_t = (d_1, d_2, d_3, \varepsilon_\lambda)$, and the sign cuts each branch further. Its cut is not bookkeeping, and it is not idle: by Remark 3.8 the sign sees the $t - 2$ directions the triple does not, from $t = 3$ on — constant in (m_i) at fixed v in all 107 slots at $t = 2$, but in only 43 of 62 at $t = 3$ and 26 of 36 at $t = 4$. What is the restriction of ε_λ to a branch? The obstruction to reading it off (10) is visible in (6): $\text{inv}(b_S)$ is a statistic of the whole beta-word, and the lattice coordinates split that word into two distinguished classes and $t - 2$ free letters, so a rule would have to reassemble what the bijection separates.

Two smaller gaps go with it. The same description for the size-three profile should be the same bookkeeping with one class of three in place of two of two; we have not written it. The degenerate profile is a different question, since there the fibre is the whole vanishing locus of Φ_t , which Corollary 3.2 and Proposition 3.5 describe by other means.

Problem 10.8 (a sieving phenomenon that keeps a parameter). At $t = 2$ Corollary 6.1 is a signed count of $\text{SSYT}(\lambda, 4)$, and a signed count of order two is what Stembridge's $q = -1$ phenomenon [Ste94] computes: on a set carrying an involution, the value at -1 counts the fixed points. That is the case $\#C = 2$ of the cyclic sieving phenomenon of Reiner, Stanton and White [RSW04], whose Theorem 4.3 evaluates the principal specialization $s_\lambda(1, q, \dots, q^{k-1})$ at a primitive d -th root of unity in terms of the d -core and the d -quotient of λ — the two invariants that carry Theorem 3.1 as well — and which Lee and Oh [LO22] have since extended to skew shapes. Every evaluation in that line is at a *number*, as a sieving statement requires: the alphabet there is the geometric progression $1, q, \dots, q^{k-1}$ specialized at a root of unity, not a plain orbit, and nothing is left free. Ours is not of that kind — the reciprocal pair keeps z , and (5) is a product formula in it. Is there a cyclic action on $\text{SSYT}(\lambda, t + 2)$, and a q -analogue of Φ_t , whose value at a t -th root of unity counts fixed points refined by

$m_{t+1} - m_{t+2}$? The case $\#C = 2$ is where to look first, and there the involution such a statement would need already exists: §6 assembles it.

Asked that way the question has a name. Sieving statements that keep an extra statistic are studied in their own right — Ahlback and Swanson [AS18] refine the cyclic sieving on words by fixing the cyclic descent type alongside the major index — and the two-variable form, in which a second grading carries a second action, is the *bicyclic* sieving of Barcelo, Reiner and Stanton [BRS08]. On semistandard tableaux in particular, Oh and Park [OP19] build a cyclic action out of the crystal structure whose sieving polynomial is $q^{-\kappa(\lambda)} s_\lambda(1, q, \dots, q^{n-1})$ — the frozen half of our alphabet, evaluated exactly as a sieving statement requires — and obtain a bicyclic statement for hooks; Alexandersson, Kantarci Oğuz and Linusson [AKL21] do the same under promotion for several families of shapes. None of these keeps a variable *free*, which is what (1) does and what makes the question here a different one, but together they say what an answer should look like. The sharp form is therefore: does each fixed z -weight fibre of $\text{SSYT}(\lambda, t+2)$ carry a cyclic sieving of its own?

The existence half of the question is decidable without exhibiting the action. Alexandersson and Amini [AA19] give a necessary and sufficient criterion for a cyclic action realizing a given sieving polynomial to exist: for every $d \mid t$, the value at ω^d must be $\sum_{j \mid d} j c_j$ with every c_j a nonnegative integer, and the c_j — the numbers of orbits of each size — are then computable by triangularity. Applied to the candidate that the alphabet itself suggests, $F(q, z) = s_\lambda(1, q, \dots, q^{t-1}, z, z^{-1})$, whose z -grading is already the refinement asked for, the criterion is met by almost every graded piece in the ranges we can reach; but so it is for a deliberately wrong refinement, grading by $m_{t+1} + m_{t+2}$ instead, which passes at the same rate. At those sizes the criterion does not discriminate, and we report the attempt as evidence in neither direction.

One narrower gap is recorded where it arises rather than here, in the same form as the problems above and with the same meaning — something we would like to know and have not settled: the involution of §6 is assembled at $t = 2$ and nowhere else, because Proposition 6.4 is a statement about one word and that word is the $t = 2$ one. The distinction we keep throughout is between a *conjecture*, which is a statement we believe and have tested, and a *problem*, which is not a claim at all.

Taken together the problems say where we think the object stops being ours. The evaluation is complete and the alphabet is minimal, so what is left is not more of the same: it is whether the two formalisms that own the two halves of (1) compose, whether a sieving statement can keep a variable, and what the coefficients C_μ do once the certificate we proposed for them is gone. Of the three, the last is the one the paper needs and the first is the one we would rather see answered.

It is worth closing where we began. The introduction named two reasons for a character evaluation to collapse — that the alphabet is an orbit, and that it is closed under inversion — and observed that no alphabet had carried one of each. This one does, and the two reasons behave quite differently on it. The orbit half is exhausted: Theorem 3.1 evaluates it in closed form for every t , and Theorem 5.2 shows that within the two-row independence problem exactly one further family behaves that way. The inversion half is not: the letter -1 moves the alphabet into the non-identity component of an orthogonal group, and that has been enough to say a good deal about *where* the character vanishes — Theorem 8.6 settles it at $t = 2$ for every number of pairs, and Theorem 8.35 gives one implication for every t and every r — but it has not told us what the character *is* there for more than one pair, and the other implication is Conjecture 8.43. The value never stops being a character

value: it is s_λ evaluated at a torus element throughout. What ceases to hold past one pair is that the folded symplectic expansion is always a genuine character: from two pairs on, properly virtual examples occur. That is a statement about the expansion and not about the object. What the paper hands over is smaller than either half and more portable: a closed form carrying its sign, which makes this family a test case for any implementation of twisted or universal characters; a vanishing test that costs two subtractions on the beta set; and an enumeration in which the parameter is still free. A reader looking for the next alphabet should look for the one that makes the second reason as complete as the first has become here; on the evidence of §7 it will not be found by adding letters to this one.

Acknowledgements. This paper exists because Arvind Ayyer asked whether the rank-one case generalizes to arbitrary t with the free variable symbolic. The question was better than the note that prompted it. We thank him for it, and for the correspondence that followed. Remark 8.8 exists because Nishu Kumari asked whether the condition $n_i(\lambda) = 0$ could be phrased in terms of the t -core; she also confirmed the reading of [AK25, Theorem 5.3] in §5, and it was a pointer of hers that sent us to [AK25, §3], where Lemma 8.9 was waiting; we are grateful for it. The involution of §6 is assembled from three lemmas of her thesis [KumT], and is hers in that sense too.

A third debt is to a tool rather than to a conversation. The proof of Theorem 8.35 turns on being able to reflect a beta set inside a determinant without expanding anything, and the way to do that — scale each row by a power to send x_i to $\bar{x}_i$, then reverse a block of columns — is the manoeuvre of Ayyer and Behrend [AB19, §3], used on the root-of-unity side by Albion [Alb23, Alb25]. We adopted it because it is plainly the right instrument for an inversion-closed alphabet, and we kept it because it turned out to be more robust than the setting it was written for: it never uses that the reflection is applied to the whole alphabet, and so it goes through for a part of one. Everything §8 says about $t \geq 4$ rests on that.

Code, data and disclosure. Every computation quoted in this paper is performed by a named script distributed as an ancillary file, and every number quoted is reproduced by an archived run of one; §9 tabulates which script answers for which claim. The same scripts and the same archived output are also kept at <https://github.com/karlesmarin/schur-orbit-and-reciprocal-pair>, which is a convenience: the ancillary files carry everything the paper appeals to. Generative AI (Claude, Anthropic) was used throughout as a research assistant — for literature search, for writing and checking the ancillary code, and on the prose. The mathematics, and the responsibility for it, are the author’s.

References

- [FG86] A. Federgruen and H. Groenevelt, *The greedy procedure for resource allocation problems: necessary and sufficient conditions for optimality*, Oper. Res. **34** (1986), no. 6, 909–918.
- [HK16] A. Hanany and R. Kalveks, *Quiver theories for moduli spaces of classical group nilpotent orbits*, arXiv:1601.04020.
- [IK88] T. Ibaraki and N. Katoh, *Resource Allocation Problems: Algorithmic Approaches*, MIT Press, Cambridge MA, 1988.
- [PvW08] K. Purbhoo and S. van Willigenburg, *On tensor products of polynomial representations*, Canad. Math. Bull. **51** (2008), no. 4, 584–592; arXiv:0706.3251.
- [Raj04] C. S. Rajan, *Unique decomposition of tensor products of irreducible representations of simple algebraic groups*, Ann. of Math. (2) **160** (2004), no. 2, 683–704.
- [RGST05] J. Richter-Gebert, B. Sturmfels and T. Theobald, *First steps in tropical geometry*, in: Idempotent Mathematics and Mathematical Physics, Contemp. Math. **377**, Amer. Math. Soc., 2005, pp. 289–317.

- [Alb23] S. P. Albion, *Universal characters twisted by roots of unity*, Algebraic Combinatorics **6** (2023), 1653–1676; arXiv:2212.07343.
- [Alb25] S. P. Albion, *Character factorisations, z -asymmetric partitions and plethysm*, to appear in Combinatorial Theory; arXiv:2501.18520.
- [AS18] C. Ahlbach and J. P. Swanson, *Refined cyclic sieving on words for the major index statistic*, European J. Combin. **73** (2018), 37–60; arXiv:1706.08631.
- [BRS08] H. Barcelo, V. Reiner and D. Stanton, *Bimahonian distributions*, J. London Math. Soc. (2) **77** (2008), 627–646; arXiv:math/0703479.
- [OP19] Y.-T. Oh and E. Park, *Crystals, semistandard tableaux and cyclic sieving phenomenon*, Electron. J. Combin. **26** (2019), no. 4, #P4.39; arXiv:1906.07420.
- [AKL21] P. Alexandersson, E. Kantarcı Oğuz and S. Linusson, *Promotion and cyclic sieving on families of SSYT*, Ark. Mat. **59** (2021), 247–274; arXiv:2007.10478.
- [AA19] P. Alexandersson and N. Amini, *The cone of cyclic sieving phenomena*, Discrete Math. **342** (2019), no. 6, 1581–1601; arXiv:1804.01447.
- [AB19] A. Ayyer and R. E. Behrend, *Factorization theorems for classical group characters, with applications to alternating sign matrices and plane partitions*, J. Combin. Theory Ser. A **165** (2019), 78–105; arXiv:1804.04514.
- [AK22] A. Ayyer and N. Kumari, *Factorization of classical characters twisted by roots of unity*, J. Algebra **609** (2022), 437–483; arXiv:2109.11310.
- [AK25] A. Ayyer and N. Kumari, *Further results for classical and universal characters twisted by roots of unity*, J. Algebraic Combin. **63** (2026), Paper No. 2, 32 pp.; arXiv:2501.00275.
- [CK09] M. Ciucu and C. Krattenthaler, *A factorization theorem for classical group characters, with applications to plane partitions and rhombus tilings*, in: Advances in Combinatorial Mathematics, Springer, Berlin, 2009.
- [GKS90] F. Garvan, D. Kim and D. Stanton, *Cranks and t -cores*, Invent. Math. **101** (1990), no. 1, 1–18.
- [HI24] M. Hidaka and M. Itoh, *The Schur polynomials in all primitive n th roots of unity*, J. Combin. Theory Ser. A **218** (2026), Paper No. 106107; arXiv:2403.10817.
- [Jan73] J. C. Jantzen, *Darstellungen halbeinfacher algebraischer Gruppen*, Bonner Math. Schriften **67** (1973).
- [Kar22] C. Karmakar, *Character factorizations for representations of $GL(n, \mathbb{C})$* , arXiv:2210.03544.
- [Kar24] C. Karmakar, *Character values at elements of order 2*, Proc. Indian Acad. Sci. Math. Sci. **135** (2025), no. 2, Paper No. 41; arXiv:2412.17324.
- [Mar26] C. Marín, *Schur functions at $(1, -1, t, t^{-1})$: a closed form on the non-identity component of $O(4)$* , preprint, 2026; doi:10.5281/zenodo.21463000.
- [JJLW24] Z. Jin, N. Jing, Z. Li and D. Wang, *Universal symplectic/orthogonal functions and general branching rules*, arXiv:2209.00767.
- [KLP09] S. Kumar, G. Lusztig and D. Prasad, *Characters of simply-laced nonconnected groups versus characters of nonsimply-laced connected groups*, Contemp. Math. **478** (2009), 99–101; arXiv:math/0701615.
- [KT99] A. Knutson and T. Tao, *The honeycomb model of $GL_n(\mathbb{C})$ tensor products I: proof of the saturation conjecture*, J. Amer. Math. Soc. **12** (1999), no. 4, 1055–1090.
- [Kum24] N. Kumari, *Factorization of classical characters twisted by roots of unity: II*, J. Pure Appl. Algebra **228** (2024), no. 11, 107714; arXiv:2212.12477.
- [KumT] N. Kumari, *Characters of classical groups twisted by roots of unity*, PhD thesis, Indian Institute of Science, Bangalore, 2023.
- [Lec09] C. Lecouvey, *Parabolic Kazhdan–Lusztig polynomials, plethysm and generalized Hall–Littlewood functions for classical types*, European J. Combin. **30** (2009), no. 1, 157–191.
- [LR34] D. E. Littlewood and A. R. Richardson, *Immanants of some special matrices*, Q. J. Math. **os-5** (1934), no. 1, 269–282.
- [LLT97] A. Lascoux, B. Leclerc and J.-Y. Thibon, *Ribbon tableaux, Hall–Littlewood functions, quantum affine algebras and unipotent varieties*, J. Math. Phys. **38** (1997), no. 2, 1041–1068.
- [LO22] S.-Y. Lee and Y.-T. Oh, *Skew Schur polynomials and cyclic sieving phenomenon*, Electron. J. Combin. **29** (2022), no. 4, Paper 4.6; arXiv:2112.12394.
- [Mac95] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, second edition, Oxford University Press, 1995.
- [RSW04] V. Reiner, D. Stanton and D. White, *The cyclic sieving phenomenon*, J. Combin. Theory Ser. A **108** (2004), no. 1, 17–50.

- [Ste94] J. R. Stembridge, *On minuscule representations, plane partitions and involutions in complex Lie groups*, Duke Math. J. **73** (1994), no. 2, 469–490.
- [Oka98] S. Okada, *Applications of minor summation formulas to rectangular-shaped representations of classical groups*, J. Algebra **205** (1998), no. 2, 337–367.
- [Pra16] D. Prasad, *A character relationship on $GL_n(\mathbb{C})$* , Israel J. Math. **211** (2016), no. 1, 257–270.
- [AF20] A. Ayyer and I. Fischer, *Bijective proofs of skew Schur polynomial factorizations*, J. Combin. Theory Ser. A **174** (2020), 105241; arXiv:1905.05226.
- [Eis07] T. Eisenkölbl, *A Schur function identity related to the (-1) -enumeration of self-complementary plane partitions*, J. Combin. Theory Ser. A **115** (2008), 199–212; arXiv:math/0602401.
- [FJKW05] B. Fauser, P. D. Jarvis, R. C. King and B. G. Wybourne, *New branching rules induced by plethysm*, J. Phys. A **39** (2006), 2611–2655; arXiv:math-ph/0508034.
- [Gav99] F. Gavarini, *A Brauer algebra-theoretic proof of Littlewood’s restriction rules*, J. Algebra **212** (1999), 240–271.
- [EW04] T. J. Enright and J. F. Willenbring, *Hilbert series, Howe duality and branching for classical groups*, Ann. of Math. **159** (2004), 337–375.
- [New51] M. J. Newell, *Modification rules for the orthogonal and symplectic groups*, Proc. Roy. Irish Acad. Sect. A **54** (1951), 153–163.
- [Sun90] S. Sundaram, *Tableaux in the representation theory of the classical Lie groups*, in *Invariant Theory and Tableaux* (Minneapolis, MN, 1988), IMA Vol. Math. Appl. **19**, 191–225, Springer-Verlag, New York, 1990.
- [IOTZ06] M. Ishikawa, S. Okada, H. Tagawa and J. Zeng, *Generalizations of Cauchy’s determinant and Schur’s Pfaffian*, Adv. in Appl. Math. **36** (2006), 251–287.
- [Kin71] R. C. King, *Modification rules and products of irreducible representations of the unitary, orthogonal, and symplectic groups*, J. Math. Phys. **12** (1971), 1588–1598.
- [KT87] K. Koike and I. Terada, *Young-diagrammatic methods for the representation theory of the classical groups of type B_n, C_n, D_n* , J. Algebra **107** (1987), 466–511.
- [Doc] For the name and the surrounding family of identities (Cassini, Catalan, d’Ocagne, Vajda) see e.g. M. Bunder and J. Tonien, *Cassini, d’Ocagne and Vajda identities for n -step Fibonacci numbers*, arXiv:2201.06269.
- [KS26] N. Kumari and A. Stokke, *Murnaghan–Nakayama rules for symplectic, orthogonal and orthosymplectic Schur functions*, Discrete Math. **349** (2026), no. 2, Paper No. 114741; arXiv:2411.11447.
- [Fro25] A. Frohmader, *Graded multiplicities in the Kostant–Rallis setting*, arXiv:2312.11295.
- [Kra05] C. Krattenthaler, *Advanced determinant calculus: a complement*, Linear Algebra Appl. **411** (2005), 68–166; arXiv:math/0503507.
- [Kup02] G. Kuperberg, *Symmetry classes of alternating-sign matrices under one roof*, Ann. of Math. (2) **156** (2002), 835–866.
- [Lit50] D. E. Littlewood, *The Theory of Group Characters and Matrix Representations of Groups*, 2nd ed., Oxford University Press, 1950.
- [SK23] V. Sathish Kumar, *Flagged skew Schur polynomials twisted by roots of unity*, arXiv:2306.10289.
- [Sta86] R. P. Stanley, *Symmetries of plane partitions*, J. Combin. Theory Ser. A **43** (1986), 103–113.

INDEPENDENT RESEARCHER

Email address: karlesmarin@gmail.com